Distributional Dynamics*

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Abstract

We develop a new framework for estimating high-frequency distributional dynamics across multiple economic variables. Our approach delivers synthetic microdata by combining existing microdata sources that can differ in frequency, variable coverage, and measurement concepts. Central to the framework is treating distributions as functional time series and representing their dynamics through a factor structure. We embed these factors in a state-space model and use Bayesian estimation to recover the underlying distributions from sparse and noisy observations. We apply the framework to the joint distribution of consumption, income, and wealth, which is central to modern heterogeneous-agent macroeconomics. Using a wide range of U.S. microdata and macroeconomic time series, we construct the first quarterly estimates of this joint distribution from 1962 to 2024. We demonstrate that our approach reliably recovers its evolution, even when direct observations are sparse or unavailable. These new data provide previously unavailable empirical counterparts to the distributional dynamics at the heart of modern heterogeneous-agent macroeconomics.

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¶The replication code and distributional time series are available at: **distributionaldynamics.com**.

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1 Introduction

Macroeconomic dynamics are the outcome of decisions made by workers, households, firms, and other economic agents. A central development in modern macroeconomics has been to build increasingly rich microfoundations for economic decision-making and its macroeconomic consequences. With heterogeneous agents, macroeconomic dynamics generally depend on the distribution of individual states: Agents' positions along multiple dimensions shape how they respond to shocks and how these responses aggregate.[1]

This development in economic theory has moved ahead of measurement. In the spirit of Prescott (1986), progress in quantitative macroeconomics rests on continuous feedback between theory and measurement: theory identifies economically relevant objects, while measurement provides the empirical discipline for further theoretical progress. Heterogeneous-agent macroeconomics has reached such a point: Models characterize rich distributional states and their evolution at business-cycle frequency, while their empirical counterparts are typically observed only infrequently and incompletely. This measurement gap limits our ability to confront these theories with the distributional dynamics at the core of their mechanisms. Quantitative models are therefore often disciplined and evaluated largely through steady-state cross-sections, selected moments, or aggregate time series rather than the distributional dynamics.[2] This paper closes this gap.

We develop a novel framework to estimate the *distributional dynamics* of the macroeconomy. The idea is to treat the joint distribution as a time series of functions and summarize its dynamics through low-dimensional factor structures embedded in a state-space model. This allows us to combine multiple sources of micro- and macro-data and recover the high-frequency evolution of the distribution even when the underlying microdata are sparse, observed at different frequencies, or cover only subsets of the variables of interest. The new framework is general and therefore applicable to a broad class of joint distributions. Its basic data requirement is that the variables defining the joint distribution be

1. See, e.g., Kaplan, Moll, and Violante (2018), Bayer, Luetticke, Pham-Dao, and Tjaden (2019), Auclert (2019), and Bilbiie (2020), and, for the cross-sectional effects of policy, Mian, Straub, and Sufi (2020), Holm, Paul, and Tischbirek (2021), Andersen, Johannesen, Jørgensen, and Peydró (2023), and Bhandari, Evans, Golosov, and Sargent (2021). Kaplan and Violante (2018) and McKay and Wolf (2023) survey this literature.

2. An exception is provided by carefully designed environments with block recursivity, where equilibrium objects can be characterized without keeping track of the full cross-sectional distribution (Menzio and Shi 2011). Our focus is the broad class of heterogeneous-agent models in which the distribution is part of the equilibrium state.

observed together in at least one source over several, potentially infrequent, periods.

A second contribution addresses a distinct measurement challenge that arises when drawing on multiple microdata sources: different datasets typically measure the same underlying economic object using different sampling designs, definitions, and measurement concepts. Rather than assuming that these sources coincide, we treat them as complementary, imperfect measurements of a common underlying distribution and explicitly allow for sampling uncertainty, measurement error, and persistent or time-varying differences in operationalization. The novel framework thereby produces *consensus estimates* of the underlying distributional dynamics from the heterogeneous information available to the researcher.

The paper's third contribution is to apply the framework to construct high-frequency estimates of the *joint distribution* of consumption, income, and wealth for the United States, delivering the first quarterly estimates of this joint distribution from 1962Q3 to 2024Q1. We make these estimates publicly available on a dedicated website, distributionaldynamics.com, as a data resource for further empirical and quantitative work. While consumption, income, and wealth constitute a particularly important application, the method can also be used to study other economically relevant joint distributions, such as liquid and illiquid assets and different forms of household debt, or productivity, employment, capital, and leverage across firms.

Turning to the details of the framework, we adopt the understanding that joint distributions are functional data and hence infinite-dimensional objects. Because describing their dynamics requires more than tracking a finite set of moments, it poses a statistical challenge. Our method, which builds on the contribution of Chang, Chen, and Schorfheide (2024), addresses this challenge by projecting these infinite-dimensional objects onto a finite set of basis functions and using functional principal component analysis to uncover a low-dimensional factor structure in their dynamics. The resulting factors form the distributional states of the state-space model. Importantly, the dimensionality of this factor structure is not imposed by the method: the number of retained factors is selected from the data and may therefore differ across applications. Nothing in the framework requires the number of retained factors to be small. If an application calls for a rich factor space, the method carries through unchanged, albeit at greater computational cost.

Heterogeneous-agent theory provides a rationale for why rich distributional

dynamics may nevertheless admit a low-dimensional representation. In our application, household decisions, and hence the distributions generated by them, respond to a relatively small set of aggregate prices such as wages, asset returns, taxes, and transfers (Auclert, Bardóczy, Rognlie, and Straub 2021; Bayer, Born, and Luetticke 2024). More generally, heterogeneous-agent models often share the feature that individual decisions depend on a limited set of aggregate equilibrium objects. Reduced-form evidence also supports a close link between distributions and a small number of aggregate prices.[3] These aggregate objects are, in turn, closely tied to the macroeconomy, which itself admits a low-dimensional factor structure (Stock and Watson 2002). Consistent with this reasoning, in our application we retain eight factors, which capture 99% of the business-cycle-frequency variation in the distributional data, and we show that the implied approximation error relative to the observed microdata is negligible.

Given this factor representation, the state-space model allows us to estimate the distributional dynamics using Bayesian techniques and link them to aggregate variables without imposing a fully structural macroeconomic model. Its measurement equation provides the flexibility needed to combine micro-datasets that measure the same underlying object but differ in sampling uncertainty, frequency, variable coverage, and measurement concepts. It also allows datasets containing only subsets of the variables of interest to inform the underlying joint distribution (Schorfheide and Song 2015; Durbin and Koopman 2012). The Kalman smoother then recovers high-frequency movements in the complete joint distribution, including periods in which direct microdata observations are unavailable.

We apply our method to the joint distribution of household consumption, income, and wealth in the United States. Observing all three variables jointly is central to confronting heterogeneous-agent theory with the data. Households with similar current income can occupy very different positions in the wealth distribution. Consider, for example, a young household with low income and little accumulated wealth and a retired household with similarly low current income but substantial wealth. The two households have very different wealth buffers and adjustment margins and may therefore respond very differently with their consumption to the same shock (Carroll, Hall, and Zeldes 1992). The joint distribution of income and wealth thus shapes the consumption response, and empirically evaluating this mechanism requires observing consumption together with

3. Papers on the price-distribution nexus include Di Maggio, Kermani, and Majlesi (2020), Chodorow-Reich, Nenov, and Simsek (2021), and Kuhn, Schularick, and Steins (2020).

both state variables. More generally, economically relevant household types are often characterized by combinations of states rather than by any single marginal distribution, as illustrated by the wealthy hand-to-mouth (Kaplan, Violante, and Weidner 2014). The three-dimensional joint distribution of consumption, income, and wealth therefore provides the empirical counterpart needed to characterize heterogeneous consumption dynamics and confront quantitative theories with their distributional predictions (Auclert 2019; Bilbiie 2020; Patterson 2023; Bilbiie et al. 2025).

Furthermore, the available U.S. microdata illustrate precisely the measurement problem addressed by the paper's second contribution. No single survey provides comprehensive information on consumption, income, and wealth at high frequency over a long time horizon. We combine the *Panel Study of Income Dynamics* (PSID), *Survey of Consumer Finances* (SCF), *Consumer Expenditure Survey* (CEX), *Survey of Income and Program Participation* (SIPP), and *Current Population Survey* (CPS), complemented by a comprehensive set of macroeconomic time series. The PSID satisfies the requirement that all variables defining the joint distribution are observed together over several periods, while the other datasets provide complementary information at different frequencies and with different variable coverage. The datasets also differ substantially in sample design, size, and measurement concepts. Our framework combines this heterogeneous information to construct a consensus estimate of the quarterly joint distribution of consumption, income, and wealth, capturing changes in both the marginal distributions and their cross-sectional dependence.

The complete quarterly distributional time series are available at distributionaldynamics.com. Researchers can access synthetic microdata—weighted decile cells ($10 \times 10 \times 10$) per quarter at the posterior mode, together with five implicates drawn from the posterior—as well as the factors with their posterior draws and the basis coefficients which allows users to generate their moment(s) of interest. We also provide quarterly balance-sheet components of household wealth (real estate, mortgages, consumer credit, stock, and pension assets) for the wealth and income groups of the Distributional Financial Accounts (Batty et al. 2020). The replication package contains the code that builds all estimation inputs from the original survey microdata, so users can also construct their own distributional series. Together, these form a ready-to-use resource for empirical work, model validation, and the development of quantitative theories of heterogeneity, and we will continue to expand it.

Our method recovers distributional states also when they are not directly ob-

served. To assess its empirical performance and establish the reliability of the resulting estimates, we therefore subject the procedure to several validation exercises. First, we show that the factor representation entails almost no loss of information relative to the observed microdata. Second, we test whether the model can recover distributional data that are deliberately treated as unobserved, exploiting their comovement with aggregate variables. We demonstrate this for the consumption distribution in the CEX and the wealth distribution in the SCF. Third, we use simulated data from a theoretical HANK model as a data generating process. In this setting, the true high-frequency distribution is known, allowing us to show that our method closely tracks its movements when supplied only with microdata samples of the size and frequency available in the U.S. data. Finally, we decompose every estimate into the contributions of the individual data sources and report, for each quarter, how much of it is anchored in observed data rather than inferred through the state-space model. This provides transparent information about the empirical provenance of the estimates, alongside measures of their uncertainty.[4]

In the last part of the paper, we return to the paper's initial motivation and use the new distributional time series to close the loop between theory and measurement. As a first use case, we provide empirical counterparts to the distributional objects that feature in theories of consumption dynamics in recessions. We ask whether consumption dynamics are similar across downturns by tracing consumption responses along the joint distribution of income and wealth during the last three U.S. recessions. These objects arise naturally in heterogeneous-agent models, but until now their empirical counterparts have not been observed at business-cycle frequency over a long horizon spanning several recessions with markedly different underlying drivers. Our estimates therefore provide empirical targets against which theories of heterogeneous consumption dynamics can be evaluated.

The results reveal clear and substantial differences across recessions. The household groups experiencing the largest relative consumption changes differ across episodes and, importantly, cannot generally be identified from either income or wealth alone. Wealth differences were particularly important during

4. We also validate the choice of priors in the Bayesian estimation, particularly with respect to measurement error, and show that the state-space model is consistent with the sampling uncertainty of the observed distributional data at the sampling points. We further compare the predicted dynamics of the income and wealth distributions with those implied by the Distributional Financial Accounts methodology (DFA, see Batty, Bricker, Briggs, Friedman, Nemschoff, Nielsen, Sommer, and Volz 2020) and those estimated by the World Inequality Database (Alvaredo, Atkinson, Chancel, Piketty, Saez, and Zucman 2016; Piketty, Saez, and Zucman 2018).

the Dot-com recession, changes in households' joint income and wealth positions around the Global Financial Crisis, and the combination of low income and low wealth during the COVID-19 recession. The distributional transmission of aggregate shocks therefore appears strongly state- or shock-dependent: which households are most affected, and which dimension of their economic position matters most, depends on the nature of the downturn. This complements Bilbiie, Galaasen, Gürkaynak, Mæhlum, and Molnár (2025), who show that automatic stabilizers render household heterogeneity largely irrelevant for aggregate dynamics on average in Norway. Our results for the United States suggest that such averages can mask substantial variation across episodes, consistent with the time- and state-dependent role of heterogeneity emphasized by McKay and Wolf (2023).

The new estimates also show that marginal distributions alone do not capture all relevant distributional dynamics. Aggregate dynamics account for close to 60 percent of the fluctuations in how consumption, income, and wealth comove, but only about a quarter of the variance in their marginal distributions. The business cycle is therefore felt more strongly in the dependence structure than in the marginals, complementing the structural findings of Bayer, Born, and Luetticke (2024). Incomplete-markets theory provides a natural interpretation: wealth provides a buffer against income risk, so the relationship between income and wealth contains information about households' ability to self-insure (Krueger, Mitman, and Perri 2016). Changes in this relationship over the business cycle therefore reveal dimensions of household insurance capacity that are invisible in the marginal distributions alone.

Recessions, moreover, differ in the balance between the forces driving these dynamics. A historical decomposition of the factor paths shows that autonomous innovations dominate during the Dot-com recession and the Global Financial Crisis, whereas aggregate innovations play a larger role in the COVID-19 recession. More broadly, this first application already illustrates some of the new possibilities opened up by measuring distributional dynamics at business-cycle frequency, as these new measurements provide previously missing evidence to distinguish mechanisms, discipline quantitative models, and study how shocks and policies reshape the distribution of economic activity across heterogeneous agents.

The remainder of this paper is organized as follows: Section 2 provides an overview of the relevant literature. Section 3 develops our estimation method and Section 4 evaluates its quality. Section 5 presents an application of the method.

Finally, Section 6 concludes the paper. An appendix follows.

2 Literature

Relevant methodological papers. The paper most closely related to ours is the important contribution of Chang, Chen, and Schorfheide (2024), which develops a Bayesian state-space approach to estimate the coevolution of technology shocks, GDP, employment, and the marginal distribution of earnings. Their earnings distribution is observed in a single, data-rich source (the CPS, with 60,000 observations), at a single frequency, and without missing observations (1989–2017)—conditions that permit carrying the functional coefficients (to be explained in Section 3) directly as states. None of this carries over to our problem, which leads our work to differ in three respects.

First, our object is the *joint* distribution of consumption, income, and wealth, together with a richer set of aggregates: for heterogeneous-agent macroeconomics, the economically relevant information lies in the dependence structure—which balance sheets absorb an income loss, whose consumption responds to a wealth revaluation—and representing it raises the dimension of the functional coefficients from dozens to thousands. Carrying them as states becomes infeasible. Following Kneip and Utikal (2001) and Tsay (2016), we instead estimate the state-space model in a lower-dimensional factor space obtained through functional PCA (see also Meeks and Monti 2023).

Second, no single dataset observes our object continuously and/or directly: We combine five surveys that differ in variable coverage, frequency, sampling design/size, and measurement concepts, so missing observations and mixed frequencies are the rule rather than the exception—requiring the mixed-frequency measurement framework with dataset-specific, design-based error structures at the core of the method.[5] Third, our output is not only a method but a public data product: a quarterly panel of joint distributions for the United States from 1962 to 2024—as factor estimates, coefficients, or microdata—with posterior uncertainty and source attribution for every estimate, currently unavailable off the shelf.

Another relevant contribution is Marcellino, Renzetti, and Tornese (2024), who nowcast the distribution of *total* household income, also from the CPS, using high-frequency macroeconomic and financial indicators: each annual distribution is represented by a kernel-smoothed log-quantile-density (Petersen and

5. While there is a nascent literature (see, e.g., Ettmeier, Kim, and Schorfheide 2024) emphasizing the role of unit-level dynamics in properly addressing the mixed-frequency problem, we abstract from this consideration in the present study and follow the standard mixed-frequency approach.

Müller 2016), reduced by a truncated Karhunen–Loève expansion, and predicted by mixed-data-sampling (MIDAS) regressions. The first two differences above apply here as well; in addition, we obtain now- and backcasts as smoothed states of the mixed-frequency state-space model of Schorfheide and Song (2015), rather than as regression predictions.

Our representation itself differs from both papers because it must accommodate a multivariate distribution and different datasets informing the *same* object. We represent that distribution using orthonormal polynomials. This choice has three important benefits. It turns the mathematical restrictions needed for a valid distribution into simple restrictions on the coefficients (again, explained later); it lets datasets with only partial information contribute to the same set of underlying coefficients as datasets with complete information; and it provides a natural way to compare distributions when extracting factors. Additionally, compared with methods based on log-densities or log-quantile densities, our rank-based representation keeps many economically relevant quantities linear and avoids kernel smoothing. This means that all of the available survey data can be combined within one linear filtering framework.

More broadly, our method builds on the literature that formulates temporal disaggregation in a state-space framework (see e.g., Harvey and Chung 2000) and on functional data methods in economics (see e.g., Kneip and Utikal 2001; Diebold and Li 2006; Chang, Kim, and Park 2016). The key advantage over (primarily static) methods such as Chow and Lin (1971) and Fernandez (1981) is that the state-space representation is inherently dynamic and lends itself to the Kalman filter for estimation and diagnostics.

Macroeconomic motivation. Our contribution relative to the growing body of work linking macroeconomic aggregates with distributional dynamics (see e.g., Baumeister et al. 2025; Koop et al. 2026; Ettmeier, Kim, and Schorfheide 2024) is to treat the *joint* distribution—marginals and their dependence structure—as the state variable itself, recovering its dynamics from multiple partially overlapping surveys at mixed frequencies. This provides the still-missing descriptions of the short-run dynamics of the consumption, income, and wealth distribution to the theoretical literature on heterogeneity and aggregate dynamics (Kaplan, Moll, and Violante 2018; Bayer et al. 2019; Bayer, Born, and Luetticke 2024) and extends the rapidly growing empirical work on the effects of policy shocks on marginal distributions.[6]

6. See, for example, Berger, Bocola, and Dovis (2023), Coibion et al. (2017), Cloyne, Ferreira, and Surico (2020), Holm, Paul, and Tischbirek (2021), Chang and Schorfheide (2024), and

Through our application, we also speak to a central debate on macroeconomic amplification from household inequality. Auclert (2019) and Bilbiie (2020) formalize the idea that amplification depends on the covariance between the marginal propensity to consume and the cyclicality of income. Patterson (2023) finds this covariance to be strongly positive for U.S. labor earnings, creating a powerful "matching multiplier"; recent evidence from Bilbiie et al. (2025) for Norway provides a counterpoint, showing that once disposable income is considered, automatic stabilizers render the covariance negligible over *average* business cycles. Our synthetic data reveal that consumption distribution dynamics are potentially *very* business-cycle-specific, which motivates examining how these mechanisms vary across episodes.

Distributional measurement. The paper also contributes to a large empirical literature, beginning with Piketty and Saez (2003), measuring trends and fluctuations in inequality. Most of this work produces high-frequency data for *marginal* distributions—income or wealth separately—but not for their joint distribution. Blanchet, Saez, and Zucman (2022) construct monthly income and wealth distributions for the United States; Smith, Zidar, and Zwick (2023), using administrative data and the capitalization method of Saez and Zucman (2016), provide high-frequency wealth estimates concentrated on the upper tail. The Distributional Financial Accounts (Batty et al. 2020) produce quarterly balance-sheet estimates since 1989 by combining SCF microdata with the Financial Accounts through Chow-Lin/Fernández-type models. Our state-space approach offers three advantages over these methods: it treats the underlying microdata as noisy samples of a time series of distribution functions, explicitly handling sampling uncertainty in a dynamic setting; it can combine multiple microdata sources with different operationalizations of the same economic concepts; and it recovers the *joint* distribution of consumption, income, and wealth—not just individual marginals—going back to the 1960s.

3 Method

This section describes a general method for generating high-frequency estimates of joint distributions of economic variables of interest over a large number of micro-units. The method proceeds in four estimation stages, followed by reconstruction, as summarized in Figure 1 and based on the idea that the dynam-

Bartscher et al. (2022). McKay and Wolf (2023) survey this literature.

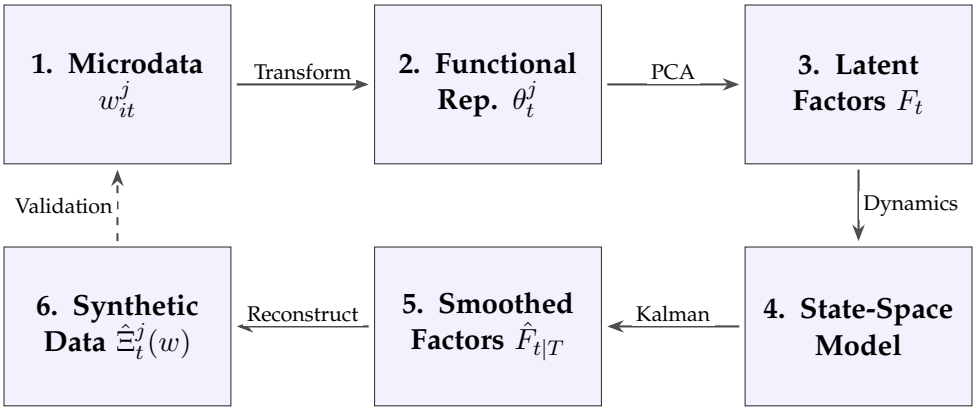


Figure 1: Visual Roadmap of the Methodology

ics of distributions are functional time-series data (Section 3.1). First, starting from microdata, we estimate and represent joint distributions by their quantile functions coupled with a copula, isolating marginal shapes from their dependence structure and projecting these functions onto Legendre polynomials (Section 3.2.1). Second, we reduce dimensionality using functional principal component analysis (PCA) (Section 3.2.2). Third, we formulate the dynamics of these latent factors as a state-space model that handles mixed frequencies (Section 3.2.3). Fourth, we estimate the system using Bayesian techniques (Section 3.2.4) and recover the full distributional time series via the Kalman smoother (Section 3.3). The reverse provides us with the high-frequency synthetic distributional data.

This method uses microeconomic and aggregate data as inputs. It requires only the joint observation of the microeconomic variables in at least one dataset over several, but potentially infrequent, time periods. The developed method treats the distributional data as functional data in a time-series state-space framework with unobserved states. In the following, we describe the method using the example of the joint distribution of consumption, income, and wealth, an important macroeconomic application.

3.1 Distributions as time series of functional data

We consider a sequence $\Xi_t(w)$ of multidimensional distribution functions, indexed by $t \in \mathbb{T} := \{1 \dots T\}$, defined over a d-dimensional *vector* $w \in \mathbb{R}^d$. In the case of our application, we have $d = 3$, where w is a vector of consumption, income, and wealth at the household level. In addition to this sequence of distribution *functions*, there is a sequence of real-valued vectors Y_t of stationary aggregate data. In the following exposition, we assume that Y_t is observed at all times $t \in \mathbb{T}$. The extension to missing observations in Y_t is standard.

From the distributions $\Xi_t(w)$, we observe only randomly drawn samples. We allow these samples to come from different sampling procedures or to have different operationalizations of the underlying theoretical variables. For example, the PSID and the SCF use different sampling procedures and slightly different concepts of wealth and income. We index each of the sampling procedures/datasets by $j = 1 \dots J$. All of these different datasets are typically not observed for all time periods. Instead, dataset j is only observed in a particular subset $\mathcal{T}^j \subset \mathbb{T}$. In addition, not all samples contain all variables of interest, but may contain only a subset $\mathcal{D}^j \subseteq \mathbb{D} := \{1, \dots, d\}$ of variables. For example, the Current Population Survey (CPS) provides only income information but neither wealth nor consumption. However, at least one dataset j must contain all variables of interest, i.e., $\mathcal{D}^j = \mathbb{D}$. In our application, this dataset is the PSID, which contains information on consumption, income, and wealth (for some years).

Our goal is to obtain estimates of the joint distribution functions, $\hat{\Xi}_t(w)$, $\forall t \in \mathbb{T}$, by efficiently combining information from the various related microdata sources and aggregate information, Y_t. We assume that there is a time-series structure such that the density $d\Xi_t$ evolves according to the functional difference equation

$$d\Xi_{t+1} = G\left(d\Xi_t, Y_{t-\ell:t}\right) + \epsilon_{t+1}, \tag{1}$$

where $Y_{t-\ell:t}$ are observed lagged aggregate data.[7] G determines the dynamics of the system, and ϵ_t are the corresponding shocks to the functional equation.[8] This time-series structure arises naturally in so-called HANK models (see e.g. Kaplan, Moll, and Violante 2018; Bayer, Born, and Luetticke 2024). It is assumed that Ξ_t is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^d$, so that a density $d\Xi_t$ exists; we additionally assume $d\Xi_t$ to be continuous. Moreover, we require that the operator G is Lipschitz-continuous in its first argument, a condition satisfied mechanically under the proposed linear model. We provide details below.

Viewing the J sampling procedures as capturing the same fundamental object Ξ_t but with some measurement error, ν_t^j, allows us to combine the data in a

7. We deliberately leave the law of motion of Y_{t+1} unspecified at this point; the system is completed in Section 3.2.3, where the aggregates follow their own transition equation and are neither assumed exogenous to ϵ_{t+1} nor to $d\Xi_t$.

8. To ensure Ξ_{t+1} is a valid distribution: $\int G(d\Xi, \cdot)(w)\, dw = 1$, $\int \epsilon_t(w), dw = 0$, and $G(d\Xi, Y_{t-\ell:t})(w) \geq -\epsilon_{t+1}(w)$ (non-negativity). Non-negativity also applies to $d\tilde{\Xi}$ and thus restricts ν in (2).

systematic way. This means that a dataset gives us an estimate

$$d\tilde{\Xi}_t^j = \int_{\mathbb{R}^{\mathbb{D}\setminus\mathcal{D}^j}} d\Xi_t + \nu_t^j \qquad \text{for} \quad t \in \mathcal{T}^j, \tag{2}$$

where the integral runs over the components of w unobserved in dataset j—those indexed by $\mathbb{D} \setminus \mathcal{D}^j$—which are thereby integrated out.[9] The measurement error then captures sampling uncertainty, differences in sampling procedures, and differences in operationalization of economic concepts, all of which vary by dataset j.

3.2 Implementing the Estimation

Estimating (1) directly is not feasible because it is an infinite-dimensional nonlinear functional difference equation and, of course, we only observe samples of the distribution functions, not the functional data itself. Our innovation is to overcome this challenge by making it possible to estimate (1) using traditional Bayesian techniques and a Kalman-filter (Section 3.2.4). This requires transforming (1) into a linearized (infinite-dimensional) state-space model (Section 3.2.3), which is estimable once we render it finite and of reduced dimensionality under an appropriate factor representation (Section 3.2.2). However, first, we need to operationalize the measurement of the distribution functions as they appear in (2) by transforming the microdata samples into estimates of the distribution functions themselves (Section 3.2.1). In doing so, we have to account for changes in the effective support of the distributions and deal with the unobservability of some of the micro-variables in some datasets. In the following, we cover these points, which ultimately define our estimation procedure.

3.2.1 Transforming the microdata

Handling changes in scale One challenge in working with distributional data is that the magnitude of the variables of interest in w, and thus the support of $\Xi(w)$, changes over time. The population support of w may well be regarded as fixed; what varies is the *effective* support—the region containing all but a negligible fraction of the mass. Secular growth in incomes and wealth shifts and dilates the distribution, and the realized extremes in any finite sample expand or contract the range over which Ξ_t can be estimated. This is particularly important when considering long time-horizons.

9. For clarification, the tilde ($d\tilde{\Xi}$) is to denote that this is an estimate from the data vs. an estimate generated from the model, which will be denoted by a hat ($d\hat{\Xi}$).

This variation in the support poses two problems. First, over time, any finite-dimensional representation of Ξ_t requires a common domain. If *instead* one would estimate Ξ_t on time-varying domains, the finite dimensional description could no longer be posed consistently in the form of (1)—our object of interest. Second, across datasets *within* a period, sampling designs differ in their coverage of the tails, so dataset-specific estimates are defined on dataset-specific domains and cannot be interpreted as noisy measurements of a common underlying object, as the measurement equation (2) requires.

We deal with this in two ways. First, to deal with level changes, we rescale the vector w observed for individual i in micro-dataset j at time t, w_{it}^j, by its dataset- and time-specific mean $\bar{w}_t^j$.[10] Second, to deal with changes in the width of the effective support, we change domain from levels to ranks. Assigning each household its rank in the cross-section—the poorest household always occupying the lowest rank and the richest the highest, regardless of the levels involved—maps every measure into the unit interval, a domain that is constant over time and across datasets by construction.

Two objects then summarize the distribution without loss of information. The mapping from ranks back to levels—the quantile function, i.e., the inverse of the marginal cumulative distribution function—recovers each marginal distribution, one function per measure. The joint distribution over ranks across measures—by definition, a copula—captures their dependence. Together, the quantile functions and the copula contain the same information as Ξ_t itself, but in contrast to Ξ_t, all of these functions are defined on the fixed domains $[0,1]$ and $[0,1]^d$: their supports are time-invariant by construction. Sklar's theorem (Sklar 1973) states this formally: any multivariate distribution function can be decomposed into its marginals and a copula:

$$\Xi_t(w) \;=\; C_t\big(\Xi_{1t}(w_1), \ldots, \Xi_{mt}(w_m), \ldots, \Xi_{dt}(w_d)\big), \tag{3}$$

where Ξ_{mt} denotes the marginal distribution of measure $m \in \mathbb{D}$ and the copula $C_t : [0,1]^d \to [0,1]$ is itself a distribution function with uniform marginals—unique under our maintained assumption that Ξ_t is absolutely continuous. The copula isolates the dependence structure (e.g., whether high-income households also tend to hold high wealth) from the shapes of the individual distributions

10. Estimating relative to dataset-specific time means also lets us flexibly align the synthetic distributions with per-capita or per-household aggregate targets: we simply rescale using the desired aggregate target rather than the dataset mean. A consensus business-cycle component can be obtained by rescaling with average fixed effects and omitting any trend.

(e.g., how skewed income is), and it is invariant to strictly increasing transformations applied componentwise to w—in particular, the inverse-hyperbolic-sine treatment of the marginals introduced below leaves the copula unchanged.

For our application, the copula is

$$\begin{aligned} C_t(u_1, \ldots, u_d) &= P_t(U_1 \leq u_1, \ldots, U_m \leq u_m, \ldots U_d \leq u_d) \\ &= P_t\left(w_1 \leq \Xi_{1t}^{-1}(u_1), \ldots, w_m \leq \Xi_{mt}^{-1}(u_m), \ldots, w_d \leq \Xi_{dt}^{-1}(u_d)\right) \qquad \forall t \in \mathbb{T} \end{aligned} \tag{4}$$

where $U_m \sim U[0,1]$ for $m \in \mathbb{D}$ are the uniform marginals generated by taking the probability integral transform of each component m s.t. $U_m = \Xi_m(w_m) \sim U[0,1]$. The second line highlights the quantile functions or the inverse transform of the univariate CDFs, $\Xi_{mt}^{-1}(u_m)$, where:

$$\Xi_{mt}^{-1}(u_m) = \inf\{w_m \in \mathbb{R} : \Xi_m(w_m) \geq u_m\} \qquad \forall t \in \mathbb{T}, m \in \mathbb{D}. \tag{5}$$

Finally, for C_t to be a copula, the constraint must hold for all $m \in \{1 \ldots d\}$ that when integrating out all but one dimension, i.e., setting all $u_j = 1$ except for $j = m$ (the marginal distribution), the copula is identical to the value of the marginal distribution:

$$C_t(1, \ldots u_m, \ldots 1) = u_m. \tag{6}$$

Dealing with the infinite-dimensionality of the distribution functions The second challenge in our endeavor is dealing with the infinite-dimensionality of the distribution functions, while maintaining a functional approach. To do so, we will rely on a series estimator for both the copula and quantile functions. Series estimators project the functions of interest onto some infinite-dimensional space of polynomials, namely the space of square-integrable functions $\mathcal{L}^2$.

In the following, we will work with the copula density dC_t, which contains the same information as C_t under absolute continuity and provides several advantages that we explain later. Treat copula densities dC_t and (transformed) quantile functions Ξ_{mt}^{-1} as elements of $\mathcal{L}^2$, and project them onto a space of shifted orthonormal Legendre polynomials $\{Q_o : o \in \mathbb{N}_0\}$ for some order o, satisfying

$$\int_{[0,1]} Q_o(x)\, Q_k(x)\; dx = \delta_{ok}, \tag{7}$$

with $\delta_{ok} = 1$ if $o = k$ (reflecting the normalization condition) and 0 otherwise by orthogonality. Classical Legendre polynomials are defined on $[-1, 1]$, so the

shift is to the same domain as the copula densities and quantile functions—in the support $[0,1]$. The orthonormal shifted Legendre polynomials Q_o themselves are defined on $[0,1]$ by

$$Q_o(x) = \sqrt{2o+1}\, L_o(2x-1), \tag{8}$$

where L_o are the classical Legendre polynomials defined by

$$L_0 = 1, \qquad L_1(x) = x, \qquad (o+1)L_{o+1}(x) = (2o+1)\, x\, L_o(x) - oL_{o-1}(x). \tag{9}$$

One concern is that the approximation method could be poorly chosen, as the underlying population functions are themselves not completely square-integrable. For example, empirical evidence for distributions of consumption, income, and wealth points to tail behaviour that violates the $\mathcal{L}^2$ condition (see, e.g., Vermeulen 2018).[11] For the copula density, the same issue arises—density at the tails may fail to lie in the fulldomain space $\mathcal{L}^2([0,1]^d)$ for copulas exhibiting non-zero tail dependence e.g., the Gumbel copula.

To ensure that these underlying functions lie in $\mathcal{L}^2$, rendering the projection apropos, we first treat the data and apply the inverse-hyperbolic-sine transformation on them. This regularizes the upper tail of the distribution. This treatment is similar in nature to working with log densities, as in Chang, Chen, and Schorfheide (2024).[12] We then estimate our quantile functions on these transformed values. Appendix A provides evidence that by using the transformed percentile functions, the projection performs well at capturing variation in the tails. For the copula reconstruction (after having estimated the model), we impose a *local* $\mathcal{L}^2$ condition on a trimmed domain $[\varepsilon, 1-\varepsilon]^d$, for some very small $\varepsilon > 0$. We assume that inside that interior cube, $dC_t(\mathbf{u})$ is bounded and hence square-integrable.[13] Appendix A also addresses the treatment of measures (e.g., stocks) in the presence of non-negligible zero-mass.

11.
$$\int_a^b \left[Q(p)\right]^2 dp = \int_a^b \Big[\underbrace{x_m\,(1-p)^{-1/\alpha}}_{\substack{\text{Pareto}\\ \text{quantile}}}\Big]^2 dp = x_m^2 \int_a^b (1-p)^{-2/\alpha}\, dp. \tag{10}$$

In particular, if the uppertail of the income distribution on $p \in [a,b]$ follows a Pareto law with shape α, then its quantile function is $Q(p) = x_m(1-p)^{-1/\alpha}$. This lies in $\mathcal{L}^2[0,1]$ (i.e. $\int_a^b Q(p)^2\, dp < \infty$) if and only if $\alpha > 2$. When $\alpha \leq 2$, the integral diverges and the second moment does not exist.

12. For notational convenience, we use $\Xi_{mt}^{-1}(u_m)$ as the treated quantile function.

13. A simulation study shows that our chosen estimator minimizes the mean integrated squared error relative to other nonparametric approaches such as, beta-kernel smoothing, empirical copulas, and checkerboard methods, especially when fitting bi- and trivariate copulas with moderate to strong tail dependence, which may be the case here. See Bakam and Pommeret (2023) for more details.

Altogether, this means the following representation:

$$\Xi_{mt}^{-1}(u_m) = \sum_{o \in \mathbb{N}_0} \xi_{mot} Q_o(u) \tag{11}$$

$$dC_t(u_1, u_2, ...u_d) = \sum_{(o_1,...,o_d) \in \mathbb{N}_0^d} \kappa_{(o_1,...,o_d),t} \prod_{m=1}^{d} Q_{o_m}(u_m), \quad u_m \in [\varepsilon, 1-\varepsilon] \tag{12}$$

where both functions are represented as infinite sums of polynomials over the order of the polynomials $o \in \mathbb{N}_0$. Because of orthonormality (7) and using (11), for some fixed o, we obtain:

$$\int_0^1 \Xi_{mt}^{-1} Q_o(u) du = \int_0^1 \sum_{k \in \mathbb{N}_0} \xi_{mkt} Q_k(u) \, Q_o(u) du = \sum_{k \in \mathbb{N}_0} \delta_{ko} \xi_{mkt} = \xi_{mot}. \tag{13}$$

This means that the coefficients can be obtained as the inner products of the functions and the Legendre polynomials of some order o. The same holds true for the copula density:

$$\int_{[\varepsilon,1-\varepsilon]^d} \prod_{m=1}^{d} Q_{o_m}(u_m) \, dC_t \; du_1, \ldots, du_d = \kappa_{(o_1,...,o_d),t}. \tag{14}$$

For the estimation of these coefficients in practice, we rely on the uniformity of ranks and that ranks are within $[0, 1]$ and replace the inner product by sample averages:[14]

$$\hat{\xi}_{mot} := N_j^{-1} \sum_i w_{mit} Q_o(u_{mit}) \tag{15}$$

$$\hat{\kappa}_{(o_1,...,o_d),t} := N_j^{-1} \sum_i \left(\prod_{m=1}^{d} Q_{o_m}(u_{mit}) \right) \tag{16}$$

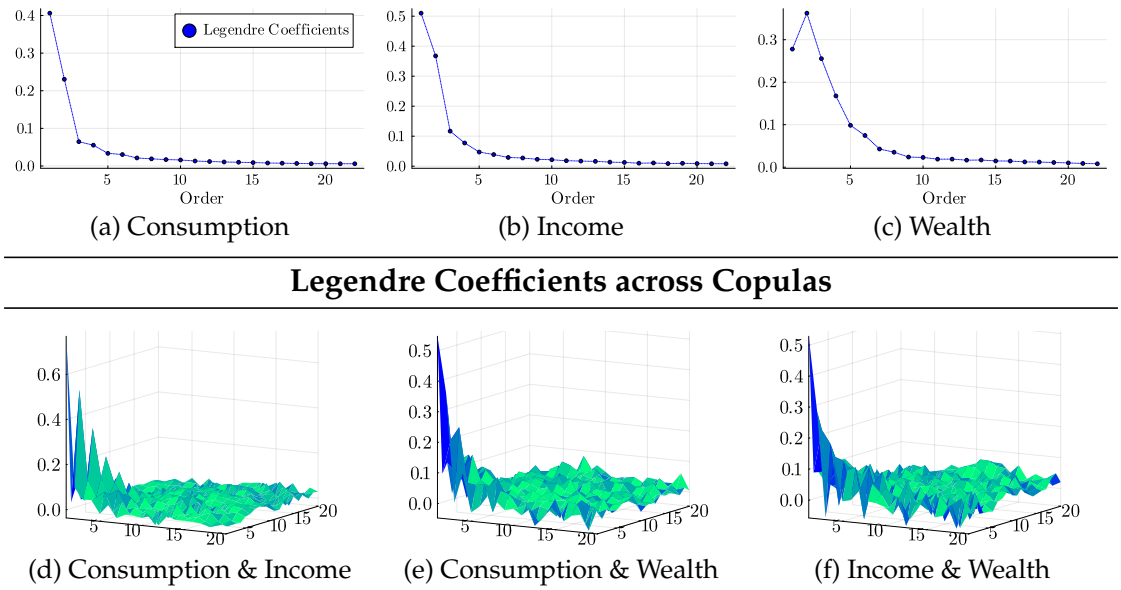
where u_{mit} is the data rank of w_{mit}, the latter being the sample analogue of Ξ_{mt}^{-1} for observation i.

In the estimation, we truncate the sums in (11) and (12) at a given maximal order O and by orthonormality of the polynomials, the kept coefficients are not affected by the truncation. The coefficients in our case indeed decrease rapidly with each polynomial as we show in Figure 2. The figure plots the coefficient value as a function of the order of its respective polynomial term. At around order 10, the coefficients all become very small.[15] Evidence from Appendix A

14. Survey weights are disregarded here for notational convenience but applied in the application

15. A small coefficient in absolute value does imply the contribution of the corresponding poly-

Figure 2: Legendre Coefficients Across the Distributional Data



(a) Consumption (b) Income (c) Wealth

(d) Consumption & Income (e) Consumption & Wealth (f) Income & Wealth

Notes: The first row shows the coefficients (dots) on the Legendre polynomials from estimating the quantile function, by polynomial order. The second row shows the coefficients (surface) on the Legendre polynomials from estimating the copula density, by the polynomial orders of the two variables. Data are from the 2019 PSID.

supports this, showing a truncation to $O = 11$ captures the variation observed in the data quite well.

Are there advantages to using the quantile function here? An alternative in the functional-data literature is to transform the distribution into an unconstrained function space before projecting—log-densities, as in Chang, Chen, and Schorfheide (2024), or the log-quantile-density transformation (Petersen and Müller 2016; Marcellino, Renzetti, and Tornese 2024). Estimation typically requires kernel smoothing, unlike our rank-based coefficients—but avoids imposing non-negativity and monotonicity constraints in a linear model. We find in the end the cost of staying in quantile coordinates is small: the only constraint on the quantile function, monotonicity, we verify ex post by evaluating the Legendre approximation on a fine grid (10,000 points), only finding violations in measures where there is a flat region in the support (e.g., bottom wealth components), with rearrangement of quantiles (Chernozhukov, Fernández-Val, and Galichon 2010) available as a remedy if needed, but avoiding the numerical stability issues of the log-transformation of densities around tails.

nomial to be small in an $\mathcal{L}^2$ sense.

Handling partial observability with the copula density A key advantage and contribution of the procedure concerns a difficulty that any application combining multiple data sources faces: the microdata may not always contain the entire vector w, but only a subset. When w is incompletely observed, we cannot estimate the full d-dimensional copula directly—only copulas with the unobserved dimensions integrated out. For the copula density, these lower-dimensional copulas are exact *slices* of the higher-dimensional coefficient array, so partially observing datasets remain informative about the full d-dimensional copula. The representation in the form of (Legendre) polynomials makes this precise. Integrating out the "missing" dimension d must return the lower-dimensional density,

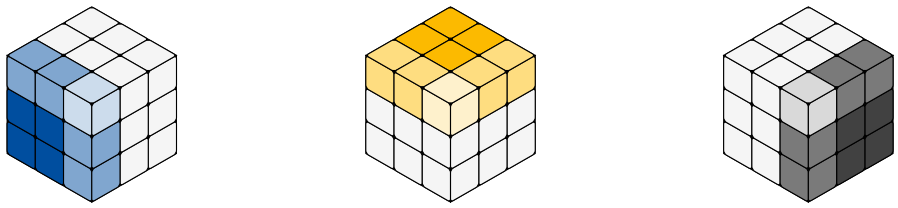
$$\int_{\varepsilon}^{1-\varepsilon} dC(u_1, \ldots, u_d)\, du_d \overset{!}{=} dC(u_1, \ldots, u_{d-1}). \tag{17}$$

Writing out the integrand using (12) and making use of the fact that the order-zero (shifted) Legendre polynomial $Q_0 \equiv 1$ integrates to one and all higher orders to zero, we obtain

$$\begin{aligned}
\int_{\varepsilon}^{1-\varepsilon} dC(u_1, \ldots, u_d)\, du_d &= \int_{\varepsilon}^{1-\varepsilon} \sum_{o_1} \cdots \sum_{o_d} \kappa_{(o_1,\ldots,o_d)} \prod_{m=1}^{d} Q_{o_m}(u_m)\, du_d \\
&= \sum_{o_1} \cdots \sum_{o_{d-1}} \sum_{\mathbf{o_d}} \kappa_{(o_1,\ldots,o_{d-1},\mathbf{o_d})} \left(\prod_{m=1}^{d-1} Q_{o_m}(u_m) \right) \int_{\varepsilon}^{1-\varepsilon} Q_{\mathbf{o_d}}(u_d)\; du_d \\
&= \sum_{o_1} \cdots \sum_{o_{d-1}} \kappa_{(o_1,\ldots,o_{d-1},0)} \left(\prod_{m=1}^{d-1} Q_{o_m}(u_m) \right).
\end{aligned} \tag{18}$$

In words, the polynomial coefficients of the lower-dimensional copula density are identical to the leading "slice" of the higher-dimensional copula. This makes it straightforward to combine datasets with different levels of information. Using the copula distribution instead would lose this property: each lower-dimensional coefficient would depend on many higher-dimensional coefficients, making the relationship much more complicated and inaccurate after truncation. Figure 3 also illustrates another advantage over the distribution. Some coefficients (shown as lighter or white cubelets) are fixed by the requirement that each variable has a uniform marginal distribution. These coefficients are known in advance, never estimated, and are enforced exactly at every step—not true for the distribution. As a result, every estimated copula density automatically satisfies the uniform-margins condition.

Figure 3: Geometric Representation of Partially Observed Copula Coefficients



(a) Only Variables 1& 3 (b) Only Variables 2& 3 (c) Only Variables 1& 2

Notes: Figure shows three cubes. A cube can be interpreted as an array of copula coefficients $\kappa^j_{(o_1,\ldots,o_d),t}$ for some dataset j at time t, for $d = 3$. Each cube corresponds to a scenario where one variable is missing in the estimation of the copula density. The light edge denotes the (0,0,0) coordinate. For each scenario, the white boxes are coefficients we cannot estimate. The slightly colored boxes correspond to the immutable coefficients, which have fixed values independent of data. The darker colored boxes are scenario specific and correspond to (time-varying) coefficients that need to be estimated.

3.2.2 Dealing with the Curse of Dimensionality

Vectorizing the coefficients for each time period, t, leaves us with sequences of coefficients, $\theta^j_t = (\xi^j_{mot}, \ldots, \kappa^j_{(o_1,\ldots,o_d),t})$, for each cross-sectional dataset, j. For example, with $d = 3$ dimensions in our application—consumption, income, and wealth and using polynomials up to order eleven ($O = 11$) to organize the data—the copula density would be represented by a vector with $(O+1)^d - (d \times O + 1) = 1694$ free entries for each time point and $d \times O + 1 = 34$ fixed. These fixed entries are not included in the estimation. In addition, there would be $d \times (O + 1) = 36$ coefficients of the polynomials (including a constant) representing the quantile functions, which we collect in θ^j_t as well.

This example makes apparent that even for a modest polynomial order, the dimensionality of $\theta^j_t \in \mathbb{R}^N, N = (O + 1)^d + (d - 1)$ for a dataset with d variables is large. Equally important is the frequency of missing observations across the N entries; in our application, the data structure is irregular and includes many gaps. Taken together, this makes it impractical to specify and estimate a time-series model directly on the raw θ^j_t coefficients. For this purpose, we postulate (and then estimate) a dynamic factor model for θ. The *orthonormal* representation offers further advantages: The variance (over time) of a coefficient is proportional to its contribution to fluctuations of the function (in the $\mathcal{L}^2$ sense). Put simply: The polynomial coefficient provides a useful form of standardization that has a natural metric and allows us to uncover the factor structure behind the time-series changes in the distributions. This factor structure finally allows us to overcome the curse of dimensionality in the distributional data.

For this purpose, all (variable) coefficients of the polynomial representation

of dC_t^j (and separately of the quantiles) are horizontally concatenated:

$$\boldsymbol{\theta}^j = \begin{bmatrix} \theta_{1,1}^j & & \theta_{1,T}^j \\ & \ddots & \\ \theta_{N,1}^j & & \theta_{N,T}^j \end{bmatrix}.$$

Define $\boldsymbol{\theta}$ as the matrix obtained by horizontally concatenating the individual blocks $\boldsymbol{\theta}^j$, retaining only those time periods for which every coefficient is available.[16] A PCA (a singular value decomposition) is then performed, which nonparametrically reduces the dimensionality of the data (Breitung and Eickmeier 2006; Chen, Er, and Wu 2005). For the PCA, we define $\tilde{\boldsymbol{\theta}}$, which is the standardized representation of $\boldsymbol{\theta}$. The standardization is done by distribution object $\zeta \in \{1, \ldots, d+1\}$ (d quantile functions and a copula).[17] We standardize by object (and not by coefficient) because the polynomial coefficients represent standardized contributions to the object's structure (e.g., the quantile function or copula). As a result, additional standardization within the same object is unnecessary, since their relative magnitudes are naturally balanced by the properties of the polynomial basis. This leaves us with the standardized observation $\tilde{\theta}_{nt}^j$ of coefficient n in data source j at time t.

The PCA of $\tilde{\boldsymbol{\theta}}$ provides us with a projection matrix $\hat{\Gamma} \in \mathbb{R}^{N \times r}$, with full column rank r, that projects $r \ll N$ factors into the N dimensional distributional data. Throughout, we normalize so that each factor has unit in-sample variance, with the loadings $\hat{\Gamma}$ absorbing the scale.[18]

More specifically, we decompose $\tilde{\boldsymbol{\theta}}$ into latent orthogonal factors $\begin{bmatrix} \hat{f}_\Gamma' & \hat{f}_\gamma' \end{bmatrix}'$ divided into "important" and "unimportant" factors according to their contri-

16. Before the concatenation, each coefficient is detrended using an HP-filter. This takes care of dataset-specific effects. We store the information needed to transform $\tilde{\boldsymbol{\theta}}^j$ back to the originally observed objects to obtain source-specific predictions. For example, the income quantiles (that would be one object ζ) in the SCF and the PSID (two sources j) may be permanently different due to differences in sample design and operationalization.

17. The normalization of coefficient n in object ζ in dataset j is given by $\tilde{\theta}_{nt}^j = \left(\frac{\theta_{nt}^j - \mu_n^j}{\sigma_{\zeta(n)}}\right)$ where μ_n^j is the coefficient-specific mean and $\sigma_{\zeta(n)}$ is the standard deviation of *all* coefficients (rows) pertaining to object ζ. This removes dataset- and coefficient-specific fixed effects, μ_n^j. This can capture, e.g., permanent differences in sampling procedure and operationalization. The order of operations is: each coefficient series is HP-detrended within dataset j (see the previous footnote), then demeaned, then scaled. The demeaning is stated for explicitness only—in sample, HP residuals sum to zero, so $\mu_n^j = 0$. The stored trend, mean, and scale are reapplied in reverse order to map model output back to source-specific observables.

18. Concretely, with P the orthonormal principal axes and Λ the diagonal matrix of the corresponding eigenvalues (the sample variances of the principal components) in decreasing order, we set $\hat{\Gamma} := P\Lambda^{1/2}$ and $\hat{f}_\Gamma := \Lambda^{-1/2}P'\tilde{\boldsymbol{\theta}}$: the product $\hat{\Gamma}\hat{f}_\Gamma$ is unchanged, each factor has unit variance, and the scale of the loadings carries the explanatory weight of each factor. The priors on the factor innovations reference this unit-variance convention (Section 3.2.4).

bution to the total variance (measured by their singular value). Their respective time-invariant loadings are $\begin{bmatrix}\hat{\Gamma} & \hat{\gamma}\end{bmatrix}$, for $\hat{f}_\Gamma \in \mathbb{R}^{r\times T}$ and $\hat{f}_\gamma \in \mathbb{R}^{(T_c-r)\times T}$ for T_c number of time periods with observed data. This decomposition is unique up to the scale (and sign) of each factor, which allows us to normalize so that all factors have unit variance, with the loadings absorbing the inverse rescaling. Altogether, we have:

$$\tilde{\boldsymbol{\theta}} = \begin{bmatrix}\hat{\Gamma} & \hat{\gamma}\end{bmatrix} \begin{bmatrix}\hat{f}_\Gamma \\ \hat{f}_\gamma\end{bmatrix} \tag{19}$$

where $\hat{f}_\Gamma$ represents the r important factors, which capture almost all of the variation in the data, and $\hat{f}_\gamma$ the $N-r$ less important factors, which can be interpreted as some measurement noise.[19]

This step delivers two objects: the dimension r of the factor space and the estimated loadings $\hat{\Gamma}$, whose columns constitute the functional basis (see Kneip and Utikal 2001; Tsay 2016). The dimension r determines the size of the state-space model set up in the next subsection. We select r on two complementary grounds: the share of business-cycle-frequency variation explained (eight factors capture 99%) and a comparison of marginal data densities of the estimated state-space model across r (Appendix B)—a comparison that is meaningful precisely because varying r leaves the observation vector unchanged and thus likelihoods comparable. The factor time paths themselves are *not* taken from the PCA: they are latent states, estimated by the Kalman smoother using all datasets—including those excluded from the PCA because of partial coverage—with $\hat{\Gamma}$ held fixed as the linear map between the high-dimensional coefficients $\tilde{\boldsymbol{\theta}}_t$ and the factors f_t. Holding $\hat{\Gamma}$ fixed is part of the modular estimation strategy discussed in Section 3.2.4; Section 3.2.3 makes the mapping precise.

The estimation of $\hat{\Gamma}$ requires time observations where every coefficient is observed, yet some data sources will not satisfy this criterion *at all*—for example, the SCF never measures consumption. Observations from such incomplete

19. By working with an approximate factor model i.e., our principal components estimator, we do not force the idiosyncratic (measurement error) terms to be cross-sectionally independent; they may display weak correlation across series. Consistency, however, requires that this residual dependence remain limited in the Bai-Ng sense (no eigenvalue of the idiosyncratic covariance can diverge with N). We therefore adopt the standard "weak cross-sectional dependence" assumptions used by Bai (2003), Bai and Ng (2006), and, more recently, Freyaldenhoven (2022), which rule out pervasive or strong common shocks beyond those captured by the retained factors; however, we retain components that explain 99% of the total variation, suggesting that any remaining idiosyncratic error is quantitatively small and therefore unlikely to contain an additional pervasive factor. We confirm this by inspecting the largest *residual* eigenvalue and perform standard cross-correlation tests, finding no evidence of strong common factors (shocks) in the residuals, hence, supplying evidence of weak cross-dependence in our context.

sources are therefore excluded from the PCA that yields $\hat{\Gamma}$. In Appendix B, we consider alternative estimators that accommodate partial unobservability in $\tilde{\boldsymbol{\theta}}^j$. Following standard practice, we select among these alternatives the estimator with the highest marginal data density.

3.2.3 Factor State Space Model and Measurement

With this preprocessing of the data, we can turn to specifying the state-space model that captures the evolution of the r latent distributional factors. Due to the possible mixed-frequency nature of the data, we postulate a mixed-frequency state space model as in Schorfheide and Song (2015), with the following augmented state vector

$$F_t \;=\; \left[f_t' \; f_{t-1}' \; \dots \; f_{t-p+1}' \right]' \in \mathbb{R}^{rp \times 1}, \tag{20}$$

where F_t collects the r latent distributional factors f_t, in addition to its $p-1$ lags. For the application of our methodology in Section 5, we set $p = 4$, with the high-frequency data of interest being quarterly in order to allow for annual aggregation in measurement when a dataset collects a sample over an entire year. To facilitate the reading across sections, we provide the example of this quarterly case herein, though the methodology easily generalizes to other frequencies.

State Equation Stacking the lags directly with the q aggregate factors Y_t[20] in the state gives the following representation

$$\begin{bmatrix} F_t \\ Y_t \end{bmatrix} = \Phi \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + \epsilon_t = \begin{bmatrix} \Phi_{FF} & \Phi_{FY} \\ \Phi_{YF} & \Phi_{YY} \end{bmatrix} \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}\big(0,\, \Omega\big) \tag{21}$$

and for the case of quarterly data

$$F_{t-1} = \begin{bmatrix} f_{t-1} \\ f_{t-2} \\ f_{t-3} \\ f_{t-4} \end{bmatrix}, \Phi_{FF} = \begin{bmatrix} A & 0 & 0 & 0 \\ I_r & 0 & 0 & 0 \\ 0 & I_r & 0 & 0 \\ 0 & 0 & I_r & 0 \end{bmatrix}, \Phi_{FY} = \begin{bmatrix} B \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Phi_{YF} = \begin{bmatrix} C & 0 & 0 & 0 \end{bmatrix}, \Phi_{YY} = D,$$

where distributional factors f_t are assumed to follow a VAR(1) process, summarized by the law of motion matrix $A \in \mathbb{R}^{r \times r}$. In addition, they are assumed to

20. Recall that the set of aggregate factors at time t depends on an information set that captures both contemporaneous and past aggregate variation. More on the set of aggregates in Section 4.1.

co-move with aggregate factors Y_t, whose relationship is summarized by loading matrix $B \in \mathbb{R}^{r\times q}$ and $C \in \mathbb{R}^{q\times r}$. The second block row of (21) thus specifies the data-generating process of the aggregate factors: $Y_t = Cf_{t-1} + DY_{t-1} + \epsilon_{Y,t}$. In particular, Y is not assumed exogenous to the distribution: it responds to the lagged distributional factors through C, which is estimated and left unrestricted, and its innovations may be contemporaneously correlated with those of the distributional factors through Ω.

The aggregate factors are principal components and hence contemporaneously orthogonal; we additionally restrict their autoregressive matrix $D \in \mathbb{R}^{q\times q}$ to be diagonal, so that each aggregate factor follows its own autoregression. This is a parsimony restriction—orthogonality does not rule out lagged cross-predictability—and it restricts neither the factors' effect on the distributional factors (B), the feedback from the distribution (C), nor the correlation of their innovations (Ω), through which cross-factor dynamics can still operate. All other matrices A, B and C are left unrestricted. As in Schorfheide and Song (2015), the remaining rows of Φ are defined to deliver the identities $f_{t-l} = f_{t-l}$ for $l = 1, ..., p-1$—hence the I_r in the example of a quarterly estimation.

While it is customary to restrict the innovations ϵ_t to be uncorrelated, both serially and between factors, so that Ω is a diagonal matrix, we do estimate both the variances and covariances of Ω. This allows for a shock structure where innovations to the distributional factors can be *contemporaneously* correlated with innovations to the aggregate factors. This is also how the linearized system accommodates within-period comovement between distribution and aggregates: no variable dated t enters the right-hand side of (21)—the system is strictly first order—and any contemporaneous comovement is captured by the off-diagonal blocks of Ω as correlated innovations, without imposing a causal direction.

Measurement Equation Since distributional factors are not directly observable, we complement the model with an observation equation for each dataset j:

$$\tilde{\theta}_t^j = H_t^j \left(\hat{\Gamma}^{MF,j} F_t + \nu_{F,t}^j \right), \quad \nu_{F,t}^j \sim \mathcal{N}\left(\mathbf{0}, \hat{\Sigma}_j^{1/2} S^j \hat{\Sigma}_j^{1/2\prime} \right) \tag{22}$$

This observation equation maps factors F_t, through the loading matrix $\hat{\Gamma}^{MF,j}$, to the standardized coefficients $\tilde{\theta}_t^j$, allowing for a Gaussian measurement error $\nu_{F,t}^j$. The selector matrix $H_t^j \in \{0,1\}^{N\times N}$ indicates whether (parts of) the distribution are observed in data source j at time t, following Durbin and Koopman (2012). The Gaussian error term $\nu_{F,t}^j$ has a variance-covariance matrix decomposed into S^j, a diagonal scaling matrix with diagonal entries s_ζ^j varying by object ζ and $\hat{\Sigma}_j$,

a positive semi-definite (covariance) matrix.

The matrix $\hat{\Gamma}^{MF,j}$ is constructed to take time aggregation into account. Specifically, these matrices map the high-frequency factors to the lower-frequency observations.

For example, when the frequency of interest is quarterly, as it is in the application:

$$\hat{\Gamma}^{MF,j}_{\zeta(n)} = \begin{cases} \frac{1}{4}\left[\hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}\right] & \text{annual flow,} \\ \left[\hat{\Gamma}_{\zeta(n)}, \mathbf{0}, \mathbf{0}, \mathbf{0}\right] & \text{quarterly flow or stocks.} \end{cases}$$

In the first case, the distributional object corresponds to an *annual flow* (e.g. the PSID income / consumption quantile function). We then approximate the distribution of annual averages by the average distribution over four quarters and set $\hat{\Gamma}^{MF,j}_{\zeta(n)} = \frac{1}{4}\left[\hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}\right]$. For *stock variables* (e.g. any wealth object for which the date is known), the measurement loads exclusively on the contemporaneous factor block and the lagged blocks are set to zero i.e., $\hat{\Gamma}^{MF,j}_{\zeta(n)} = \left[\hat{\Gamma}_{\zeta(n)}, \mathbf{0}, \mathbf{0}, \mathbf{0}\right]$, for $\mathbf{0}$ the same size as $\hat{\Gamma}_{\zeta(n)}$. The same holds for the third case, *quarterly flows* (e.g., SIPP income, CEX), which again loads on the contemporaneous block only.

For the copula, since multiple variables are necessary for its construction, the different measurements of the variables could be mixed *within* e.g., annual income flows and stock wealth. In these cases, for some point in time t, the lowest frequency is chosen among the set of measurements as an approximation to the temporal aggregation.

Measurement Error The measurement error for the distributional data, $\nu^j_{F,t}$, is composed of sampling uncertainty and other errors that reflect the fact that a given dataset has its specific operationalizations of the common economic variables being measured. Differences in operationalizations can not only shift the level of a particular measurement (which we capture through fixed effects, see Footnote 17), but can also become differentially important over time.[21] In principle, to fully capture these time-varying differences, we would need to estimate a measurement error variance for each coefficient and dataset. However, this

21. For example, the PSID and SCF differ in the way they ask respondents about their business wealth (Pfeffer, Schoeni, Kennickell, and Andreski 2016). PSID and CPS differ in the sampling unit, making household/family income sensitive to labor supply patterns (Gouskova, Andreski, and Schoeni 2010). Similarly, differences in the propensity to sample business owners between the different datasets make income sensitive to relative changes in business and labor income (Kim and Stafford 2000). Finally, the CEX and the PSID differ in the consumption categories covered in the survey, with the PSID being much coarser (Insolera, Simmert, and Johnson 2021).

would be too large a number of parameters to estimate within the time-series model; a common problem when estimating approximate dynamic factor models with maximum likelihood (see, e.g., Bai and Li 2016). The standard approach in the literature uses quasi-maximum likelihood with EM algorithms (Doz, Giannone, and Reichlin 2012; Banbura and Modugno 2014; Barigozzi and Luciani 2019). EM is a local method: it converges to a stationary point that depends on its starting values (Wu 1983), and its guarantees hold only when initialized near the true parameters (Balakrishnan, Wainwright, and Yu 2017), which is hard to ensure in high-dimensional settings such as ours.

For this reason, we do two things, the second of which will be discussed in Section 3.2.4, when we cover our sampling procedure. First, we assume that the covariance of all measurement errors is generated by scaling the whitened directions of the sampling uncertainty (and in that sense preserves the dependency structure). Under this assumption, the matrix Σ_j can be estimated outside the time-series model using bootstraps or supplied replication weights to estimate the covariance from sampling uncertainty by data source j.[22]

This achieves differently the diagonality assumed in the aforementioned literature. This leaves N diagonal elements of S^j to be estimated within the time-series model. To reduce the parameter count further, we constrain these elements to vary only by dataset j and variable ζ, yielding just $d + 1 = 4$ parameters per dataset. Consequently, each diagonal element s^j_ζ indicates by what factor the sampling-based variance for ζ must be inflated (or deflated) in dataset j; the standard error correspondingly scales with $\sqrt{s^j_\zeta}$. Therefore, the overall measurement error variance in (22) is $\hat{\Sigma}_j^{1/2} S^j (\hat{\Sigma}_j^{1/2})'$.

Thus, similarly to feasible generalized least squares, we transform observations $\tilde{\theta}^j_t$ by pre-multiplying the coefficients with the external estimate $\hat{\Sigma}_j^{-1/2}$[23]

$$\begin{aligned} \hat{\Sigma}_j^{-1/2}\tilde{\theta}^j_t &= H^j_t(\hat{\Sigma}_j^{-1/2}\,\hat{\Gamma}^{MF,j} F_t + \tilde{\nu}^j_{F,t}) \qquad \tilde{\nu}^j_{F,t} \sim \mathcal{N}(0, S^j) \\ \tilde{\tilde{\theta}}^j_t &= H^j_t(\tilde{\hat{\Gamma}}^{MF}_j F_t + \tilde{\nu}^j_{F,t}), \qquad \tilde{\hat{\Gamma}}^{MF}_j := \hat{\Sigma}_j^{-1/2}\hat{\Gamma}^{MF,j}. \end{aligned} \tag{23}$$

22. We draw bootstrap samples / use the supplied replication weights for each dataset j, $\{\tilde{\boldsymbol{\theta}}^j_{t,b}\}_{b=1}^B$, for each period t. Then, we demean the bootstrap samples b for each j and t and compute the average within-time variance-covariance matrix $\hat{\Sigma}_j$ pooling the demeaned bootstrap samples of the dataset j. If an object ζ is unobserved in dataset j, we set the covariance terms to zero and the diagonal elements to one to still be able to compute $\Sigma_j^{-1/2}$. To regularize $\hat{\Sigma}_j$ we add a small variance term $\lambda\,\bar{\sigma}^2_j\,I$. Here $\bar{\sigma}^2_j$ is the average bootstrap variance of the coefficients survey j observes and we set $\lambda = 0.05$. This preserves the ranking between measurements in terms of precision.

23. The whitening is applied blockwise to the subvector observed in each period—i.e., $\hat{\Sigma}_j^{-1/2}$ is taken conformably with the selection H^j_t—so the transformation and the selection commute.

With the measurement error treatment in place, denoted by the additional tilde, we can stack the J measurement equations, and with the aggregates, arrive at the final measurement equation

$$\begin{bmatrix} \tilde{\tilde{\theta}}_t \\ \hat{Y}_t \end{bmatrix} = H_t \left(\begin{bmatrix} \tilde{\hat{\Gamma}}^{MF} & \mathbf{0} \\ \mathbf{0} & I_Y \end{bmatrix} \begin{bmatrix} F_t \\ Y_t \end{bmatrix} + \begin{bmatrix} \tilde{\nu}_{F,t} \\ \nu_{Y,t} \end{bmatrix} \right) \qquad \nu_{Y,t} \sim \mathcal{N}(\mathbf{0}, \Sigma_Y). \tag{24}$$

Note that the terms are free of the j index to indicate they have been vertically stacked and note the additional appendage for the aggregates Y_t. Measurements of the aggregate factors are observed with noise Σ_Y.

In sum, Equations (21) and (24) form a linear state-space system that can be estimated using standard Bayesian techniques and a Kalman filter.

Two-step estimation vs. a unified Bayesian approach. Our procedure is deliberately modular: we first summarize the microdata by estimating the distributional coefficients $\tilde{\boldsymbol{\theta}}_t^j$ and their uncertainty Σ_j, and then use these estimates as noisy observations in the state-space model. An alternative would be to estimate everything jointly by placing a household-level likelihood directly inside the state-space model. For the simpler case of independent cross-sectional data, Chang, Chen, and Schorfheide (2024) show that this distinction is largely one of implementation: the exact joint model is computationally difficult, while a practical approximation leads back to the same two-step approach, where period-by-period estimates enter the state-space model with uncertainty that shrinks as survey size increases. Our setting differs because household surveys are not simple random samples. Sampling weights, stratification, non-response, and multiple imputation affect the uncertainty of the estimated coefficients, so a naive independent household likelihood would not correctly capture their precision. Instead, we estimate Σ_j directly from the survey design using replicate weights or design-based bootstraps, allowing the state-space model to weight each coefficient according to its actual sampling uncertainty. This modular approach also prevents errors in the dynamic model from affecting the cross-sectional estimates (Plummer 2015; Jacob et al. 2017).

3.2.4 Bayesian Estimation

We need to estimate seven objects. From the state equation, we need to estimate the factor-autoregression matrix A, the loading matrix B from aggregate controls to distributional factors, the loading matrix C from distributional factors

to aggregate factors, the vector autocorrelation of the aggregate factors themselves, the diagonal of D, the variance-covariance matrix of the shocks to the factors Ω, the variance-covariance matrices $\Sigma_j^{1/2} S^j \Sigma_j^{1/2'}$ and Σ_Y of the measurement errors. In a first step, the covariance structure $\Sigma^{1/2}$ is estimated outside the time-series model, as noted above, leaving only the scaling matrices S^j to be estimated within the model. Given the size of the A, B, C, D, Ω, S^j, and Σ_Y matrices, we use a Bayesian approach to estimate the system. This imposes regularization via priors on the parameters, as well as hyperpriors on the hyperparameters, to avoid overfitting the data.

This results in a hierarchical Bayesian state-space estimation with mixed-frequency data conditional on the first-stage estimates. We collect all parameters and hyperparameters into a single parameter vector $\boldsymbol{\psi} = (\boldsymbol{\psi}_{\text{par}}, \boldsymbol{\psi}_{\text{hyper}})$, estimated jointly. The remainder of this section details the specific prior distributions on $\boldsymbol{\psi}$, first for the state equation parameters and subsequently for those of the measurement equation.

State Equation For the state equation, we define the following prior:

$$\begin{pmatrix} vec(A) \\ vec(B) \\ vec(C) \\ diag(D) \end{pmatrix} \sim \mathcal{MN}(\mu_{\text{Minn}}, V_{\text{Minn}}) \qquad \begin{aligned} L_{chol} &\sim LKJChol(r+q, \eta) \\ T^{-1}(\Omega_{F,ii}) &\sim \mathcal{N}(\mu_{\Omega_F}, \sigma_F^2) \\ T^{-1}(\Omega_{Y,ii}) &\sim \mathcal{N}(\mu_{\Omega_Y}, \sigma_Y^2) \end{aligned} \tag{25}$$

First, we impose a Minnesota prior on A, B, C, and $diag(D)$. This means we set the vector of expected values μ_{Minn} so that all *but* the autocorrelation terms in A and D have an expected value of zero. This implies that B and C are shrunk to zero and so are the off-diagonals of A if not needed to describe the data. The remaining values, the expected values for the autocorrelations (main diagonal of A and D), are governed by hyperparameters described later in the section.

For the prior on Ω, we impose a *separation strategy* (see e.g., Barnard, McCulloch, and Meng 2000) and decompose Ω into a diagonal of standard deviations Ω_σ and a correlation matrix Ω_{corr} i.e., $\Omega = \Omega_\sigma \times \Omega_{corr} \times \Omega_\sigma{}'$. This avoids any a priori correlation between variances and covariances, normally imposed in Inverse-Wishart setups (Barnard, McCulloch, and Meng 2000). For the shocks, Ω_σ is further decomposed into Ω_F and Ω_Y, imposing different priors on the standard deviations of aggregate shocks and those of distributional shocks. It will be on the diagonal elements $\Omega_{\sigma,ii}$ for which we specify the priors.

For the parameters $\Omega_{\sigma,ii}$, we first draw from a set of Normal distributions

and map these parameters to $\mathbb{R}^+$ using the soft-plus transformation, denoted as $T(\cdot)$. This provides flexible exploration for the optimizer by avoiding exponential jumps from small perturbations in these parameters and avoids acceptance rates being driven by out-of-bounds draws (Chapter 55, Stan User Guide). For $\Omega_{F,ii}$, the mean μ_{Ω_F} is set such that the post-transformation entry—the innovation standard deviation—is centered at $1 - \kappa_{2,i}^2$. Together with the prior on the autocorrelation (in the matrix A), this centers the a-priori long-run variance of each distributional factor at $1 - \kappa_{2,i}^2$, below the unit-variance normalization of the factors: a conservative centering that shrinks factor volatility rather than presuming the normalization holds a priori. The same construction applies to the standard deviations of the shocks to the aggregate factors, $\Omega_{Y,ii}$, with κ_3. Hyperparameters $\kappa_{2,i}$ and κ_3 are discussed in more detail later.

For the estimation of the correlation matrix, Ω_{corr}, the matrix must satisfy several constraints simultaneously which may be difficult to enforce in any optimization: it must be symmetric, have a unit diagonal, and be positive definite. To circumvent this, we employ a reparameterization strategy. The key insight is to work with the Cholesky factor, L_{chol} where $\Omega_{corr} = L_{chol}\, L'_{chol}$. Following the methodology of Lewandowski, Kurowicka, and Joe (2009, hereafter: LKJ), we construct the Cholesky factor, L_{chol}, from an unconstrained vector of real numbers, $v \in \mathbb{R}^{(r+q)(r+q-1)/2}$ and it is precisely on this Cholesky factor that we evaluate our prior. The LKJ-Cholesky prior will be governed by a single shape parameter, $\eta \geq 1$, which will control the expected correlation strength between aggregate and distributional shocks.

For the state equation, there are $|\boldsymbol{\psi}_{hyper}| = 5 + r$ hyperparameters that are jointly estimated with $\boldsymbol{\psi}_{par}$. One hyperparameter shapes V_{Minn}; $r + 1$ hyperparameters govern the autocorrelation in A and D; and the remaining three hyperparameters govern Ω. See Appendix C for further information on these hyperparameters.

Measurement Equation First, for the distributional measurements, we specify independent priors for the measurement error variances for each dataset and object, s_ζ^j. Recall that estimating the diagonal of S^j would imply one variance parameter per coefficient n. As such, we impose the restrictions explained in the previous section, assigning one variance parameter s_ζ^j per object ζ and dataset j. This means a maximum (since some measures are unobserved) of $J \times (d + 1)$ variance parameters. To ensure strict positivity of s_ζ^j while maintaining efficient sampling, we again parameterize each variance using a softplus transformation

and place Normal priors on the unconstrained parameters $T^{-1}(s_\zeta)$. For the distributional data, we assign a weakly informative prior centered such that the median variance is 1.0, with a standard deviation of 2.0. This says, first, on average, there is only sampling uncertainty and no additional measurement error reflecting conceptual differences, but that, second, $\hat{\Sigma}_j$ is still itself an estimate, additionally perturbing each s_ζ^j away from 1.[24]

$$T^{-1}(s_\zeta^j) \sim \mathcal{N}(0.54, 2^2), \qquad T^{-1}(\Sigma_{Y,ii}) \sim \mathcal{N}(-7.6, 0.02^2). \tag{26}$$

For the aggregate factors, we impose highly informative priors on variances $\Sigma_{Y,ii}$, reflecting the data underlying these factors are precisely observed aggregates: Priors are centered at a near-zero variance with a very narrow standard deviation of 0.02.

Likelihood and sampling With this prior on $\boldsymbol{\psi}$, we obtain the model-likelihood $p(\tilde{\tilde{\boldsymbol{\theta}}}, \hat{Y}|\boldsymbol{\psi})$ of the stacked observation vector in (24) using a Kalman-filter. The log posterior density is then calculated as the sum of the hyperprior log-probability, the prior log-probability and the data log-likelihood. As previously mentioned, the standard approach of using quasi-maximum likelihood with EM algorithms may converge to a local optimum in our high-dimensional setting. We instead initialize with a global differential-evolution search and, to sample from the potentially complex, multi-modal, high-dimensional posterior distribution, we employ the Differential-Independence Mixture Ensemble (DIME) sampler from Boehl (2024), whose proposals are designed to move between modes of the posterior. No method guarantees locating the global mode in finite time, but this reduces the risk of reporting a local optimum and, unlike a point estimate, characterizes parameter uncertainty. Details and convergence results are in Appendix D. For stability, we check that draws of Φ satisfy a spectral radius within 0.995; draws outside this radius are rejected.

3.3 Reconstructing and Using the Distributional Data

After finding the posterior mode of our model parameters, we apply the Kalman smoother to recover the full path of latent factors $\{\hat{F}_t\}_{t=1}^T$. Given these mode-conditional smoothed factors, we then reconstruct the sequence of joint distributions, $\hat{\Xi}_t^j$, from the implied standardized series of polynomial coefficients.

24. Identical priors across datasets mean that we do not a priori prioritize a conceptual measure for object ζ in one dataset over another. Setting different priors is generally possible if a particular measurement concept should be prioritized on theoretical grounds.

Appendix E outlines the procedure. Drawing from the posterior distribution allows us to capture parameter uncertainty and uncertainty about the latent factors given the parameters.

This estimation approach allows practitioners to use the above data in three principal ways: First, in its factor representation, $\hat{F}_t$, second, in its functional representation, $\hat{\theta}_t^j$, or, third, in its observational representation, e.g., quantile functions and copula—$\left(\{\hat{\Xi}_{jmt}^{-1}\}_{m=1}^d,\ d\hat{C}_t^j\right)$. All three representations are provided for download at distributionaldynamics.com.

For example, if one is interested in replicating the exercise of Chang and Schorfheide (2024), one can use the second representation and use the coefficients related to the marginal distributions of interest in their *distribution*-augmented VAR. If one is only interested in augmenting a VAR with a low-dimensional summary containing virtually all variation in the joint distribution of consumption, income, and wealth—in the style of these factor-augmented VARs—one can use the first representation.

Lastly, one can use the coefficients to generate economically interpretable data and work with this economically interpretable/meaningful data directly. One can generate the quantile functions and copula densities, but also other common statistics of interest for the marginal distribution: e.g., Gini coefficients of consumption, income, or wealth; levels, quantiles, shares, etc. and objects that arise from the correlational structure (the copula density): e.g., correlation measures, conditional probabilities (e.g., population shares), and Kendall's tau.

For the application in Section 5, we adopt (3) and use the quantile functions and copula densities to generate a synthetic time-series of micro-datasets (cross-sections) with entry X_{it}^j. Each (representative) household i is defined by a cell of ranks $\mathbf{u} \in \mathcal{U}_i \equiv \prod_{m=1}^d U_i^m$, with one range $U_i^m \subseteq [0,1]$ per variable (e.g., deciles). Integrating the copula density over the cell gives the household's probability weight. Integrating each quantile function, after undoing the inverse-hyperbolic-sine transformation, against the copula density over the same cell and dividing by the cell's weight gives the mean of that variable among the households in the cell, in levels.[25] Below is precisely this combination, for the example of the distribution of consumption, income, and wealth, which together forms a single

25. Here μ_{mt} denotes the scale used in the transformation of variable m; see Appendix E. Because the copula density is a polynomial, the integrals factor into one-dimensional integrals over the cell's rank ranges. Cells with negligible mass carry the average of each variable over its own rank range instead.

row of the time series of weighted synthetic cross-sections:

$$X_{it}^{j} = \begin{pmatrix} \omega_{it}^{j} \\ c_{it}^{j} \\ y_{it}^{j} \\ w_{it}^{j} \end{pmatrix} = \begin{pmatrix} \int_{\mathcal{U}_i} d\hat{C}_t^j(\mathbf{u})\, d\mathbf{u} \\ \frac{1}{\omega_{it}^{j}} \int_{\mathcal{U}_i} \mu_{ct} \sinh\left(\hat{\Xi}_{jct}^{-1}(u_c)\right) d\hat{C}_t^j(\mathbf{u})\, d\mathbf{u} \\ \frac{1}{\omega_{it}^{j}} \int_{\mathcal{U}_i} \mu_{yt} \sinh\left(\hat{\Xi}_{jyt}^{-1}(u_y)\right) d\hat{C}_t^j(\mathbf{u})\, d\mathbf{u} \\ \frac{1}{\omega_{it}^{j}} \int_{\mathcal{U}_i} \mu_{wt} \sinh\left(\hat{\Xi}_{jwt}^{-1}(u_w)\right) d\hat{C}_t^j(\mathbf{u})\, d\mathbf{u} \end{pmatrix}. \tag{27}$$

In this example, we have a synthetic (representative) household i, where $(U_i^c \times U_i^y \times U_i^w)$ is the quantile combination that defines household i, e.g., the first decile in consumption c, the third decile in income y, and the seventh decile in wealth w. The mass, ω_{it}^j, of the households in that cell defines a weight for that synthetic household. This implies that, without the need for sampling, we obtain an output that can be immediately interpreted as synthetic microdata. The group means in the application (Figure 11) are the same object, integrated over larger groups. As illustrated above, X_{it}^j is dataset specific, but one can obtain a consensus estimate of the business cycle component across datasets as a simple average of the $d\hat{C}_t^j$ and $\hat{\Xi}_{jmt}^{-1}$ over datasets j or, alternatively, as a weighted average by using, e.g., the inverse measurement error standard deviations (by object and dataset) as weights.

4 Credibility Checks

Before turning to the application, we evaluate the reliability of our procedure step-by-step using actual (Section 4.1) and simulated data. This addresses three potential concerns: (i) Does the factor representation lose information? (Section 4.2); (ii) Does the state-space model predict accurately when data are missing? (Section 4.3); and (iii) whose data do the estimates rest on, and how much of the synthetic panel is interpolation rather than observation? (Section 4.3.2). We first show that the factor representation preserves the information needed for state-space estimation, validating the approach described in Sections 3.2.1 and 3.2.2. Without this reduction step, we would need to track and propagate a high-dimensional set of distributional coefficients directly in the state-space model (as in Chang, Chen, and Schorfheide 2024), which is computationally infeasible in our setting. The mapping between latent distributional states and the noisy coefficients then defines our measurement equation, whose priors and hyperparameter choices are validated in Appendix F (validating in part Section 3.2.3 and Section 3.2.4), showing that the estimated path of the functional data falls

reasonably well into the bounds given by sampling uncertainty.

With the functional state-space model in place, we perform three further experiments to address the second concern. The first two experiments show that the model predicts omitted survey waves well using actual data. The third experiment evaluates the procedure in a controlled environment with a known data-generating process, using the heterogeneous-agent model of Bayer, Born, and Luetticke (2024). Results from these experiments show the resulting synthetic microdata closely match the omitted observations, indicating that the estimated state-space model successfully captures time-series fluctuations in the distributional data (thereby validating the procedure in Sections 3.2.3 and 3.2.4) and in particular, when the data-generating process is known. In Appendix G, we further show that our reconstructed distributional data agree well with the cyclical fluctuations (for the coverage of measures) found in the World Inequality Database (WID) and the Distributional Financial Accounts (DFAs).[26]

4.1 Data

To test and apply our method, we estimate the joint distribution of consumption, income, and wealth at the household level for the United States from 1962 to 2024. We rely on commonly used microdata for the U.S.: the *Consumer Expenditure Survey* (CEX), *Current Population Survey* (CPS, Flood et al. (2023)), *Panel Study of Income Dynamics* (PSID), *Survey of Consumer Finances* (SCF); including the historical backfiles (SCF+), and the *Survey of Income and Program Participation* (SIPP).[27] Table 1 lists the distributional objects from each dataset, together with the sampling period. Appendix H visualizes more elaborately the underlying scarcity, charting each source's reference periods, variable coverage, sample sizes, and publication lags.

We do not impose any demographic household restriction and only mild sample selection for consistency (see Appendix H) in all of these datasets. For the CEX consumption and SIPP income, which are at quarterly frequency, we remove seasonality using X-13 ARIMA-SEATS. We date CPS to quarter 4; the PSID is assumed to reflect quarter 2; and the SCF is dated to quarter 3. SIPP income data are aggregated to quarterly level and then naturally assigned to the

26. Note that all trends will be fitted by construction, see Section 3.2.1.

27. The methodology accommodates permanent and time-varying differences in measurement, but major survey redesigns require intervention. For the CPS (1994) and SIPP (1996, 2013), we treat pre- and post-break data as separate surveysi.e., distinct measurements of the distribution to avoid spurious dynamics from seam bias.

Table 1: Micro Data Sources and their Sample Periods

Object	CEX	CPS	SCF	PSID	SIPP
Consumption quantiles	1984Q4 - 2021Q4	-	-	1999Q2-2021Q2	-
Income quantiles	1984Q4 - 2021Q4	1967Q4-2022Q4	1962Q3-2022Q3	1968Q2-2021Q2	1983Q3-2022Q4
Wealth quantiles	-	-	1962Q3-2022Q3	1984Q2-2021Q2	1983Q3-2022Q4
Copula densities	1984Q4 - 2021Q4	-	1962Q3-2022Q3	1984Q2-2021Q2	1983Q3-2022Q4

Notes: The table reports sample periods for different micro datasets across the different objects.

respective quarter; wealth assignment depended on the survey vintage.[28] Further details on the microdata and in particular how the respective measures are defined can be found in Appendix H. Note again that we require at least one dataset j that includes all objects, which in our case is the PSID from 1999 to 2021.

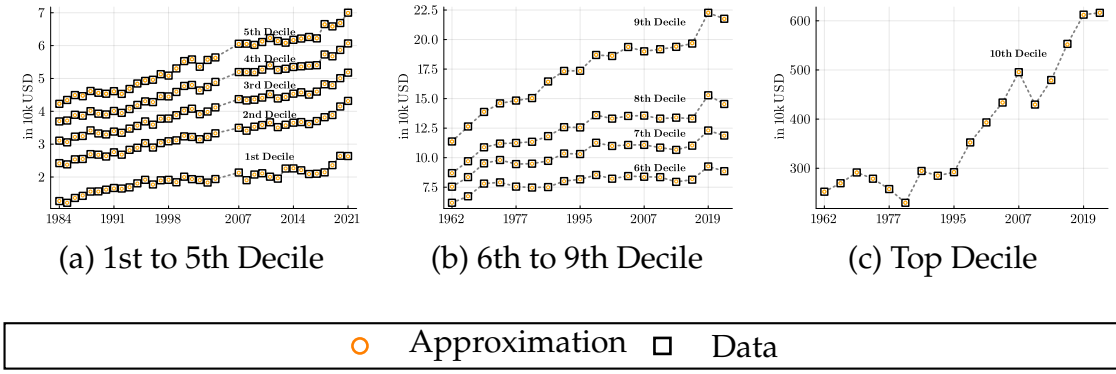
In terms of aggregate data, we use a wide range of standard business cycle data (GDP, consumption, employment, etc.) as well as data on household balance sheets, expectations, asset prices, and interest rates from McCracken and Ng (2021). We include data from 1962Q3 to 2024Q1. The starting point of the aggregate data determines the earliest date for the sample periods of the microdata used. From these aggregate time series, we extract the 11 most important factors. Details on the macro-data can also be found in Appendix H. Further details on factor selection and estimated parameters are available upon request.

4.2 Precision of the Factor Model

The first step in our procedure is to estimate Legendre polynomial coefficients for the functional representation of the distribution. Then, we estimate the factor structure in these coefficient data. Since we only retain "important" factors, we potentially introduce an approximation error (relative to the data) resulting from forcing "unimportant" factors f_t to take time-averaged values. The size of the approximation error can be controlled by choosing how many factors to keep. We choose to retain the eight most important factors, which explain 99% of the

28. Documentation on the timing of the CPS can be found here, pg. 6. For the timing of the SCF, see here, pg. 33. For the PSID, interview dates are provided as variables, with the vast majority of interviews falling in quarter 2. Given the PSID reports annual income and the model is estimated with mixed-frequency, this quarter 2 assignment only affects the timing of wealth, which is always dated to the interview date. For data on consumption and income, which is for the year prior, it is scaled by the growth rate of their respective aggregate to proxy the contemporaneous income and consumption level.

Figure 4:
Quantile Functions: Raw vs. Approximation (Important Factors)



(a) 1st to 5th Decile (b) 6th to 9th Decile (c) Top Decile

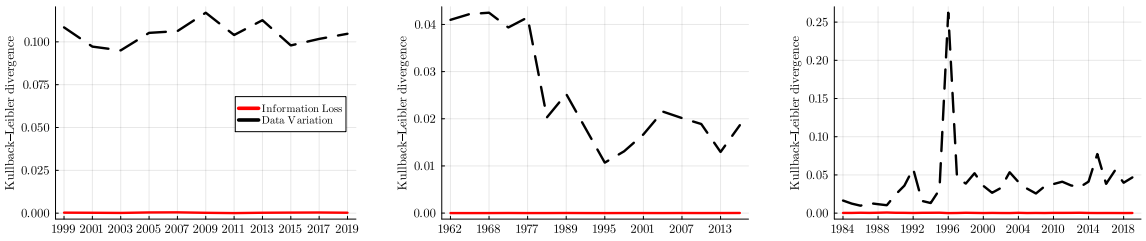
Notes: Figure shows the quantile functions (mean within decile) for consumption, income, and wealth deciles from the survey data (squares) and an approximation (circles) using only the fluctuations in the most important factors in (19). Consumption panel shows quantile functions for CEX consumption. Income panel shows quantile functions for SCF income. Wealth panel shows quantile functions for SCF wealth. Dotted lines show linear interpolation between survey waves.

(business cycle frequency) variation of the distribution (i.e., of $\tilde{\boldsymbol{\theta}}$ to be precise).[29] The different panels in Figure 4 visualize the approximation error in our application by showing some of the implied deciles. The figure compares the observed conditional decile means for consumption (bottom five), income (next four), and wealth (top) (squares) with their approximated counterparts (circles). The comparison for all deciles and variables looks similar and is available upon request.

We find that the factor model with its eight main factors is very close to the distributional dynamics over time. The circles are typically centered around the midpoint of the squares. Figure 5 compares the copula over time between the approximation and the raw data. We do this in terms of the Kullback-Leibler divergence. The dashed black line shows how distant the actual distribution is from its long-term average (how much variation is there to capture), and the solid line shows the difference between the actual distribution and the approximation based on the important factors only (how much the factors do not capture). The Kullback-Leibler divergence of the various copula from its long-run average is between 0.01 and 0.11 in most cases, while the divergence between the approximation and the actual distribution is almost two orders of magnitude smaller. Put simply: There are significant fluctuations in the copulas over time, but the

29. A PCA over other distributional data (World Inequality Database / Distributional Financial Accounts) is in the same neighborhood of number of factors.

Figure 5: Copulas: Raw vs. Approximation (Important Factors)



(a) C, Y, and W (b) Y and W (c) C and Y

Notes: Figure shows the Kullback-Leibler divergence between the raw-data copula and two reference copulas, by survey year. The black dashed line represents the divergence between the time-averaged copula and the raw data copula for each survey year. The red solid line represents the divergence between the copula obtained by allowing only the most important factors in (19) to fluctuate and the raw data copula for each survey year. Panel (a) is from the PSID (Consumption, Income, and Wealth), Panel (b) is from the SCF (Income and Wealth), and Panel (c) is from the CEX (Consumption and Income).

factors are able to capture them well.

4.3 Predictability of Distributional Data

To validate how well the method can predict distributional dynamics, we perform three experiments. The first experiment fixes the model parameters at their baseline estimates and removes some waves of microdata as inputs when running the Kalman smoother. This experiment answers the question how informative the model structure, aggregate data, and remaining microdata are for pinning down the distributional series when some survey waves are missing, holding parameters fixed. The second experiment repeats the omission exercise while *re-estimating* the model. This experiment takes into account that changes in the data also imply changes in the model parameter estimates. In the final experiment, we evaluate our model's ability to recover distributional dynamics generated from a heterogeneous agent model. We show the results of these experiments in Figures 6, 7, and 10.

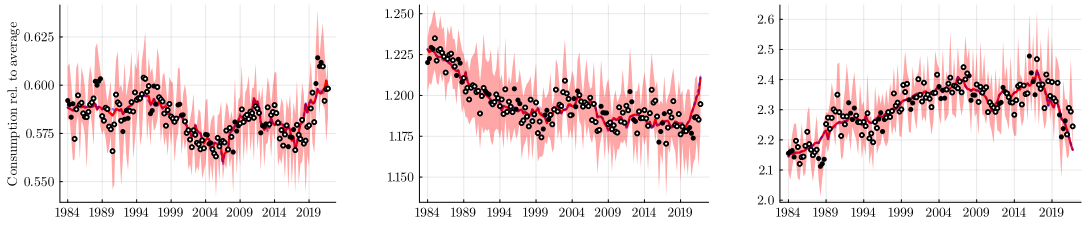
4.3.1 Predicting Omitted Survey Waves

For the first experiment, we include only every fourth CEX survey year in the Kalman smoother, holding the model parameters fixed at their baseline estimates—mimicking the fact that countries in Europe only survey consumption every four years and reducing the number of CEX survey years entering the smoother from 38 to 10. The resulting estimates from this smoother run with less data are shown

Figure 6: Predictability of Distributional Data (given the Baseline Parameters)

Consumption by Consumption Percentile

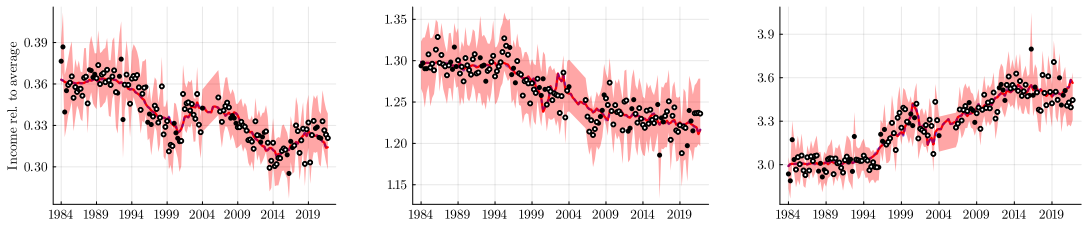
CEX data vs. estimates from using CEX every 4 years



(a) Bottom 50 Percent (b) 50-90 Percent (c) Top 10 Percent

Income by Income Percentile

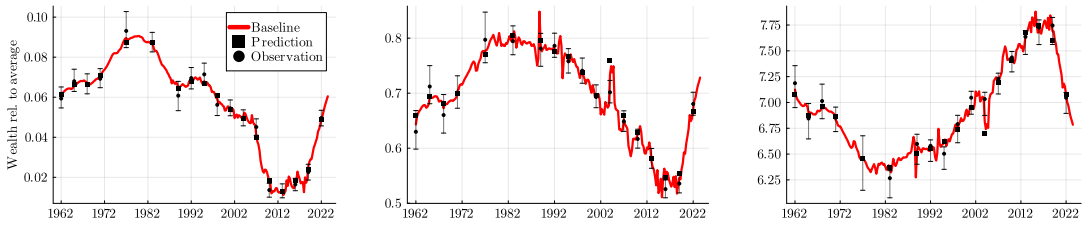
CEX data vs. estimates from using CEX every 4 years



(d) Bottom 50 Percent (e) 50-90 Percent (f) Top 10 Percent

Wealth by Wealth Percentile

Removing one SCF wave at a time



(g) Bottom 50 Percent (h) 50-90 Percent (i) Top 10 Percent

Baseline Less Data

Notes: The figure shows baseline model estimates for consumption (Panels (a)–(c)), income (Panels (d)–(f)), and wealth (Panels (g)–(i)) for different samples. Baseline estimates using all data are always shown as a solid red line. Panel (a)–(f): *less data* (dashed blue line) shows smoothed estimates when CEX microdata enter the smoother only every fourth year (black *solid dots*). *Empty dots* denote the observations removed from the Kalman smoother. Panel (g)–(i): shows smoothed estimates when only a single SCF wave has been dropped in the Kalman smoother. The black squares show the prediction of the dropped data at the survey wave and the dots show the empirical estimate from the survey data in that wave. Error bars/bands in all figures indicate 95% confidence bounds for each individual survey sample.

in Figure 6. We show the average of the top 10%, next 40% and bottom 50% for the measures in the CEX: consumption (Panels (a) to (c)) and income (Panels (d) - (f)). Note the large sampling uncertainty around the CEX data, whose 95% confidence intervals are displayed as a red error band. For this reason, even in the data-rich specification (with all CEX data), the smoothed estimate regularly deviates from the raw distributional data, while staying within the 95% confidence band of the survey estimates in 78–95% of quarters (Appendix Table 5). The correlation of the smoothed data using only every fourth survey year with the data using every survey year is very high, meaning that the model can predict the consumption distribution well.

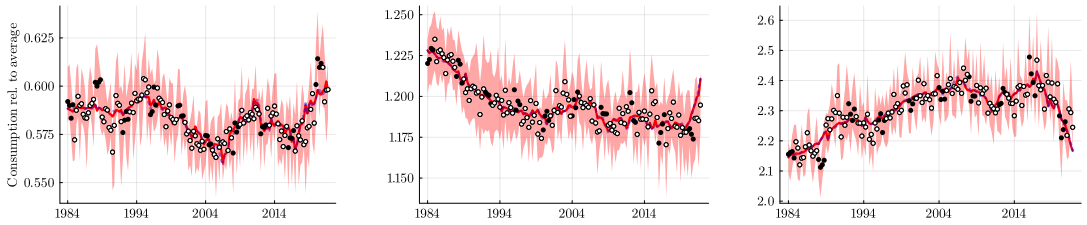
The bottom row of Figure 6, Panels (g) to (i), presents a different exercise. We run 18 mini experiments, each time dropping one SCF wave and generating new smoother estimates. This gives us 18 smoothed distributional data series alongside the one using all SCF waves. The solid lines are estimates from using all the SCF data. The squares represent the smoothed predictions corresponding to the timing of each omitted survey wave. The circle shows the direct empirical estimate of the corresponding survey (with its confidence limits). For example, for Panel (g), the square for 1992 is the prediction for the wealth distribution where the 1992 SCF survey *did not* enter the smoother. The fact that the squares are virtually on top of the solid line implies that, conditional on the model, a single observation of the distributional data has little effect on the smoothed series. In other words, the model structure, aggregate factors, and other surrounding micro-data collaborate well in imputing the missing wave. Two readings should be kept apart: the proximity of the squares to the all-data line shows that a single wave has little leverage on the smoother, while predictive accuracy is assessed by comparing the squares to the withheld survey estimates (circles) and their confidence bounds—the predictions generally lie within these bounds. An exception to highlight happens in 2004 during the housing boom, where the model (squares) predicts greater cyclicality than shown in the baseline (red line — with all data).

The first row of Figure 7 repeats the CEX exercise of Figure 6 but now re-estimates the entire model. The first row of Figure 7 shows the resulting reconstruction. The difference to Figure 6 is small, reconfirming our claim that the estimation procedure can impute well distributional data. The second and third rows show this in a different vein. In the second row, we remove a contiguous block of *all* microdata and ask whether the model can still recover distributional dynamics from aggregate data and the remaining (surrounding) micro-

Figure 7: Predictability of Distributional Data (Re-estimating the Model)

Consumption by Consumption Percentile

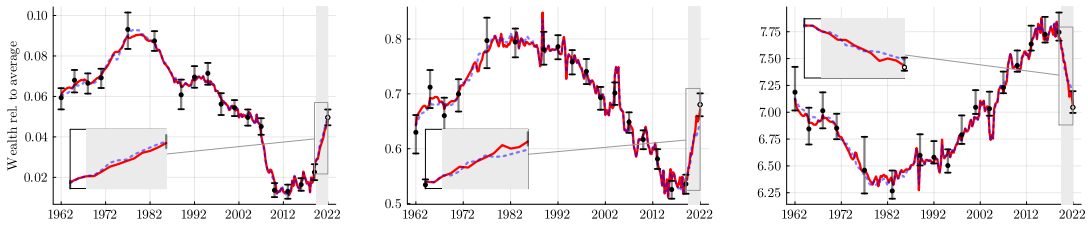
CEX data vs. estimates from using CEX every 4 years



(a) Bottom 50 Percent (b) 50-90 Percent (c) Top 10 Percent

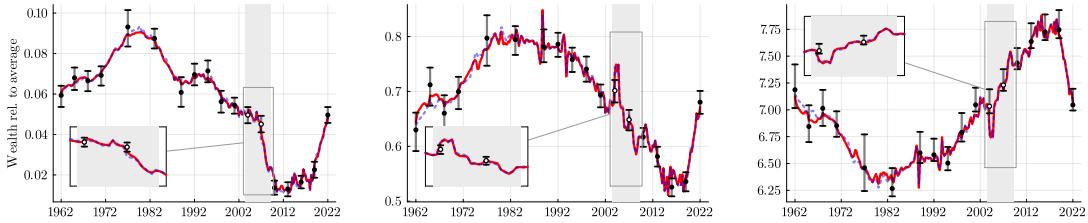
Wealth by Wealth Percentile

Removing microdata from last 4 years of estimation



(d) Bottom 50 Percent (e) 50-90 Percent (f) Top 10 Percent

Removing microdata from the housing cycle



(g) Bottom 50 Percent (h) 50-90 Percent (i) Top 10 Percent

Baseline Less Data

Notes: The figure shows baseline model estimates for consumption (Panels (a)–(c)) and wealth (Panels (d)–(i)) for different samples. Baseline estimates using all data are always shown as a solid red line. Panel (a)–(c): *less data* (dashed blue line) shows the smoothed estimate that results from a re-estimation of the model (and Kalman smoother) when CEX microdata enters only every fourth year (black *solid dots*). Panel (d)–(f): *less data* (dashed blue line) shows smoothed estimates when the last four years of *all* microdata have been dropped in model estimation (2020Q1 to 2024Q1) and the Kalman smoother. *Empty dots* denote the observations removed from the re-estimation and Kalman smoother. Panel (g)–(i): same exercise as (d)–(f) but dropping the *wealth* observations over the house price cycle (2004Q1 to 2009Q4). Error bars in all figures indicate 95% confidence bounds for each individual survey sample.

data. Here, we omit the last four years of the sample (2020Q1-2024Q1) to evaluate "nowcast" performance for the wealth distribution. In the third row, we omit all *wealth* microdata during the housing cycle (2004Q1-2009Q4), a period with large swings in wealth inequality (Kuhn, Schularick, and Steins 2020).[30]

Across all exercises, the reconstructed series obtained with withheld microdata closely track the baseline estimates over the full sample and remain informative within the withheld windows, indicating that the model captures wealth distribution dynamics even during periods without microdata, consistent with evidence that aggregate factors account for an important share of distributional fluctuations (Kuhn, Schularick, and Steins 2020; Bayer, Born, and Luetticke 2024), and by information from surrounding microdata, since smoother estimates incorporate the full set of observations.

4.3.2 The Anatomy of the Synthetic Data

The experiments above assess model reliability by removing observations and comparing model estimates and data to the resulting predictions. The state-space structure permits complementary, exact answers to two further questions, for every quarter at once: *whose* data is an estimate resting on, and how much does it rest on data at all? Both exploit the fact that, holding parameters and the observation pattern fixed, the Kalman smoother is an affine function of the observations (Koopman and Harvey 2003). Collect the state of (21) as $x_t = (F_t', Y_t')'$ and partition the stacked observation vector of (24) into source blocks: y_s^b denotes the transformed coefficients $\tilde{\tilde{\theta}}_s^j$ of the datasets j belonging to source b, with the aggregate observations $\hat{Y}_s$ forming a block of their own. The smoothed state then satisfies

$$\hat{x}_{t|T} \;=\; c_t \;+\; \sum_{b} \sum_{s=1}^{T} W_{t,s}^{b}\, y_s^b, \tag{28}$$

where the weights $W_{t,s}^b$ are implied by the filtering recursions and c_t collects the contribution of deterministic terms and the prior. We compute the decomposition exactly by exploiting linearity: setting the *values* of all blocks except b to zero—while leaving the selector matrices H_t in (24) unchanged, so that the missingness pattern and hence all filter gains are unchanged—and re-running the smoother isolates one term in the sum: $c_t + \sum_s W_{t,s}^b y_s^b$. A run with *all* blocks

30. Appendix I, Table 7 reports correlations between the baseline estimates and all missing-data estimates. Over the entire sample, they lie between 0.95 and 0.99; restricted to the withheld periods themselves, they range from 0.75 to 1.00, with the lowest values for income during the last four years of the sample.

zeroed returns c_t alone, which we subtract from each source-specific run, so the contributions defined below are free of the common deterministic-and-prior term. Additivity of the resulting components holds by construction and is verified to machine precision. The information sources b in our application are the PSID, SCF, CEX, CPS, SIPP, and the macroeconomic aggregates.[31]

The *information share* measures the relative reliance of the estimate on each source at each date. Isolating source b as above yields its contribution to the smoothed state,

$$\chi_t^b \;=\; \sum_{s=1}^{T} W_{t,s}^b \, y_s^b, \qquad \hat{x}_{t|T} = c_t + \sum_b \chi_t^b, \tag{29}$$

which collects source b's data from *all* periods, since the smoother spreads each observation's influence across time through the state dynamics. Normalizing the contributions componentwise—$\chi_{k,t}^b$ denotes the entry of χ_t^b corresponding to factor k—gives the share

$$s_{k,t}^b \;=\; \frac{|\chi_{k,t}^b|}{\sum_{b'} |\chi_{k,t}^{b'}|}. \tag{30}$$

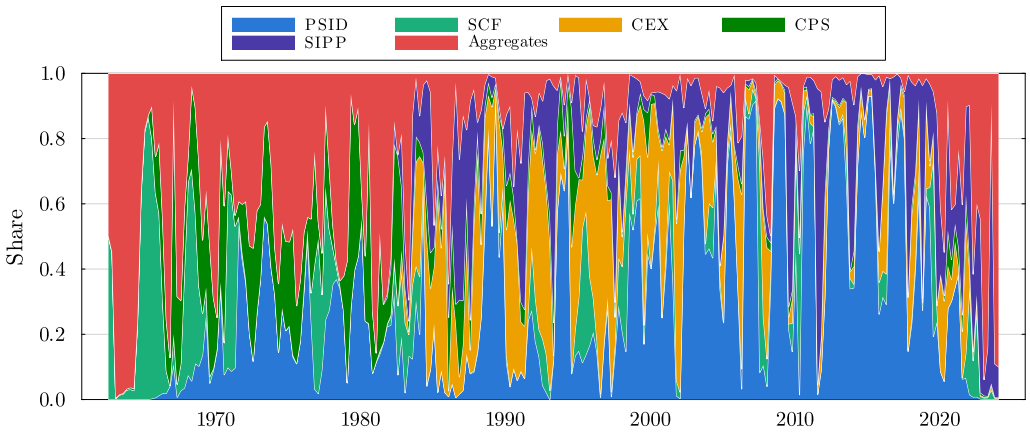
Note the deterministic-and-prior term c_t is negligible by construction—the coefficient observations $\tilde{\tilde{\theta}}_t$ are detrended and the prior mean of the state is zero—so it is omitted from the shares. Figure 8 shows the shares for the first factor: before the mid-1980s, when only the PSID, the SCF precursors, and CPS waves are available, the estimates lean heavily on the aggregate series; once the CEX and SIPP enter the sample, the survey sources take over and the aggregate share recedes. Results from the other factors will have strictly less reliance on aggregates and different reliance on the various microdata e.g., the factor emphasizes more wealth measurements than consumption; these results are available upon request.

The shares tell the researcher *whose data* the estimate rests on, but not how much the model itself is carrying. Let P^∞ denote the stationary covariance of the state x_t implied by the estimated law of motion (21), obtained from the Lyapunov equation $P^\infty = \Phi P^\infty \Phi' + \Omega$, and let $\sigma_{k,t|T}^2$ denote the posterior variance of the distributional factor $f_{k,t}$ from the Kalman smoother. The *interpolation share*

$$\iota_{k,t} \;=\; \frac{\sigma_{k,t|T}^2}{P_{kk}^\infty} \;\in\; [0,1] \tag{31}$$

31. A level decomposition is also possible and is available upon request.

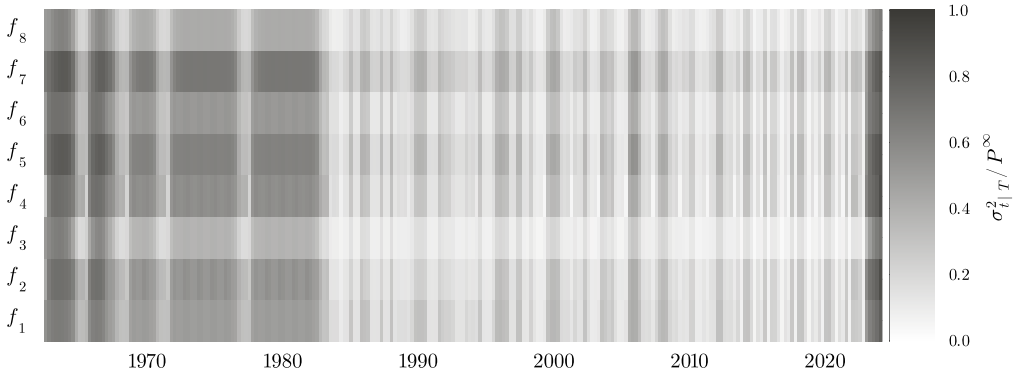
Figure 8: Information Shares



Notes: Stacked shares $s^b_{k,t}$ of each data source in the smoothed factor at each date, for $k = 1$.

is the fraction of the factor's unconditional uncertainty that *remains* after conditioning on all observations. Values near zero indicate that the data pin the factor down at that date; values near one indicate that the estimate rests on the model's dynamics—interpolation in the precise sense. Figure 9 reports $\iota_{k,t}$ for all factors, from f_8 at the top to f_1 at the bottom. The pattern mirrors the arrival of the data: interpolation shares are high throughout the survey-sparse decades before the mid-1980s, though less for f_1 to f_4, drop sharply once the CEX and SIPP enter the sample, and rise again at the very end of the sample, where the two-sided smoother runs out of future survey waves to condition on. The vertical striping within the data-rich era traces individual survey waves. A user of the synthetic data can thus read off, quarter by quarter, where the panel is data-rich and where it is model-driven. Results above, as well as validation exercises (Appendix G), show model-driven estimates, despite greater scarcity of data, find cyclical variation in the neighborhood of other estimators.

Figure 9: Interpolation Shares



Notes: Each cell reports $\iota_{k,t} = \sigma^2_{k,t|T}/P^\infty_{kk}$, the smoothed posterior variance of the distributional factor $f_{k,t}$ at date t relative to its stationary variance from the Lyapunov equation. Dark cells indicate dates at which the estimate rests on the model's dynamics (interpolation); light cells indicate dates at which the observations pin the factor down. The measure conditions on past and future observations alike and thus describes the historical panel, not a real-time estimate.

4.3.3 Predicting Dynamics from a simulated HANK model

Thus far, we have examined the model's ability to predict intentionally omitted data. While informative, these tests do not fully resolve a central concern: in empirical settings, the underlying distributional dynamics are only imperfectly observed and agreement with these sparse microdata (or external benchmarks, see Appendix G) is not definitive proof that the model recovers the true latent process. To assess this, we study the model in a controlled environment with a known data-generating process. For this purpose, we simulate a heterogeneous agent business cycle (HANK) model (concretely, from Bayer, Born, and Luetticke 2024) as a data-generating process. This model is well suited to our setting, as it generates business-cycle fluctuations in general equilibrium in response to a set of aggregate shocks and their interaction with the joint distribution of households, also characterized by a copula and a set of marginals. The simulation produces functional distributional data, which we can use to draw micro-data comparable to ones actually available.

We simulate one economy for $T = 2200$ periods (quarters). For each period t, we buffet the HANK model economy with a series of aggregate shocks, whose effects propagate through linearized policy functions. From the resulting joint distribution, we generate **five** samples: *A*, *B*, *C*, *D* and *E*. Each sample is drawn from the copula histogram, ensuring that the samples preserve the correlational structure between economic measures. Sampling is stratified along each dimension to ensure coverage of the marginal distributions. Each simulated household

defines a unit of observation and is characterized by its consumption, income, and wealth and a sampling weight. We abstract from unit-level measurement error.

Samples *A*, *B*, *C*, *D* and *E* are designed to mimic one dataset from the U.S., each representative, but with differences in their degree of missingness and precision. Dataset *A* can be considered a modern PSID-like dataset: 9,000 household observations per survey, containing consumption, income, and wealth, and released every two years. Dataset *B* can be considered CPS-like: 60,000 observations per survey (very precise), with only income, released every year. Dataset *C* can be considered a CEX-like dataset: 6,000 observations per survey, with income and consumption, released every quarter. Dataset *D* can be considered an SCF-like dataset: 5,000 observations per survey, with only wealth and income observed, released every three years. Dataset *E* can be considered a SIPP-like dataset: 25,000 observations per survey, with income observed every quarter, and wealth measured only in the fourth quarter of each year. We abstract from differences in operationalization across the different sample measurements.

The aggregate shocks driving the economy are also recorded at each t, whose factor representation will be used in our state-space system.[32] Before proceeding, the initial 1200 periods are discarded as burn-in, ensuring that the simulated economy has converged to its ergodic distribution independently of initial conditions. The remaining 1000 periods are kept for the exercise and split into 10 chunks, obtaining 10 simulation blocks (economies) observed for 100 quarters each. Accordingly, we can run 10 separate estimation exercises that are comparable to the actual data also in terms of the time that they span (25 years, similar to the available PSID waves containing consumption). In the end, the procedure suggests five distributional factors and eight aggregate factors.

We use the five survey-style samples (*A*–*E*) as microdata inputs to our dynamic factor method alongside the factor representation of the aggregate shocks from the HANK model. The resulting estimates are labeled as *Distributional Dynamics Factor Model (DDFM)* and compared to two benchmarks. The first benchmark consists of the values obtained directly from the model simulation, the *Truth*. The second benchmark is a linearly interpolated series of (conditional) cross-sectional sample moments for some variable and dataset X. The interpolation points would be the survey release dates. This *naive* Sample X benchmark ignores that the samples (*A*–*E*) are, in fact, just samples of an underlying distri-

32. The HANK economy is subject to shocks in total factor productivity, investment-specific technology, consumption wedges, labor wedges, monetary policy, government spending, fiscal policy, and household preferences. For a full description of the model, see Bayer, Born, and Luetticke (2024) and the associated `BASEtoolbox.jl` code repository.

Table 2: Time-Series Correlations: Model Estimates vs. HANK Simulated Truth

	Consumption		Income		Wealth	
	DDFM	Sample C	DDFM	Sample A	DDFM	Sample D
			By Consumption groups			
Bottom 40%	0.92	0.99	0.96	0.63	0.91	–
Middle 40–80%	0.91	0.94	0.70	0.34	0.86	–
Top 20%	0.94	0.91	0.96	0.64	0.91	–
Top 10%	0.94	0.90	0.93	0.64	0.89	–
			By Income groups			
Bottom 40%	0.93	0.99	0.96	0.57	0.74	0.67
Middle 40–80%	0.84	0.90	0.82	0.29	0.75	0.64
Top 20%	0.92	0.89	0.95	0.61	0.79	0.74
Top 10%	0.79	0.90	0.94	0.64	0.66	0.71
			By Wealth groups			
Bottom 40%	0.92	–	0.94	0.39	0.93	0.93
Middle 40–80%	0.89	–	0.85	0.36	0.90	0.81
Top 20%	0.94	–	0.89	0.46	0.91	0.37
Top 10%	0.93	–	0.86	0.58	0.91	0.48

Notes: The table shows correlations between two benchmarks and the HANK simulated truth. Column *DDFM* shows correlations between the HANK simulated truth and our model estimates (*DDFM*). Column *Sample X* shows correlations between the HANK simulated truth and a linearly-interpolated Sample $X \in \{C, A, D\}$. As in the figure above, conditional moments are averages by *Percentile Group* (Bottom 40%, Middle 40-80%, Top 20%, and Top 10%) of the distribution of consumption, income, and wealth and all time series are relative to the economy-wide average. Sample *B*, which contains only income, is omitted for brevity. Sample *E* as well. Correlations are averages over the ten simulated economies.

bution that has some dynamic structure.

Table 2 compares two benchmarks to the HANK simulated truth. First, column *DDFM* reports correlations between the HANK simulated truth and our DDFM estimates generated from using the incomplete samples. Column *Sample X* reports correlations between the HANK simulated truth and the linearly-interpolated sample $X \in \{C, A, D\}$. These are time correlations, averaged over the ten simulated economies. We find that our DDFM estimates are an improvement over the correlations of the linear-interpolation with the true distributions. In most cases, the correlation between the truth and DDFM implied distributional summary statistics are above 0.90. Sample *C* would be a demanding benchmark, observing consumption every quarter, making its interpolation almost the truth itself (0.89–0.99). The DDFM comes close to this benchmark and

does better for the top consumption and income groups. This suggests overall that the model approximates well the marginal distributions and correlational structure across variables, as well as their tails.[33]

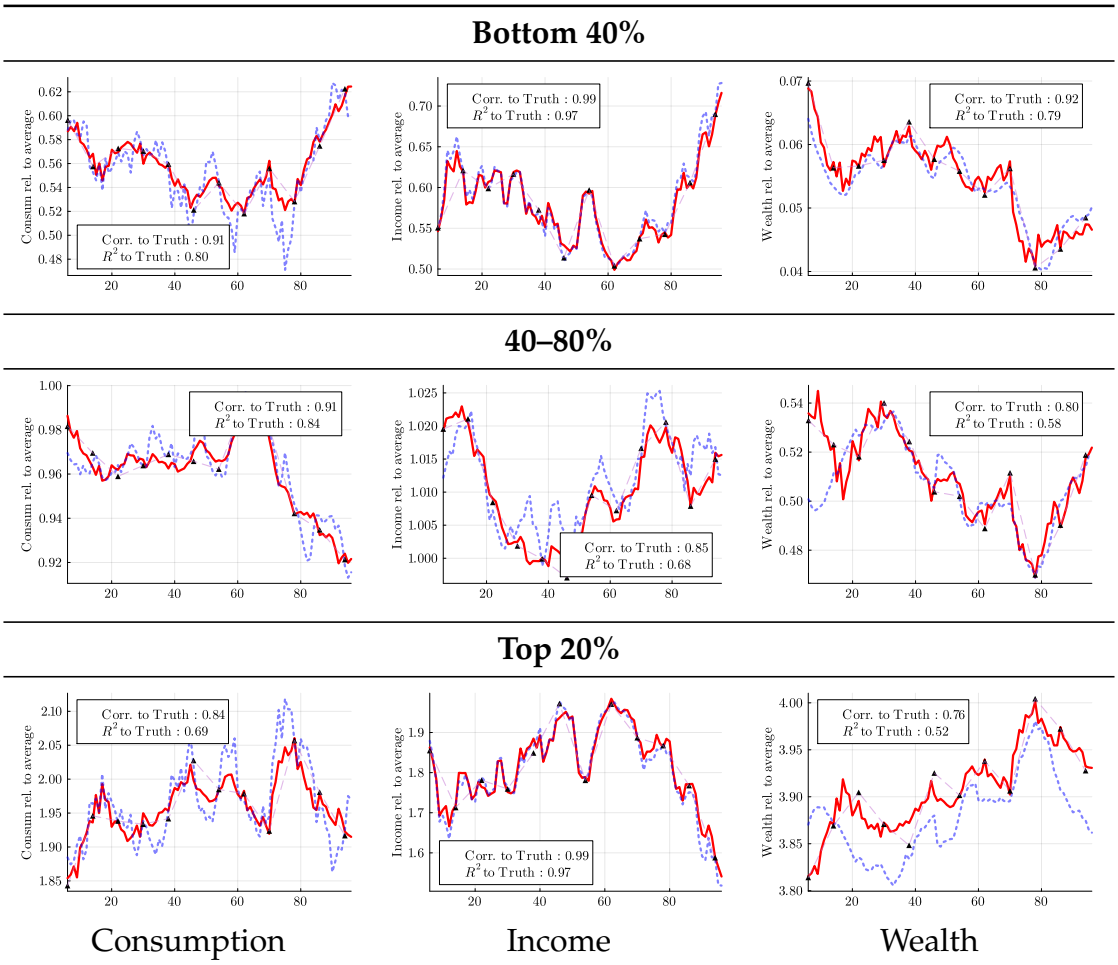
Figure 10 provides the results of this exercise visually (for economy 1), plotting the evolution of consumption, income, and wealth by groups, relative to the economy-wide average. The figure compares the *DDFM* (solid red) estimates to the HANK simulated truth (dotted blue). For comparison, we plot *naive* estimates using Sample A in a light-pink line; simply a linear-interpolation of the black triangles which mark the observations in Sample A. This also visualizes the sparseness of the data that enters our estimation. In general, the movements across the DDFM and the HANK simulated truth are in sync, i.e. they are highly correlated, and around the same magnitudes. Using the naive linear-interpolation would miss key distributional movements, especially for consumption and income.

5 Distribution Dynamics over the Business Cycle

The focus of our paper is to provide a method for synthesizing dynamic distributional data from micro and macro data inputs. In the previous section, we have established the reliability of that procedure. Before concluding, we show the usefulness of the synthetic data. We document characteristic differences across the last three U.S. recessions in terms of their consumption distribution impact, tracing consumption dynamics across the income and wealth distribution. This speaks to the emerging literature studying how consumption distributions and consumption risk move over the business cycle (see, e.g., Berger, Bocola, and Dovis 2023; Bilbiie et al. 2025; Patterson 2023). Such heterogeneity is central to modern theories of aggregate fluctuations (e.g., Kaplan, Moll, and Violante 2018; Bayer, Born, and Luetticke 2024), yet high-frequency evidence by income and wealth remains limited; our reconstructed series help fill this gap and provide new empirical targets for disciplining and calibrating heterogeneous-agent business-cycle models. A companion question—how much of the movement in the distribution is spanned by aggregate dynamics in the first place—we treat in Appendix J: Innovations to the aggregate block account for 15–31 percent of the five-year forecast-error variance of seven of the eight distributional factors, and for 69 percent of the remaining one, which captures the business cycle.

33. Wealth by income is the exception. The wealth of the middle 40%-80% income varies as much as the tails, and its comparatively lower correlation reflects the steep income profile of wealth, which the truncated basis captures less well. Truncating at order 20 or higher improves correlations in exchange for computation time.

Figure 10: Model Estimates (in red) vs. HANK Simulated Truth (in blue)



Notes: The figure reports estimates for consumption, income, and wealth relative to their economy-wide average. Estimates are shown for three groups: the bottom 40%, the next 40%, and the top 20%. Each panel compares model estimates with the HANK simulated truth. Model estimates (*DDFM*) from incomplete samples are shown as a solid red line; *HANK Simulated Truth* is shown as a dotted blue line. The black triangles labeled *HANK Data Used* mark the observations of Sample A (calculated based on the polynomial quantile estimator)—one of the five samples entering the DDFM estimates—and serve as the interpolation nodes of the naive benchmark (light-pink dashed line); the objects displayed follow Sample A's measurement. *Corr. to Truth* reports the correlation between the red and blue lines. R^2 *to Truth* is the coefficient of determination from regressing detrended HANK simulated truth on detrended model estimates.

They account for close to 60 percent of the movements in the copula. Aggregate dynamics thus matter most for how consumption, income, and wealth co-move, complementing Bayer, Born, and Luetticke (2024).

5.1 Consumption Dynamics over the last three recessions

The relative distribution of consumption losses in recessions has received considerable attention in the macroeconomic literature (e.g., Coibion et al. 2017; Cloyne, Ferreira, and Surico 2020; Bilbiie et al. 2025; Holm, Paul, and Tischbirek 2021; Fagereng, Holm, and Natvik 2021). The motivation is partly normative, in the sense of identifying groups that benefit or are adversely affected, and partly theoretical because a greater cyclicality of incomes of high-MPC households can destabilize the economy (Bilbiie 2020). Recessions are also where the balance between the aggregate and autonomous forces vary the most in our estimates: while autonomous innovations account for most of the fluctuations in the distributional factors and marginals on average, a historical decomposition of the factor paths shows that their role differs sharply across NBER recession windows—dominant for all factors during the Dot-com recession and the Global Financial Crisis, but outweighed by aggregate innovations for several factors in the COVID-19 recession (Appendix J).

Using our synthetic data, we trace consumption dynamics along both the marginal and joint distributions of income and wealth across the last three U.S. recessions. We find that their distributional consequences are state-dependent: wealth position shaped the Dot-com bust, balance-sheet distress the financial crisis, and the combination of low income and low wealth the COVID-19 recession. The most affected households are not always those the marginal distributions single out—in the financial crisis, the largest and most persistent relative decline is among asset-rich, low-income households, a group defined only in the joint distribution—challenging research strategies that rely on marginal distributions and unconditional cyclical elasticities.

What is known about these episodes comes from a variety of settings and econometric approaches. For the Financial Crisis, county-level variation ties consumption declines to collapsed housing net worth and leverage (Mian, Rao, and Sufi 2013), and household surveys show the wealth-poor cutting expenditure the most (Krueger, Mitman, and Perri 2016; Petev, Pistaferri, and Saporta-Eksten 2011). For COVID-19, card and bank-account data show high-income households cutting spending the most while transfers propped up spending at the bottom (Chetty, Friedman, and Stepner 2024; Cox et al. 2020). For the Dot-com recession, the evidence is limited to boom-era wealth-effect estimates for stockholders (Dynan and Maki 2001; Maki and Palumbo 2001) and to annual consumption-inequality series spanning the episode (Krueger and Perri 2006).

Relative to this literature, our contribution is threefold. First, the synthetic

data reproduce the observations above, but within the lens of a single model—an additional, external validation of the method. Second, we extend the data to the *joint* income–wealth position: consumption groups like "low-income but indebted" or "asset-rich but low-income," which drive our results below, are unobservable in spending data (which lack wealth) and hard to isolate in single-survey studies. Third, our approach covers all episodes within a common framework, at quarterly frequency—and reaches back to episodes that predate modern spending data—the cross-recession comparison itself becomes possible, and it is this comparison that documents variation across recession episodes. Existing distributional evidence spanning comparably long horizons concerns secular trends (Heathcote, Perri, and Violante 2010; Meyer and Sullivan 2023); our estimates supply the business-cycle counterpart.

To begin, for the three recessions, we first consider the consumption dynamics of households along the marginal distributions of income and wealth, creating four groups: (1) the bottom 20%, (2) the 20% to 40%, (3) 40% to 80%, and (4) the top 20%.[34] These are the first two rows of Figure 11. For the final row, we then consider the joint distribution and estimate consumption dynamics for four key household groups. Group (i) consists of liquidity-constrained hand-to-mouth households, defined as the bottom 20% income *and* bottom 20% wealth; group (ii) is a population of asset-rich, low-income households which offers some resemblance to the wealthy hand-to-mouth, defined here as the bottom 40% of income but top 30% of wealth; group (iii) is the middle class, presumably prudent, patient "buffer-stock" households, located between the 40th and 80th percentiles in both income and wealth; and group (iv) comprises the richest households in terms of income and wealth, largely unconstrained, presumably Permanent Income Hypothesis households, residing in the top 20% in both income and wealth.[35]

For all groups, absolute consumption (not displayed) falls in a recession and rises in a recovery; therefore, we compare the business cycle dynamics of con-

34. The income measure we use includes transfers, but is not net of taxes.

35. For our analysis, groups are defined by contemporaneous ranks, so membership can change from quarter to quarter. This implies some *churning* relative to a true panel. Absent large asset price movements that affect portfolios, compositional changes within our broad bins can be expected to be small (cf. Kuhn, Schularick, and Steins 2020).

This interpretation is consistent with evidence that mobility is limited: positive shocks required for upward mobility are rare at the bottom, while large negative shocks are infrequent at the top (Guvenen et al. 2021); hand-to-mouth status is persistent (Aguiar, Bils, and Boar 2025); wealth positions are especially sticky due to persistent heterogeneity in returns (Fagereng et al. 2020); and earnings mobility is largely confined to adjacent deciles (Ettmeier, Kim, and Schorfheide 2024). Using coarse groups mitigates these concerns, though they become more salient when conditioning on multiple dimensions.

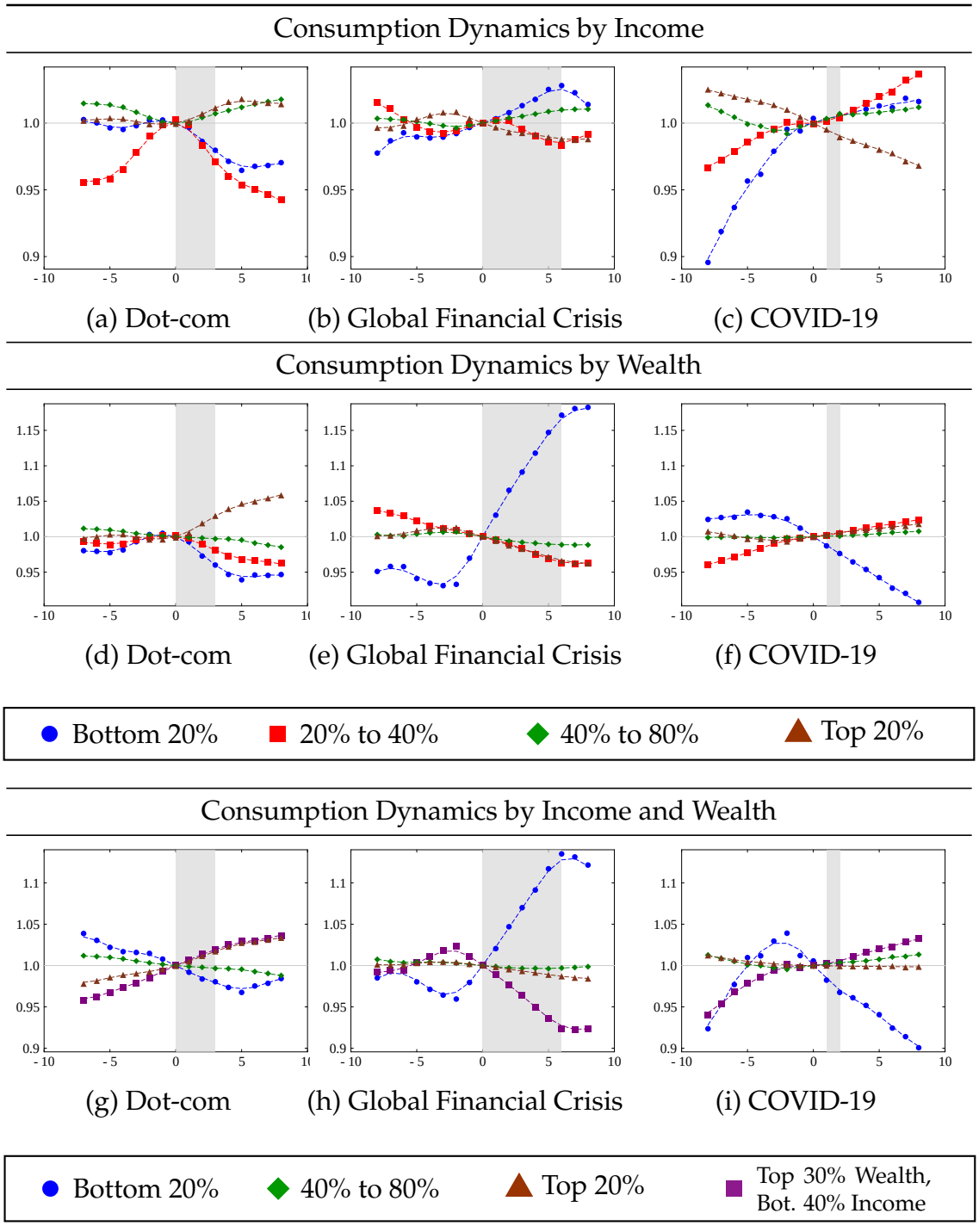
sumption for the different income and wealth groups, in each period t, *relative* to the household-wide consumption average at period t and then index these relative consumption dynamics to the quarter preceding the recession (the peak).[36]

In this sense, our goal is to understand how a group's consumption shifts relative to the aggregate (the average household) over a business cycle. The observable deviations from the aggregate provide a reduced-form summary of the distributional channels at work, which may in turn amplify or dampen the corresponding macroeconomic shocks. Figure 11 summarizes how these deviations are distributed across households depending on the nature of each recession.

Dot-com recession (Panels a, d, g). This recession resembles a financial wealth shock, though its relative burden falls on the bottom of the wealth distribution rather than on the wealthy: along income (Panel a), the bottom two income groups lose a moderate 3–6% relative to average while the top half of the distribution gains slightly. Along wealth (Panel d), the top 20% pull away, gaining about 6% relative to average, while all other wealth groups decline by 1–6%. The sharpest relative losses accrue to the bottom 20% wealth group—likely indebted households with high MPCs (Havranek and Sokolova 2020)—whereas the wealthiest, whose consumption is least tied to current income, absorb the equity correction without cutting relative consumption. The asset-rich, low-income group (purple, Panel g) climbs steadily throughout the window, from below its pre-recession level to about 4% above it two years later, and the joint bottom group falls below the average. To our knowledge, no episode-specific account of consumption dynamics along the income and wealth distribution exists for this recession—the contemporaneous literature instead measured consumption wealth effects for stockholders during the preceding boom (Dynan and Maki 2001; Maki and Palumbo 2001)—so these estimates provide the first such account.

36. We construct symmetric 3-quarter moving averages and use this moving average for the normalization to the pre-recession quarter.

Figure 11: Comparison of Consumption Dynamics during Recessions



Notes: Relative consumption dynamics during recessions along the income and wealth distribution. Consumption dynamics of each income/wealth group are shown relative to average household consumption. These relative consumption time series for each group are indexed to the peak. The vertical axis shows changes of consumption over time relative to the change of average consumption over time. The horizontal axis shows the time relative to the peak ($= 0$). The recessions are the Dot-com recession in 2001Q1, the Global Financial Crisis in 2007Q4, and the COVID-19 recession in 2019Q4. Shaded areas mark the NBER peak-to-trough quarters of each recession; the thin horizontal line marks the aggregate benchmark.

Global Financial Crisis (Panels b, e, h). The defining feature of that recession is its balance-sheet character. Correspondingly, consumption shares by income groups do not show much dynamics, within 2-3 percent in each direction. Yet by wealth, this differs. Compared to the onset of the crisis, the consumption share of the wealth poorest was lower before and increases afterwards (Panel e). One might be tempted to interpret this as a result of lower interest payments freeing up resources for net borrowers. Yet, the rise is mainly compositional: the correlation of income and wealth weakens as leveraged home-owners with relative high incomes lose their net worth; the center of the balance-sheet channel of Mian, Rao, and Sufi (2013).[37] Among the remaining groups, the wealthiest 20% and the 20–40% wealth group cut relative consumption by 3–4%, and the asset-rich, low-income group (purple, Panel h) declines most persistently, by about 8%, consistent with wealth-dependent consumers bearing the asset-price collapse. Low-income households fared relatively well (Panel b), possibly supported by fiscal transfers.

COVID-19 recession (Panels c, f, i). During this last recession, the top 20% of the income distribution is the only income group that cuts relative consumption, by about 3-4% two years out (Panel c), in line with spending-data evidence that high-income households cut the most (Chetty, Friedman, and Stepner 2024). The other income groups gain 1–4% relative to average, the bottom 20% about 2%, likely due to the large government transfers (cf. Cox et al. 2020). By wealth (Panel f), the dispersion is even larger: low-wealth households lose persistently, about 9% relative to average with no recovery two years out, while every other wealth group holds at or above the average. The asset-rich, low-income group (purple, Panel i) gains about 3% relative to average, more than any other group in the joint panel, perhaps benefiting from transfers, asset-price rebounds, and insulation from labor-market shocks. The bottom 20% in income and wealth are the most exposed.

The key finding is *state-dependence*: the channels through which recessions affect households—financial, balance-sheet, or labor-market—vary with the nature of the shock. Our findings suggest that during the Dot-com bust, wealth mattered; during the Global Financial Crisis, debt mattered; and during COVID-19, the combination of low income and low wealth mattered, partly offset by

37. A fixed-composition (Laspeyres) decomposition (by income rank) attributes essentially all of the group's relative rise to this reshuffling (results available upon demand). Household-panel evidence that the wealth-poor cut expenditure the most (Krueger, Mitman, and Perri 2016; Petev, Pistaferri, and Saporta-Eksten 2011) is not in conflict: those studies follow fixed households, whereas our groups are wealth ranks whose membership churns precisely in balance-sheet recessions.

fiscal transfers. Neither income nor wealth alone is a sufficient statistic for understanding consumption dynamics. This state-contingency cautions against research strategies that rely on uniform unconditional cyclical elasticities. It underscores that stabilization policy needs to diagnose the primary channel through which a crisis operates and take a holistic view of the sum of all stabilization measures.

6 Conclusion

In this paper, we present a new method to derive synthetic distributional consumption, income, and wealth data. The method contributes to the modern theory of macroeconomic dynamics that has the joint distribution of consumption, income, and wealth as a key determinant of aggregate dynamics. Our framework supplies quarterly empirical counterparts to distributional objects in heterogeneous-agent models. We have shown that the method can incorporate information from various microdata sources regardless of their frequency and coverage of variables. By reconstructing deliberately withheld survey observations, we show that our method can generate joint distributional information at high frequency with good precision. Using a HANK model as a laboratory, we validated that the synthetic data from our model closely follow the "true" distributional dynamics when supplied with micro data samples as rich and frequent as in the U.S.

To illustrate the method's usefulness, we further traced consumption dynamics across the last three U.S. recessions along the income and wealth distribution. The application suggests that distributional consequences of recessions are state-dependent, with different channels dominating in different episodes—financial, balance-sheet, or labor-market. Although these findings are only suggestive, they demonstrate how synthetic data can inform debates about the role of heterogeneity in macroeconomic dynamics. Our estimates also show that the business cycle acts on the distribution mainly through its dependence structure. Aggregate dynamics account for close to 60 percent of the fluctuations in how consumption, income, and wealth co-move, but only about a quarter of the variance in the marginal distributions. The business cycle is therefore felt more strongly in the dependence structure than in the marginals, complementing the structural findings of Bayer, Born, and Luetticke (2024).

Beyond what we present in this paper, our methodology provides a flexible tool for researchers: it can readily accommodate distributions based on, e.g., liq-

uid and illiquid assets, financial income, or secured and unsecured debt, and even beyond households (e.g., firms and banks) and, in principle, be of higher frequency. Another promising direction is to apply the method to short panels where the joint distribution spans current and lagged values of the same variable. This would allow researchers to estimate high-frequency dynamics of household-level autocorrelation structures as in Almuzara et al. (2025), but incorporating a richer set of variables. We leave these extensions for future work.

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Appendix

A Economic Fit

The series estimator fits a single set of coefficients $\theta_t^j = \left(\xi^j_{m,o_1,t}, \ldots, \kappa^j_{(o_1,\ldots,o_d),t}\right)$ by projecting the entire distribution of the variable(s) of interest onto an orthogonal polynomial basis; it is therefore a *global* method in the sense that one parameter set governs all quantile/copula regions simultaneously. If distributions were exactly composed of finitely many orthogonal polynomials, the global estimator would be exact. In practice, however, the truncation we impose may systematically over- or underestimate certain segments, potentially smoothing over economically relevant features.[38] Also, even without truncation, we might fit polynomials to regions with zero mass in the measure—generating, potentially, spurious variation. Both concerns are naturally important in our setting, where our task is to recover the cross-sectional and time-series variation in the entire distribution, *Economic Fit*, not just segments best approximated by our estimator.

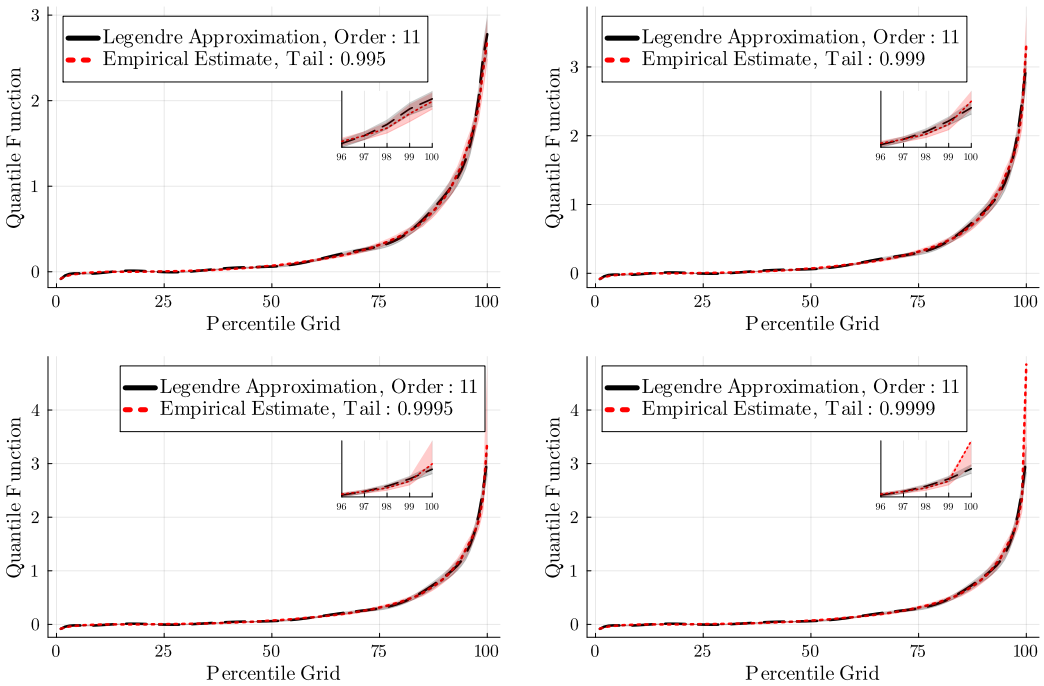
Two natural questions arise: (1) where does our estimator systematically fall short and (2) can these weaknesses be accommodated. For (2), this means (a) can the current (global) polynomial weights be adjusted to better capture specific regions (without compromising other regions), such as the upper tail of the wealth distribution or do they already carry enough flexibility to capture these specific regions (and the others)? and (b) can the estimator accommodate distributions with point-zero mass? In this spirit, the appendix addresses two points. First, it evaluates the degree to which we miss local variations. Second, it shows how an economist with a theoretical prior about which parts of the distribution to emphasize (e.g., the top of the wealth distribution) could adjust the coefficients accordingly. As a byproduct, for a given estimator, it also provides detailed evidence of variation within the distribution via observing weight changes within the distribution. More on this latter point later.

38. Our projection forces parts of the distribution onto the squareintegrable function space $\mathcal{L}^2$, even though those parts themselves do *not* belong to $\mathcal{L}^2$. Good examples are the upper tails of the consumption, income, and wealth, which are often modeled by Pareto laws. Vermeulen (2018) shows that wealth displays extremely heavy tailsso heavy that its variance is undefinedbecause the Pareto "alpha" parameters fall below 2. Likewise, Atkinson, Piketty, and Saez (2011) document that the U.S. incometail Pareto beta rose from 1.82 to 3.42 between 1976 and 2007, implying a shift in the corresponding alpha from 2.22 down to 1.41. As Toda and Walsh (2015) emphasize, analyses involving fat-tailed data require great care: when higherorder moments fail to exist, the Central Limit Theorem breaks down and conventional confidence intervalsfor instance, those on the top10 percent wealth sharelose validity. We are grateful to a referee for highlighting this issue.

A.1 Performance of the Estimator at the Tails

To investigate the first question, we begin with Figure 12, which compares two estimators: (1) the polynomial approximation as in the paper and (2) the empirical percentile function, shown with 99% interval bands. Each panel repeats the same percentile curve but varies its upper-tail cutoff, using final grid points $p \in \{0.995, 0.999, 0.9995, 0.9999\}$. This addresses, in one panel, where (1) falls short, with particular focus on the tails, motivated by the footnote. As such, an inset for each panel is provided, zooming into the top 5% of the respective distribution. For wealth, we observe near coincidence of the two estimators up to about $p = 0.9995$ in this diagnostics. Only beyond the 99.95th percentile—the top one-twentieth of one percent—does the series estimate begin to pull away from the empirical percentile function (itself subject to sampling error). Hence, any inference inside the top 0.05 percent of the distribution should be treated with particular caution.[39]

Figure 12: Wealth Tails



Notes: Figure shows four panels plotting the quantile function for wealth, evaluated on a grid $G = \{0.01, \ldots, 0.99, p\}$. Wealth, as defined, is scaled by its respective aggregate and transformed via an inverse hyperbolic sine function. The four panels differ in the plotted estimate of the tail, denoted by p, which appears in the legend. The inset of each panel zooms in on the tail, covering the last five percentiles. No estimator is evaluated for $p = 1.0$.

39. Results on consumption and income are readily available and can be provided upon request.

A.2 Local vs. Global Estimation

To answer the second question, we introduce a *local* estimator that computes a set of coefficients within a moving kernel window. By sliding the window along a percentile grid G, we obtain $|G|$ coefficient vectors, each tailored to the local variation within that window. The local estimator is flexible and well-suited for capturing local structure.[40] By comparing the global estimator's single set of coefficients with the collection of locally estimated coefficients, we can assess where and how they differ—especially around the tails of the distribution. Below, we describe the comparison procedure in detail and show that, for the economic data analyzed in this paper, our global estimator performs comparably to the local approach with minimal loss.

Data Let $w_i \in \mathbb{R}$ denote the economic variable of interest for household i as described in the main body of the text. The variable, as before, is first scaled by the corresponding average (quarterly, perhousehold level) and stabilized with an inverse hyperbolic sine transform. Results presented here rely on the 2019 wave of the PSID and are presented for the different measures consumption, income, and wealth. For this reason, we drop all time indices t and measure indices m.

Global Legendre-series estimator To have the appendix section self-contained, we provide again our estimator. Truncating the polynomial order O to 11, the shifted basis of orthogonal Legendre polynomials, evaluated on the empirical cumulative distribution u_i^m, is $Q_o(u_i^m)$ for $o = 0, \ldots, 11$. Explicitly, $u_i^m = \frac{s_i}{\sum_i s_i}$ for s_i the survey weight corresponding to household i after sorting households based on variable w. Projecting w_i on the basis yields coefficients

$$\hat{\xi}_o^w := N^{-1} \sum_i w_i Q_o(u_i^w) \qquad \text{for } o = 0, \ldots, 11.$$

The estimated quantile function with these coefficients is

$$\hat{\Xi}_{glob}^{-1}(p) \;=\; \sum_{o=0}^{11} \hat{\xi}_o \, Q_o(p), \qquad p \in [0,1], \tag{32}$$

where we use *glob* to denote that this is a global estimator and p some grid point in $[0,1]$ for which we can evaluate the percentile function $\hat{\Xi}_{glob}^{-1}(p)$.

40. Naturally, the estimator will perform poorly outside the window, but we make clear that we use the corresponding set of coefficients for a certain point when evaluating the percentile function at this point, i.e., the optimal coefficients in that window in a least squares sense.

Local kernel Legendre estimator For the local estimation, let $p_g \in G$ be some probability grid point and $G = \{0.01, 0.02, \ldots, 0.995\}$. Let $K_h(u)$ denote a Gaussian kernel with bandwidth h[41]

$$K_h(u) \;=\; \frac{1}{h\sqrt{2\pi}}\,\exp\Big\{-\tfrac{1}{2}\big(u/h\big)^2\Big\}.$$

For a given survey weight of the PSID, s_i, we can generate the local weight $s_i^{(g)} = s_i\,K_h(u_i - p_g)$. Let $W_g = \mathrm{diag}(s_1^{(g)}, \ldots, s_n^{(g)})$ and $\Phi_{io} = Q_o(u_i)$. Solving the weighted leastsquares system

$$(\Phi^\top W_g \Phi)\,\boldsymbol{\xi}^{local}(p_g) \;=\; \Phi^\top W_g \mathbf{y}$$

produces the *local* coefficient vector $\boldsymbol{\xi}^{local}(p_g) = (\xi_0(p_g), \ldots, \xi_{11}(p_g))^\top$, which is estimated separately for each grid point. The estimated *local* quantile function then uses, for each grid point evaluation, a new set of coefficients $\xi_o^{local}(p_g)$, generating

$$\hat{\Xi}^{-1}_{local}(p_g) = \sum_{o=0}^{11} \xi_o^{local}(p_g) Q_o(p_g).$$[42]

Thus, we have the two extremes: the global approach of one set of coefficients to estimate the percentile function and the local approach of many sets of coefficients—localized throughout the entire distribution—to estimate the percentile function.

To see how the coefficients change across the distribution, independently of overall scale, each coefficient vector (from the different estimators) is transformed to unit ℓ^1 length:

$$\tilde{\xi}_o^{local}(p_g) \;=\; \frac{|\xi_o^{local}(p_g)|}{\sum_{o=0}^{11}|\xi_o^{local}(p_g)|}, \qquad \tilde{\xi}_o^{glob} \;=\; \frac{|\widehat{\xi}_o^{glob}|}{\sum_{o=0}^{11}|\xi_o^{glob}(p_g)|}.$$

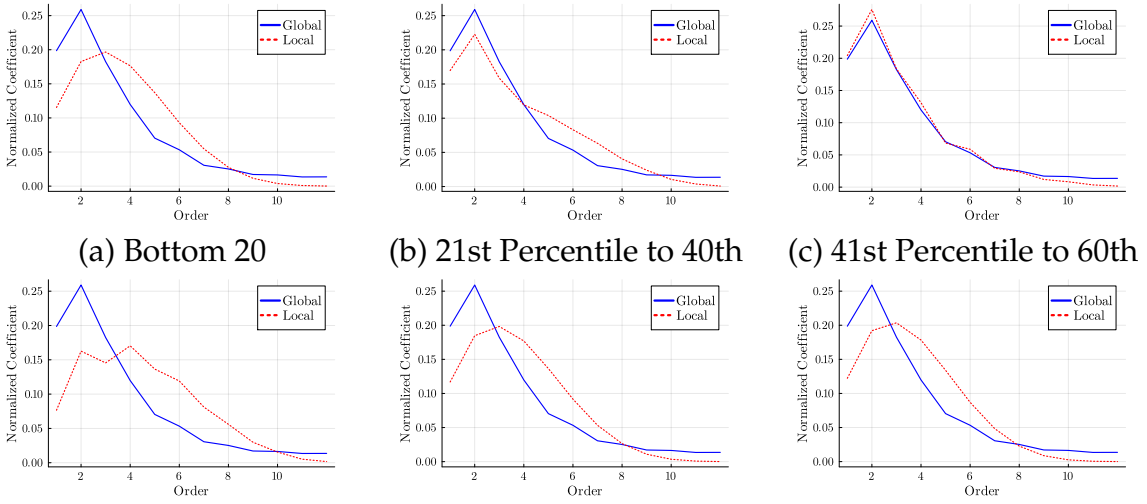
One can then average over the vector of coefficients (over intervals of G) to assess how these weights fluctuate for, for example, the bottom of the wealth distribution versus the top. Figure 13 shows precisely this (results are analogous for consumption and income). In short, our estimator assigns weights almost identically to a local estimator that focuses on the 40th to 60th percentile. In every other case, it tends to assign more weight to the polynomials of order 4 to 8 in exchange for less weight on the first 3 polynomials. One might therefore

41. Results below are unchanged for various smaller bandwidths $h \in \{.005, .01, .05, .10\}$

42. Alluding to our comment before, one can alternatively estimate $\hat{\Xi}^{-1}_{local}(p) = \sum_{o=0}^{11} \xi_o^{local}(p^*) Q_o(p)$, where $\xi_o^{local}(p^*)$ are the weights associated for a selected grid point p^* e.g., the 99th percentile.

conclude that we should instead use a different set of coefficients. Figure 14 provides evidence to the contrary.

Figure 13: Local vs. Global Estimator: Coefficients

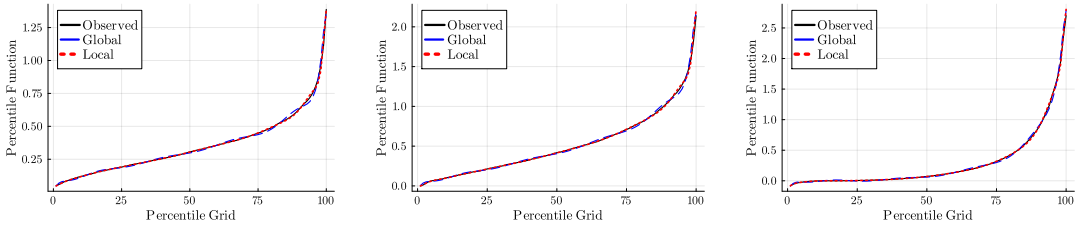


(a) Bottom 20 (b) 21st Percentile to 40th (c) 41st Percentile to 60th

(d) 61st Percentile to 80th (e) 81st Percentile to 100th (f) 91st Percentile to 100th

Notes: Figure shows panels comparing the global estimator to the local estimator for different segments of the wealth distribution. Each panel compares the contribution of each weight across the two estimators. Weights for the global estimator are the same across panels. Weights for the local estimator are calculated by averaging over the set of coefficients in that segment e.g., the Bottom 20 has 20 vectors of coefficients to average over.

Figure 14: Local vs. Global Estimator: Percentile Function



(a) Consumption (b) Income (c) Wealth

Notes: Figure shows three panels of percentile functions, comparing the global estimator to the local estimator, along with the empirical percentile function estimator. Percentile functions are evaluated over a set of grid points $G = \{0.01, \ldots, 0.99,\ 0.995\}$. The data are from the 2019 PSID wave.

A.3 Treatment of non-continuity of the marginal CDF

Some margins (e.g., earnings or financial assets) exhibit an atom at zero in addition to a continuous distribution over $\mathbb{R}_+$. Let Z_{mt} denote marginal m at

time t and

$$\pi_{mt} := P_t(Z_{mt} = 0).$$

Write the CDF as a mixture of the non-differentiable part and (c)ontinuous part

$$\Xi^{Z}_{mt}(z) = \pi_{mt}\mathbf{1}\{z \geq 0\} + (1 - \pi_{mt})\,\Xi^{Z,c}_{mt}(z), \qquad \Xi^{Z,c}_{mt}(z) := P_t(Z_{mt} \leq z \mid Z_{mt} > 0).$$

The unconditional quantile function is

$$\Xi^{Z,-1}_{mt}(u) = \begin{cases} 0, & u \leq \pi_{mt}, \\ \Xi^{Z,c,-1}_{mt}\Big(\dfrac{u - \pi_{mt}}{1 - \pi_{mt}}\Big), & u > \pi_{mt}, \end{cases} \tag{33}$$

where it is 0 when the household's rank is below π_{mt}, positive otherwise, where the positive part is rescaled to mass $1 - \pi_{mt}$. Any strictly increasing transform satisfies these cases e.g., the inverse hyperbolic sine—the transformation we use in practice. Empirically, we estimate the mixed distribution in two steps: (i) estimate the point mass

$$\hat{\pi}_{mt} = \frac{\sum_i s_{it}\mathbf{1}\{Z_{mit} = 0\}}{\sum_i s_{it}},$$

and (ii) using only $Z_{mit} > 0$, estimate the conditional quantile function of $X_{mit} = T(Z_{mit})$ via the Legendre expansion

$$\Xi^{X,c,-1}_{mt}(u) \approx \sum_{o=0}^{O} \xi^{c,X}_{mot} Q_o(u).$$

The coefficient vector entering the factor extraction carries only the conditional coefficients $\xi^{c,X}_{mot}$, so its dimension — and the entire state-space apparatus — is unchanged relative to a continuous marginal. The point mass $\hat{\pi}_{mt}$ is precisely measured by the participation surveys and is treated as observed: it enters at the reconstruction stage through (33), where the flat segment and the rescaling of the conditional quantile function reassemble the mixed distribution.

A.4 Treatment of the Copula with semi-continuous margins

Let Z again denote the semi-continuous margin. For the three-dimensional copula, this means there is a flat region along the Z margin, which renders differentiability in that neighborhood infeasible. To deal with this, there are three sets of dependence coefficients to keep track of: those among holders ($Z > 0$),

those among non-holders ($Z = 0$), and those of the population copula. This may seem daunting to estimate, but due to properties of the estimator, this is estimated quite efficiently. Throughout, let u_Y, u_W and u_Z denote population ranks (computed on the full sample, with the zeros of Z occupying $[0, \pi_t]$), and let $u'_Z = (u_Z - \pi_t)/(1 - \pi_t)$ denote the corresponding rank within holders.

For example, when expanding the copula in the orthonormal shifted Legendre basis, the coefficients would be $\kappa_{jkl,t} = E_t[Q_j(u_Y)\, Q_k(u_W)\, Q_l(u_Z)]$. Since $Q_0 \equiv 1$, the coefficients with $l = 0$ *are* the population (Y, W)-copula coefficients κ_t^{YW} — the same objects measured by surveys that do not observe Z. This is exemplified in our Section 3.

The holder and non-holder objects $\kappa^{YWZ}|_{YW}$ and $\kappa^{YW,0}$ are therefore coefficients of the densities of population ranks conditional on holder status, not copulas of their own, and (34) below is the law of total expectation applied to $E_t[Q_j(u_Y)Q_k(u_W)]$. The population, holder, and non-holder (Y, W) coefficients satisfy the following identity

$$\kappa_t^{YW} = \pi_t\, \kappa_t^{YW,0} + (1 - \pi_t)\, \kappa_t^{YWZ}\big|_{YW}, \tag{34}$$

where $\kappa_t^{YWZ}|_{YW}$ collects the (Y, W) coefficients among holders and $\kappa_t^{YW,0}$ those among non-holders.

Estimation therefore splits the coefficient array by the order of the marginal Z: the $l = 0$ entries are estimated on the full sample (the population copula), while the $l \geq 1$ entries are estimated on holders only, with Z entering through the within-holder rank u'_Z — the same rescaling as in (33). Like $\hat{\pi}_{mt}$, the holder (Y, W) slice $\kappa_t^{YWZ}|_{YW}$ is measured directly from the participation surveys and treated as observed. At reconstruction, we can recover the non-holder rank density, which follows from (34),

$$\kappa_t^{YW,0} = \frac{\kappa_t^{YW} - (1 - \pi_t)\, \kappa_t^{YWZ}\big|_{YW}}{\pi_t},$$

and the copula density on the unit cube is reassembled as the mixture

$$c_t(u_Y, u_W, u_Z) = \begin{cases} c_t^{YW,0}(u_Y, u_W), & u_Z \leq \pi_t, \\ c_t^{YWZ}\Big(u_Y, u_W, \dfrac{u_Z - \pi_t}{1 - \pi_t}\Big), & u_Z > \pi_t, \end{cases}$$

with values on the Z-margin from the conditional quantile function. Within the zero block, u_Z is assigned uniformly on $[0, \pi_t]$, independently of (u_Y, u_W); this is the standard randomized probability-integral transform for an atom. The non-

holder coefficients are formed only when $\hat{\pi}_t > 10^{-10}$ (otherwise the zero block carries no mass), and, like all truncated series densities in the paper, the recovered density need not be pointwise nonnegative; we report its exact integrals over cells.

B Estimation of Factor Structures and Marginal Data Densities

A key input to our procedure is the mapping from factors to observables, defined by the projection matrix Γ. To estimate Γ, one method would be based on complete data alone. Complete here will mean there is no missing data in the spatial sense i.e., the data reports consumption, income, and wealth. For our case, this means using exclusively the PSID data to estimate Γ based on the PCA of $\tilde{\boldsymbol{\theta}}^{PSID}$. We consider different scenarios where we retain a different number of factors each time. Specifically, for this approach, we estimate the model with 3, 6, 7, and 8 distributional factors, obtained from the 12 PSID waves; however, this approach runs the risk of not fully representing the entire subspace that characterizes the evolution of the joint distributions because of the infrequent releases of the PSID data.

The literature has suggested alternatives for taking into account *incomplete* data in factor models. Here, this would mean surveys that only collect data on some, but not all marginals e.g., only income and wealth as in the SCF. In particular, we explore two alternative estimators for Γ: we consider the Tall-Wide algorithm of Bai and Ng (2021) and an EM algorithm. We compare the quality of the model estimates under the alternative methods using marginal data densities for model comparison.

B.1 Tall-Wide Algorithm of Bai and Ng (2021)

An alternative approach to incorporating an additional block of data to the complete data is provided by Bai and Ng (2021). The paper tackles this problem by identifying two blocks of data within some larger T by N matrix— a tall block (data observed for all periods) and a wide block (time periods for which entire distribution is observed). For our setting, this means we add a tall block to our estimation of Γ, while the wide block would be what we call the complete

data.[43] The tall block we add incorporates data on lower dimensional copulas of (income, consumption) and (income, wealth), as well as additional data on the marginals. This would be variation coming from both the CEX and SCF.

This alternative estimator for Γ, since it relies on more data, may improve model fit. For our application, we extract the factors that explain 80% of the summable variation, as well as 85%. This corresponds to 10 and 12 factors; however, as mentioned before, it is unclear the degree of variation shared among this set of factors relative to the original set of factors. Furthermore, keeping more factors from this estimator runs the risk of over-parameterizing the model. Table 3 presents the marginal likelihoods of these *TW* projection matrices. Results suggest these models have inferior fit than models that only rely on complete data.

B.1.1 EM-completed loadings.

As a second alternative to the complete-case projection matrix, we estimate the loadings on the full, unbalanced panel of survey coefficients. Missing entries are filled by the expectationmaximization (EM) algorithm for principal components: starting from the unconditional mean, the algorithm alternates between imputing the missing coefficients from the current low-rank fit and re-estimating the factors and loadings by principal components on the completed panel, until the imputations converge. Banbura and Modugno (2014) develop the likelihood-based counterpart for arbitrary patterns of missing data. Unlike the complete-case estimator, which uses only the survey waves in which every coefficient is observed, the EM-completed loadings draw on all waves of all surveys. The data reject this alternative clearly. With eight distributional and eleven aggregate factors, the EM-completed specification attains a log marginal data density of $-86{,}799.3$, against $-57{,}237.9$ for the complete-case baseline, both evaluated on the common observation vector (Table 3).

B.2 Marginal Data Densities and Optimal Factor Structures

We estimate various versions of the state space model for different factor loadings Γ, which differ both in the number of factors and in the estimation of Γ itself, and for different numbers of aggregate factors. For each estimated model, we calculate the marginal data density (MDD) in order to discriminate between

43. It is important to note that, given some data, such an estimator can consistently estimate the common component without making any assumptions on the nature of missingness. We refer the reader to the paper for more details on how the exact procedure is performed.

Table 3: Model Comparison

Model	Own observation vector		Common observation vector	
	Harmonic Mean	Bridge	Harmonic Mean	Bridge
		TW Projection Matrix		
12 distributional factors, 11 agg. factors	−88195.0	−87999.6	−89543.2	−89347.9
10 distributional factors, 11 agg. factors	−88484.4	−88348.9	−89832.6	−89697.2
		Standard Projection Matrix		
3 distributional factors, 3 agg. factors	−85428.7	−85391.7	−89351.1	−89314.2
6 distributional factors, 11 agg. factors	−73636.3	−73526.6	−74984.6	−74874.8
7 distributional factors, 11 agg. factors	−64945.3	−64817.9	−66293.6	−66166.1
8 distributional factors, 3 agg. factors	−52647.6	−52481.6	−56570.0	−56404.0
8 distributional factors, 11 agg. factors	−56040.2	−55889.6	−57388.4	−57237.9
8 distributional factors, 15 agg. factors	−58214.3	−58022.0	−58214.3	−58022.0
		EM-completed Projection Matrix		
8 distributional factors, 11 agg. factors	−85597.4	−85451.1	−86945.7	−86799.3

Notes: The table reports the log of the marginal data densities (MDD) across different model specifications. The MDD is estimated using two estimators: (1) Harmonic Mean and (2) a Bridge sampler. TW Projection Matrix are models estimated using the Tall-Wide algorithm of Bai and Ng (2021). Standard Projection Matrix is from a PCA estimator on complete cases; EM-completed uses EM-completed loadings. Since the aggregate block is observed through its first k principal-component scores, models with different numbers of aggregate factors are densities over different observation vectors (*Own observation vector*). The *Common observation vector* columns add, for a model with $k < 15$ aggregate factors, the log marginal density of the scores $k+1, \ldots, 15$ under independent AR(1) processes with conjugate priors, so that all rows are densities over the same data. Higher values indicate greater marginal likelihood.

these alternatives. The models considered rely on mostly the same data, but differ in the size of the parameter space.[44] The MDD effectively internalizes these two features: it selects the model for which we can expect the best forecasting performance, while penalizing models with larger parameter spaces through a lower density.

The marginal data densities $p(\boldsymbol{y})$ are estimated using Geweke's modified harmonic mean estimator. Although standard, this estimator raises two concerns. First, given the dimensionality of the model, the posterior may not be well approximated by a Gaussian distribution. Second, the estimator may not be numerically stable, as it requires the inversion of large matrices. We therefore also estimate the density using a bridge sampler (Meng and Wong 1996), which is numerically more stable and may better internalize the shape of the posterior.

44. Since aggregate factors enter as measurements every quarter, their number changes the observation vector and hence the likelihood contributions. Figure 8 provides suggestive evidence on the strength of this influence. We therefore also report MDDs evaluated on a common observation vector; see below.

Table 3 covers the span of proposed models that were ultimately assessed, as well as the marginal data densities over the different estimators. First, we find that the MDD is most elastic to the number of distributional factors, as opposed to the number of aggregate factors. Among models with 11 aggregate factors, the model with the projection matrix from the complete PSID data, labeled *Standard Projection Matrix*, achieves the highest MDD with the maximum number of 8 factors.[45] Using the factor loadings Γ from the TW-algorithm of Bai and Ng (2021), or loadings completed by the EM algorithm, yields a lower MDD and thus a worse model fit.

C Minnesota Prior

Given the size of the model, we estimate it in a Bayesian framework and regularize the state equation parameters $(A, B, C, \mathrm{diag}(D), \Omega)$ using a Minnesota prior.

Stacking the coefficients as $\psi := (\mathrm{vec}(A)', \mathrm{vec}(B)', \mathrm{vec}(C)', \mathrm{diag}(D)')'$ and setting

$$\psi \sim \mathcal{MN}(\mu_{\text{Minn}}, V_{\text{Minn}}),$$

with mean

$$A_{ij} = \begin{cases} \kappa_{2,i}, & i = j \text{ (own first lag of distributional factor } i), \\ 0, & i \neq j, \end{cases}$$

$$B = \mathbf{0}, \quad C = \mathbf{0}, \quad D_{ii} = \kappa_3,$$

and diagonal variance

$$V_{\text{Minn},ii} = \begin{cases} \kappa_0 / l^2, & \text{own lags}, \\ (\kappa_0 \kappa_1 / l^2) \cdot (\hat{\sigma}_{ii}^2 / \hat{\sigma}_{jj}^2), & \text{cross-lags}. \end{cases}$$

we have our Minnesota prior. Here $\kappa_{2,1}, \ldots, \kappa_{2,r}$ set the prior persistence of each distributional factor separately, and κ_3 that of the aggregate factors. A single persistence parameter shared by all factors is pulled toward the fast-moving,

45. Models with different numbers of aggregate factors have different observation vectors, so their MDDs are not directly comparable. The columns labeled *Common observation vector* add, for a model with $k < 15$ aggregate factors, the log marginal density of the aggregate scores $k+1, \ldots, 15$ under independent AR(1) processes, so that all models are evaluated on the same data. On this common basis, the model with 3 aggregate factors attains a higher MDD than the model with 11.

data-rich factors (e.g. those loading on quarterly income), and the slow-moving, sparsely observed ones (e.g. those loading on wealth) inherit their jitter. A separate target per factor lets each find its own persistence. We work with unconstrained parameters and map them to $(-0.99, 0.99)$ via a scaled inverse-logit. The hyperprior on the aggregate target is diffuse, $\kappa_3 \sim \mathcal{N}(0, 5)$. The factor-specific targets get $\kappa_{2,j} \sim \mathcal{N}(3, 1.5)$ on the unconstrained scale, which centers the implied persistence near 0.9 and puts 0 about two standard deviations below. Where the data say little about a factor, its persistence defaults to a smooth value; where they are informative, they can pull it anywhere in $(-0.99, 0.99)$.

Prior for Ω. For the innovation covariance Ω, we place weakly informative priors on the diagonal elements (with variances governed by κ_4 and κ_5) and an LKJ prior on the correlation matrix. We parameterize the LKJ shape as

$$\eta = 1 + \log(\exp(\kappa_6) + 1),$$

so that $\eta \geq 1$, with $\kappa_4, \kappa_5 \sim \mathcal{N}(0, 1)$ and $\kappa_6 \sim \mathcal{N}(0, 2)$. Larger η shrinks correlations toward zero unless strongly supported by the data. We estimate $\{\kappa_1, \kappa_3, \kappa_4, \kappa_5, \kappa_6, \{\kappa_{2,i}\}_{i=1}^{6}\}$ jointly with $\boldsymbol{\psi}_{par}$ (except κ_0, fixed at 0.05).

D Details on MCMC

To estimate and sample from the posterior distribution, we employ the DIME sampler from Boehl (2024). The sampler is particularly advantageous for potentially complex, high-dimensional posterior distributions with ex-ante unknown properties.

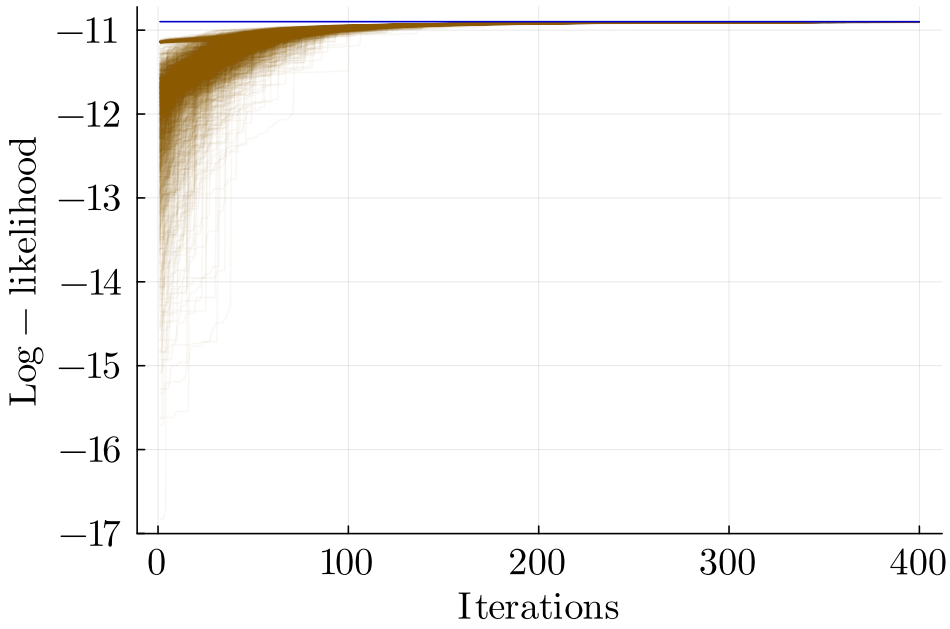
We run an ensemble of $4n$ chains (for n the size of the parameter vector). To speed up adaptation, 10% of chains are initialized at a tentative mode from a separate optimization procedure; the remainder are initialized from the priors. The ensemble runs for 400 iterations, and we keep the last 25% of draws as posterior samples.[46] The sampler uses a single tuning parameter χ that governs the mixture between the local and global transition kernel; we set $\chi = 0.1$.

Figure 15 shows the traces of the (scaled) log-likelihood for all chains and shows stabilization of the sampled log likelihood.[47] Additional diagnostics are reported in the figure notes.

46. For the baseline $|\boldsymbol{\psi}| \approx 500$, this implies $4(500) \times 400 = 800,000$ chain iterations, of which 25 percent are retained.

47. Traces of all estimated models can be provided upon request.

Figure 15: Converging Chains



Notes: Figure shows the evolution of the ensemble of chains in terms of (scaled) log-likelihood; the last 25% of draws of each chain are retained. Chains initialized at the tentative mode start at higher likelihoods and generate the "cliff" at initialization. The sampler also reports the log-weight on the history of the proposal distribution and the standard deviation of likelihoods. Early on, log-weights are positive, indicating adaptation; after convergence they are close to zero (around 10^{-7}). Standard deviations are about 1% of the mode log-likelihood. Acceptance rates are in the 40%–45% range, consistent with effective exploration. Refer to the text for model specifications.

E Reconstructing Distributional Data

Following the procedure below, one can obtain the high-frequency distributional data. Given the estimated coefficients $\left(\hat{\xi}_{mot}, \hat{\kappa}_{o_1 \cdots o_d, t}\right)$ and the associated polynomials, the user only needs to specify $\mathbf{u} \in [0,1]^d$, the domain of integration. In our application, we integrate to construct deciles. In practice, the integration bounds exclude the exact endpoints 0 and 1, which are replaced by 10^{-6} and $1 - 10^{-6}$, respectively. This is in accordance with our analysis in Appendix A and properties of our estimator.

Note. The quantile function shown is still a transformed quantile. It is $\text{asinh}(x/\mu)$ for x some quantile. Taking an inverse asinh transformation and multiplying by the appropriate aggregate scale μ gives you levels/conditional means.
Procedure:

$$\hat{\tilde{\theta}}_t = \hat{\Gamma}^{MF}\, \hat{F}_t \qquad \text{(Project factors)}$$

$$\hat{\theta}^j_{nt} = \sigma_{\zeta(n)} \times \hat{\tilde{\theta}}_{\zeta(n)t} + \mu^j_n + g(t)^j_n \qquad \text{(Unstandardize and add trend)}$$

$$\hat{\theta}^j_t = (\hat{\xi}^j_{m,o_1,t}, \ldots, \hat{\kappa}^j_{(o_1,\ldots,o_d),t}) \qquad \text{(Decomposition)}$$

$$\hat{\Xi}^{-1}_{jmt}(u_m) = \sum_{o=0}^{O} \hat{\xi}_{mot}\, Q_o(u_m) \qquad \text{(Reconstruct quantile function)}$$

$$d\hat{C}_{jt}(u_1, \ldots, u_d) = \sum_{o_1,\ldots,o_d=0}^{O} \hat{\kappa}_{o_1 \cdots o_d, t} \prod_{m=1}^{d} Q_{o_m}(u_m) \qquad \text{(Reconstruct copula density)}$$

Table 4: Description of Equation Components

Symbol	Description
$\hat{F}_t$	Estimated high-frequency factors
$\hat{\Gamma}^{MF}$	Factor loadings mapping factors to distributional observables
$\hat{\tilde{\theta}}_t$	Projected (standardized) factor representation
$\sigma_{\zeta(n)}$	Standard deviation term, varies by object
μ_n^j	Mean adjustment for coefficient n
$g(t)_n$	Non-parametric trend at time t
$\hat{\theta}_{nt}^j$	Unstandardized, trend-adjusted coefficients
$\hat{\xi}_{m,o,t}^j$	Marginal polynomial coefficients
$\hat{\kappa}_{(o_1,\ldots,o_d),t}^j$	Copula polynomial coefficients
$Q_o(u_m)$	Polynomial basis function of order o, O=11
$\hat{\Xi}_{jmt}^{-1}(u_m)$	Reconstructed quantile function (inverse CDF) for margin m
$d\hat{C}_{jt}(u_1,\ldots,u_d)$	Reconstructed copula density (dependence structure)

Notes: Table summarizes the role of each symbol in the above equations, moving from factor projections to reconstruction of marginal quantiles and copula density.

F Validation of Hyperparameter Choices

To evaluate our hyperparameter choices on the Bayesian estimation priors for the measurement error variances, we compare the series resulting from the Kalman smoother after estimation with the actual point estimates and their confidence bounds from the survey data.

Intuitively, if the prior mean for the measurement error variance is too low, it will force the estimator to exactly match each survey estimate of the distribution, despite the fact that each survey estimate is itself subject to measurement error. Thus, we should expect the smoother estimate to fall within the confidence bounds of each sample estimate at most with the corresponding confidence level of the bounds. The fact that the confidence level is an upper bound reflects that the estimated measurement error captures not only the sampling uncertainty that the confidence bounds capture, but also conceptual differences.

Choosing narrow measurement errors would overstate precision and potentially limit comovement with aggregates. Another reason not to be conservative with the measurement errors is that allowing for measurement error also accounts for the fact that we combine data from different sources to produce a consensus estimate. These different data sources, despite their individual detrend-

ing, may produce some temporarily divergent estimates of the distributions. Without sufficient measurement error, the consensus estimate is then forced to oscillate between these different distribution estimates over short time intervals, rather than capturing their co-movements.

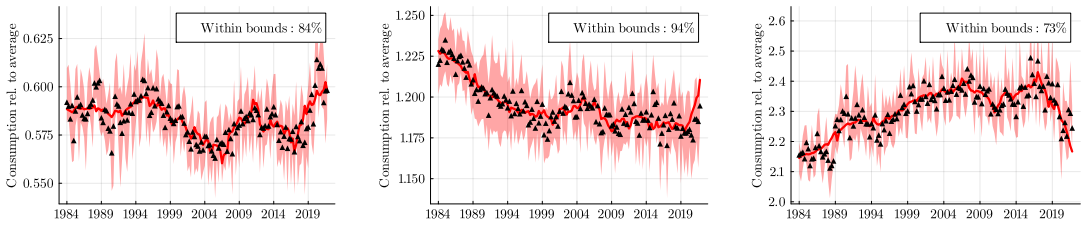
On the other hand, if the prior mean for the measurement error variance is too high, the estimator will treat the data as uninformative, and the smoother will miss the survey estimates more often and to a much greater extent than implied by its confidence bounds. We validate the choice of hyperparameters graphically for income and wealth in the SCF data and provide comprehensive summary statistics across all datasets and estimates. This visual inspection helps confirm that our smoothed series respect the noise inherent in the available microdata.

Figure 16 shows average consumption (top row), income (middle row) and wealth (bottom row) for the bottom 50 percent (first column), the next 40 percent (second column), and the top 10 percent (last column) of the respective distributions. It shows the point estimates from the surveys (black triangles), along with their 95% confidence limits, and the results from the Kalman smoother based on our estimates of the parameters of Equation (21). Overall, the smoothed estimates fall inside their respective confidence bounds in 87 out of the 108 observations (from (d) to (i)).

Table 5 provides a comprehensive summary of this validation approach. For all quantile functions and copulas, we report for each dataset and survey year how often the respective smoother estimate is within the confidence limits. Again, we use a confidence level of 95%. For the copulas and most quantile functions, the fraction of smoothed estimates within the confidence bounds is close to the confidence level (83–100 percent). The exceptions are the SIPP (69–70 percent), the SCF income quantiles (72 percent), and the CEX consumption quantiles (78 percent), for which our estimator suggests measurement error beyond sampling uncertainty, reflecting differences in sample design and income/wealth measurement.

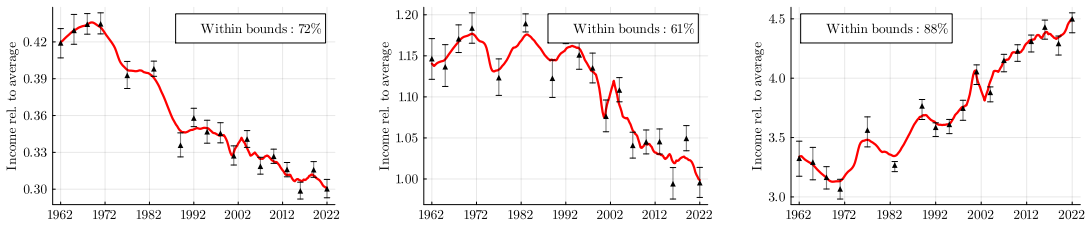
Figure 16: Comparison of Smoothed Distributional Data and Direct Survey Estimates

Consumption by Consumption



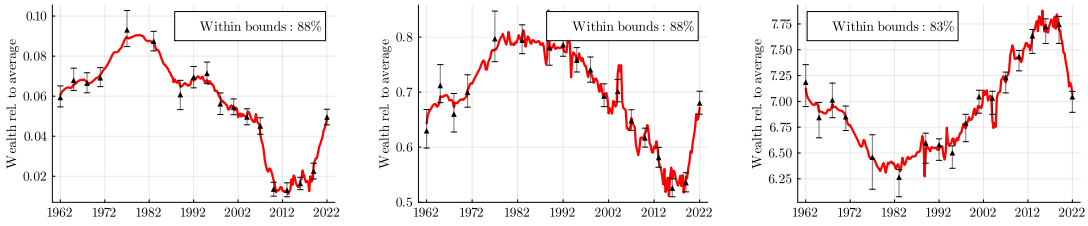
(a) Bottom 50 Percent (b) 50-90 Percent (c) Top 10 Percent

Income by Income



(d) Bottom 50 Percent (e) 50-90 Percent (f) Top 10 Percent

Wealth by Wealth



(g) Bottom 50 Percent (h) 50-90 Percent (i) Top 10 Percent

Model ▲ Data

Notes: Figure shows the average consumption, income, and wealth for the bottom 50 percent, 50-90 percent, and top 10 percent of households of the respective distribution. Triangle markers show the estimates from the individual survey waves together with 95% confidence bounds. The solid red line shows the baseline estimate from the Kalman smoother at the posterior mode. Consumption shows CEX data and reconstruction. Income and wealth show SCF data and reconstruction. The legend reports for each Panel the share of smoothed estimates within the confidence bounds of the survey waves.

Table 5: Deviations of Smoothed Estimates and Microdata: Fraction within Confidence Bounds

Measure	CEX	CPS	SCF	SIPP	PSID	Overall
Consumption quantiles	78%	—	—	—	100%	80%
Income quantiles	90%	93%	72%	70%	98%	84%
Wealth quantiles	—	—	83%	69%	99%	80%
Copula densities	95%	—	91%	88%	100%	96%

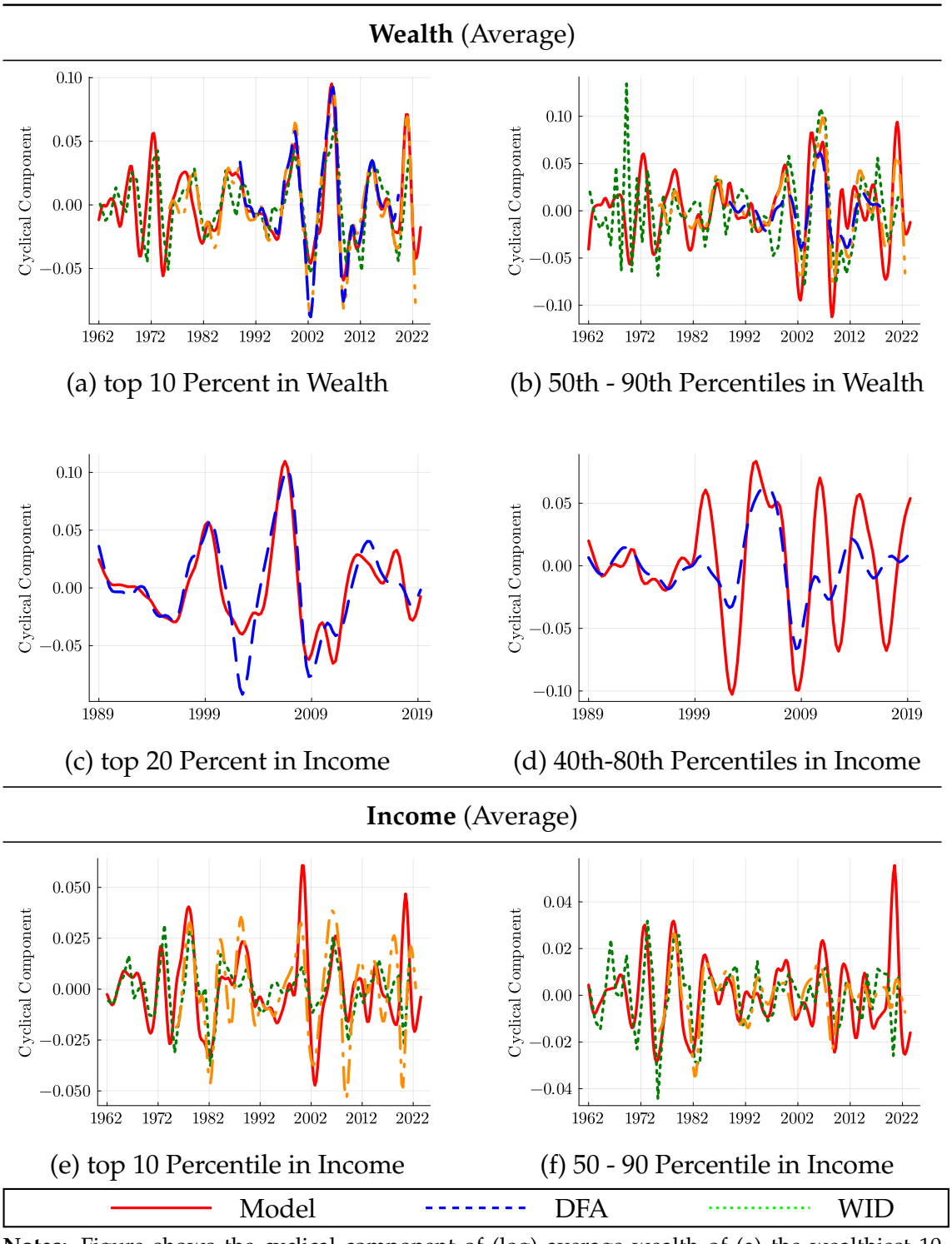
Notes: The table reports, by microdata and object, the fraction of estimates from the Kalman smoother at the posterior mode that fall within the 95% bootstrapped confidence intervals for the respective microdata. Quantile and copula estimates are defined on a decile grid.

G Comparison with External Estimates

The preceding sections established that our method predicts well out of sample and that the estimated distributions lie within sampling variability. We now assess whether the resulting high-frequency income and wealth dynamics align with external benchmarks. Since true distributional dynamics are unobservable, we compare our cyclical estimates to the *Distributional Financial Accounts (DFA)* (Batty et al. 2020) and the *World Inequality Database (WID)* (Piketty, Saez, and Zucman 2018). The DFA produces quarterly wealth distributions from the SCF using a different estimation approach, providing an "SCF flavor" benchmark. The WID provides annual estimates from administrative tax data without a time-series model, offering a robustness check for our assumption of stable and approximately linear distributional dynamics and their link to macro factors. To assess dependence dynamics (copula), we additionally compare our implied wealth-by-income series to the DFA.

Figure 17 reports the cyclical components (in logs). Panels (a)–(b) compare wealth by wealth group, focusing on the top 10% and the next 40% since the bottom half has wealth close to zero (Kuhn and Ríos-Rull 2016). Panels (c)–(d) compare wealth by income groups (informative about the joint distribution). Panels (e)–(f) compare income by income group (available in the WID). Across all cases, our estimates comove closely with DFA and WID and fall within the range of the external series.

Figure 17: Cyclical Component of Distributional Data vs. External Sources

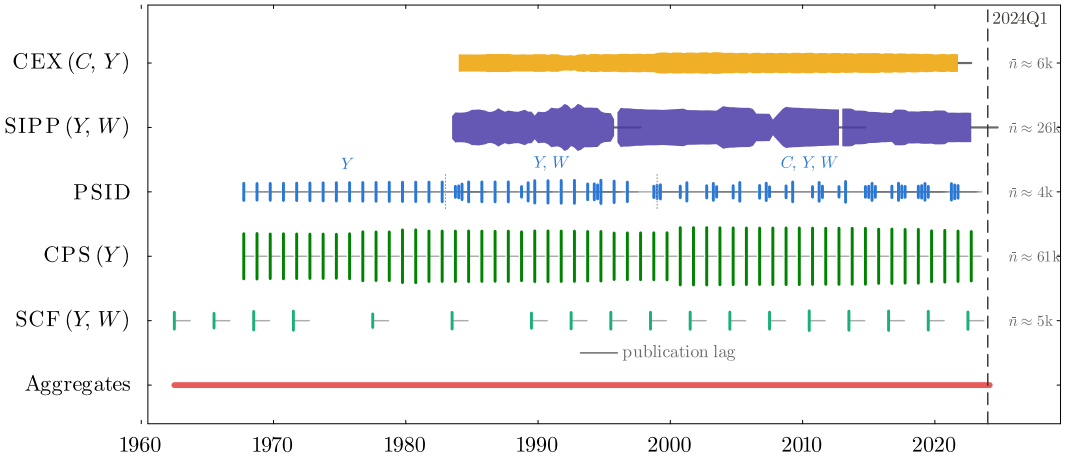


Notes: Figure shows the cyclical component of (log) average wealth of (a) the wealthiest 10 percent, (b) the next wealthiest 40 percent, (c) the 20 percent income-richest households, and (d) the next 40 percent income-richest households. Bottom row shows the cyclical component of (log) average income of (e) the income-richest 10 percent and (f) the next income-richest 40 percent. Red lines show cyclical components from baseline model at quarterly frequency. Dotted green lines show annual data from the *World Inequality Database* (WID). Dashed blue lines show quarterly data from the *Distributional Financial Accounts* (DFA). Cyclical components are obtained by an HP-filter with smoothing parameter $\lambda = 6$ for annual data and $\lambda = 1600$ for quarterly data.

H Data

The construction of these estimates relies on a great deal of data. An advantage of our method, however, is that it can incorporate these different microdata and their various differences in generating consensus estimates of the distributional data. Below, we describe the data, all expressed in 2019 dollars, and explain the mappings across data to ensure measures are broadly comparable (Curtin, Juster, and Morgan 1989; Czajka, Jacobson, and Cody 2003; Pfeffer et al. 2016). See Table 1 for information on their availability, and Figure 18 for a graphical summary of reference periods, variable coverage, sample sizes, and publication lags across sources.

Figure 18: Coverage, Sample Sizes, and Publication Lags across Microdata



Notes: Each row is one data source; parentheses list the variables it contributes (consumption C, income Y, wealth W). Colored marks show reference periods: one vertical bar per survey wave, or a solid band for sources available at least quarterly. The vertical extent of the marks is proportional to the square root of the average sample size per release, reported at the right margin. Gray whiskers indicate the typical lag between the end of a reference period and the publication of the data. For the PSID, the covered variables change over time: income throughout, wealth from the 1984 wave (a stock measured at the interview date, which we date to 1984Q2), and consumption from 1999. For the SIPP, only income is continuous at (sub)quarterly frequency; wealth is collected at topical-module waves before 2013 and once per reference year (as of December 31, dated to Q4) thereafter. Sample sizes are household counts computed from our estimation inputs (for the SCF, per implicate); $\bar{n}$ reports the average per release. Publication lags are approximate. The dashed line marks the end of our estimation sample (2024Q1).

H.1 SIPP

The SIPP panel is a nationally representative, individual-level survey known for providing high-frequency dynamics on employment, earnings, wealth, household composition and program participation in the U.S. For the data cleaning, the data are aggregated to the household-level, at quarterly frequency. Due to structural breaks induced by changing survey designs, we treat the SIPP as if it were three separate surveys (1983Q3-1995Q4, 1996Q1-2012Q4, 2013Q1-2022Q4). This is to minimize spurious fluctuations (e.g., seam bias) that obscure actual economic phenomena.

Income. We use total household income as reported by the Census Bureau in all releases: `THTOTINC` from the 1996 panel onward and its predecessor `H*TOTINC` (one variable per reference month) in the 1984–1993 panels. Income is reported monthly; we sum it to a quarterly figure only when all three months of the quarter are observed, treating the quarter as missing otherwise, and then annualize (multiplying by four) and deflate to real dollars.

Wealth. For the 2014 releases and onward, we use the `THNETWORTH` variable for wealth. For data releases prior, wealth is defined as total assets (`hhtwlth`) net total liabilities (`hhusdbt, hhscdbt`).

Note: The SIPP has undergone several major redesigns, and ignoring these seams, even with our prescribed data treatment, would lead to spurious swings in measured inequality—an issue documented by, among others, Moore (2008) and Czajka, Jacobson, and Cody (2003) and confirmed in our own diagnostics. To deal with this issue, we treat the SIPP as if it were three different surveys: (1) SIPP before 1996 (2) SIPP 1996-2012 and (3) SIPP 2013–2022. We found that around these seams, estimates tend to exhibit large jumps, which was at odds with their aggregate counterpart and general business cycle conditions. Additional observations were dropped as well ($2003Q4, 1999Q4, 1988Q1, 1985Q4$): all four are panel-boundary quarters in which mean income dips 11–20% below neighboring quarters while the sample collapses to as little as a third of its usual size (only exiting rotation groups remain in the field), with no counterpart in aggregate income at any of these dates. For the post-2013 SIPP, assets and liabilities are measured once per reference year (as of December 31) and replicated by the Census across all monthly records; we accordingly date the wealth-side variables to the fourth quarter of the reference year — the quarter the measurement refers to — rather than carrying the identical value in all four quarters,

while income, which is genuinely monthly, remains quarterly. Fortunately, the estimation framework allows each of the three SIPP subsurveys to carry its own measurementerror process, so we retain almost all available information while preventing seam effects from distorting the results.

H.2 SCF+

The SCF provides detailed household income and wealth data; SCF+ extends the modern surveys with historical waves compiled by Kuhn, Schularick and Steins (2020). The SCF+ is also augmented with the Forbes 400, from Forbes, for the years 2021 to 2023. For the years 1985 to 2020, we use the per capita dataset of Fernholz and Hagler (2023), whose observations originate from families of the Forbes 400.[48] Below we describe the measurements we used from the SCF+.

Income. Our definition of income follows Kuhn, Schularick, and Steins (2020), which consists of the following components: (1) labor income (i.e., earnings) (2) income from public transfers (3) income from professional practice and self-employment (4) income from rents (5) dividend income and (6) business/farm income. A different taxonomy that illustrates these components are taxable and transfer income.

Assets. Total assets include (1) liquid assets such as a household's checking and savings account, CDs, call/money market accounts, short-term government bonds, and mutual funds (2) illiquid assets such as housing and other real estate, automobiles (3) defined-contribution retirement plans (4) the cash value of life insurance (5) stocks and (6) business equity.

Debt. We define debt of a household as the sum of personal (mostly unsecured) debt and housing (mortgage) debt. Housing debt includes debt from all prop-

48. Accessing the data via the usual link (e.g., `http://www.forbes.com/ajax/list/data?year=2005&uri=forbes-400&type=person`) returns Forbes 400 data, but unfortunately incomplete from the 1990s until the late 2000s—with only around 200 of the 400 observations there. Using Fernholz and Hagler (2023), with the same procedure imposed by Bricker, Hansen, and Volz (2019), generates the same result as in Bricker, Hansen, and Volz (2019) (increases the top wealth share by 1.5%), but is more complete (has almost all 400 every year). Implementation details: Forbes 400 observations are dated to the third quarter, matching the timing convention of the SCF; each Forbes 400 member is entered into all five SCF implicates and assigned the smallest sampling weight observed in the SCF, so that each represents (approximately) a single household. To avoid double counting the top tail, we follow Bricker, Hansen, and Volz (2019): within year-specific wealth bins defined by multiples of the Forbes 400 minimum $F_y^{\min}$ ($[1, 1.5)$, $[1.5, 2.5)$, $[2.5, 5)$, and $[5, \infty) \times F_y^{\min}$), SCF and Forbes observations are reweighted in proportion to their relative frequencies; observations below $F_y^{\min}$ keep their original weights. Forbes records carry net worth only; their income and consumption are treated as missing.

erties and any loans made against the housing e.g., through HELOCs. Personal debt includes car loans, education loans, any loans from relatives, credit card debt, medical debt and legal debt.

Wealth. Wealth is total assets net of total debt of a household.

H.3 PSID

The Panel Study of Income Dynamics complements the SCF+ extraordinarily well, as they take our estimations over more than half a century. In comparison to the post-1983 SCF, a deeper analysis of their similarity can be found in Pfeffer et al. (2016).

Income. The PSID has collected family income annually from 1968 to 1996 and then biennially from 1997 to 2021. Its measure of income is the sum of taxable income, transfers, and social security for the reference person, the spouse/partner (if any) and other members of the family.[49]

Assets. Data collection on household wealth took place in 1984, 1989, 1994, and then every wave beginning in 1999. The data on assets is split into liquid and illiquid assets. Although minor, the definition of liquid assets will vary between datasets, so careful attention is warranted here. Liquid assets for the PSID include checking and savings accounts, short-term instruments such as money-market accounts, certificates of deposit, and treasury bills. Illiquid assets include business equity, financial assets held in mutual funds, stocks, bond funds, investment funds; real assets held in real estate, vehicles like motor homes, boats, trailers, and cars; and retirement wealth in private annuities or IRAs.

Debt. For the PSID, we achieve the same debt split: personal and mortgage debt. This includes all kinds of real-estate debt, and unsecured debt such as credit card debt, student loans, medical debt, legal debt, and loans from relatives.

Wealth. Wealth is total assets net total debt of a household.

Consumption. Studying papers such as Skinner (1987), Cutler et al. (1991), Flavin and Yamashita (2002), Attanasio, Hurst, and Pistaferri (2014), and Attanasio and Pistaferri (2014), we define consumption as the sum of these expenditures: food, rent (for renters), housing rental equivalence (for home-owners), utilities, health,

49. In the PSID, a family is a group of people living together who are economically interdependent.

public transport, education, and childcare. We set the housing rental equivalence to be 6% of the home market value reported by households in the PSID. Consumption data is only available from 1999 in a biennial interval.

H.4 CPS

We use the Current Population Survey (CPS) Annual Social and Economic Supplement (ASEC). The sample is designed primarily to produce estimates of the labor force characteristics and runs from 1967 to 2022. Similar to the SIPP, we treat the CPS as a combination of two separate surveys to avoid fluctuations driven by survey design: 1967Q4-1993Q4 and 1994Q4-2022Q4.

Income. Income data are collected as part of the ASEC for the months of February, March and April as a supplement to the regular CPS monthly labor force interviews. The ASEC asks each person in the sample who is 15 years old and over about the amount of income received from a list of sources in the previous calendar year. We treat these observations as being observed in quarter four of the previous calendar year. For details on top-coding, see https://cps.ipums.org/cps/topcodes_tables.shtml.

H.5 CEX

The Consumer Expenditure Survey (CEX) is the most comprehensive household survey in the U.S. for recording the consumption habits of households. The CEX has two components: the interview survey (IS) and the diary survey (DS). The interview survey has sufficiently rich data on what we need, so we only use data from this component. Within this component, there are several files, each of which pertains to a topic, from which we can extract information. The following table breaks down each category of consumption, defining which `UCCs` belong to which category and which file it can be found in. All of these categories will combine to make the consumption variable. The table will also define wealth concepts of the CEX we use in our study. Since each household consumption record is with respect to a `UCC`, we find this presentation most apropos.

Item	UCCs / FMLI label	File
	Consumption	
Food	190904, 790220, 190901, 190902, 190903, 790410, 790430, 200900, 790330, 790420, 800700, 790230, 790240	MTBI
Rent	210110, 800710	MTBI
Utilities	250111, 250112, 250113, 250114, 250211, 250212, 250213, 250214, 250221, 250222, 250223, 250224, 250901, 250902, 250903, 250904, 250911, 250912, 250913, 250914, 260111, 260112, 260113, 260114, 260211, 260212, 260213, 260214, 270211, 270212, 270213, 270214, 270411, 270412, 270413, 270414,270101, 270102, 270104, 270105, 270310, 270311, 690116, 270901, 270902, 270903, 270904	MTBI
Health	570110, 570111, 570210, 570220, 570230, 560110, 560210, 560310, 560330, 560400, 340906, 540000, 550110, 550320, 550330, 550340, 570901, 570903, 570240, 580111, 580112, 580113, 580114, 580311, 580312, 580901, 580903, 580904, 580905, 580906, 580400, 580907	MTBI
Public Transport	520531, 520532, 530311, 530312, 530501, 530902, 530210, 530411, 530412, 520511, 520512, 520521, 520522, 520542, 520902, 520903, 520904, 520905, 520906, 520907, 530110, 530901, 520110, 520310	MTBI
Education	210310, 370903, 390901, 660110, 660210, 660310, 660900, 670110, 670210, 670901, 670902, 800802, 800804, 690111, 690112, 660410, 660902, 670410, 670903, 690114, 690310	MTBI
Child care	340210, 340211, 340212, 670310, 660901	MTBI
Rental Equivalence	910050, 800721 (market value of home), SIMHOUSX, RENTEQVX	FMLI, MTBI

Gas & Vehicle Repairs	470111, 470112, 470113, 470220, 470211, 470212, 480110, 480212, 480213, 480214, 490110, 490211, 490212, 490221, 490231, 490232, 490311, 490312, 490313, 490314, 490318, 490319, 490411, 490412, 490413, 490501, 490502, 490900, 520410, 480215, 620113	MTBI
	Other Concepts	
Housing Debt	QBLNCM1X, QBLNCM2X, QBLNCM3X, QBLNCM1G, PRINAMTX	MOR
Personal Debt	6001, 6002 (1990–2013), 5400, 5500, 5600, CREDITX, STUDNTX, OTHLONX, CREDITX1, CREDITX5, QBALNM1X	MTBI, ITBI, FMLI, FN2
Liquid Assets	SAVACCTX, CKBKACTX, USBNDX, 920010, 920020, 920030, 5100, LIQUIDX	FMLI, ITBI
Financial Assets	5800, 920040, STOCKX, SECESTX, OTHASTX	FMLI, ITBI
Income	FINCBTAX	FMLI

Table 6: Table shows, by item, the identifiers necessary to construct each component of consumption, income and wealth for the CEX. The location of these identifiers can be found under the *File* column.

H.6 Observation frequency and seasonal adjustment

Not every series we extract from the microdata is a quarterly flow, and only quarterly flows can carry a seasonal component. In the SIPP, income is genuinely quarterly (monthly reports aggregated to quarters), while assets and liabilities come from topical modules: in the post-2013 panels they are measured once per reference year and dated to the fourth quarter (see the SIPP note above), and in the earlier panels the asset modules rotate across interview waves, so the calendar quarter of a wealth observation reflects rotation-group composition rather than a seasonal cycle. In the CEX, consumption is a genuine quarterly flow (three-month recall), whereas income is collected as a twelve-month recall—a rolling annual concept that is season-free by construction. The PSID, CPS, and SCF+ are annual or lower frequency; their moment series are never seasonally adjusted. In summary:

Series	Concept
SIPP (all eras), income	quarterly flow
SIPP 1983–2012, wealth	rotating topical module
SIPP 2013–2022, wealth	annual, dated Q4
CEX, consumption	quarterly flow
CEX, income	twelve-month recall

Seasonal adjustment of the quarterly moment series (X-13ARIMA-SEATS, X-11 adjusted component) is therefore applied series by series, subject to two screens. First, an observation-pattern screen: a series qualifies only if it is observed in at least three distinct calendar quarters; a series measured only in the fourth quarter (post-2013 SIPP wealth) has no identifiable seasonal component, and adjustment would merely perturb the genuine observations. Second, an identifiability test: X-13's M7 statistic must indicate identifiable, stable seasonality ($M7 < 1$); the diagnostic is a by-product of the same X-13 run that produces the adjusted series, so the test is costless. Series failing either screen enter the estimation unadjusted. The screening decisions are made once on the point data and reused in the bootstrap that constructs the measurement-error distributions and confidence intervals. Model-simulated validation datasets (the HANK replicas) are never seasonally adjusted, as the model has no seasonal component.

H.7 Aggregates

Together with the microdata, we specify a model component that captures the various aggregate shocks that potentially buffet the joint distribution of consumption, income, and wealth. This is represented in the state equation of the state-space model. The aggregate data we rely on to extract this information come from the FRED-QD (McCracken and Ng 2021). This has various macrodata that has already been seasonally-adjusted on industrial production, employment, housing, inventories, prices, earnings, productivity, household expectations, household balance sheets, interest rates, credit, etc. You can find more information on https://research.stlouisfed.org/econ/mccracken/fred-databases/.

Before performing the PCA on the aggregates, we transform each series to stationarity following the codes proposed by McCracken and Ng (2021), with one exception: price indices enter as inflation rates ($\Delta\log$) rather than changes in inflation.[50] We stack the series with four lags of each series, standardize each

50. Under the second-difference code, the price block is dominated by high-frequency noise. Inflation is treated as stationary by assumption; ADF rejects a unit root for most but not all inflation series, a standard finding given its high persistence.

column, and extract 11 principal components, a number suggested by marginal data density. The loadings are estimated excluding 2020Q12021Q1, so that the pandemic quarters do not dominate the components, and all quarters, including the pandemic ones, are then projected onto them. The 11 components enter the state vector as latent states with their own dynamics and measurement error, rather than as exogenous inputs.

I Out of Sample Correlations

Table 7: Out of Sample Performance

Condition	Bottom			Middle			Top		
	C	I	W	C	I	W	C	I	W
Excluding Housing Cycle Wealth									
Entire Series	-	0.98	0.98	-	0.99	0.96	-	0.99	0.95
Specific Timeframe	-	0.92	0.98	-	0.94	1.00	-	0.91	0.99
Excluding Last 4 Years									
Entire Series	-	0.98	0.95	-	0.98	0.96	-	0.98	0.95
Specific Timeframe	-	0.84	0.94	-	0.75	0.85	-	0.77	0.88
Every 4 Years									
Entire Series	0.98	0.99	-	0.99	0.98	-	0.99	0.99	-
Specific Timeframe	0.97	0.99	-	0.98	0.97	-	0.98	0.98	-

Notes: Table reports correlations between the baseline cyclical estimates and the missing-data models, split by panel. Entire Series reports correlations for all estimates in the estimation timeframe. Specific Timeframe reports correlations of estimates in periods where data was intentionally left out of the estimation of the specific missing-data model. **Bottom**, **Middle**, and **Top** are the bottom 50, next 40, and top 10 of the respective distribution, denoted by **C, I,** and **W**. **C** is for consumption, **I** is for income, and **W** is for wealth. For the first two panels, correlations are made between SCF model estimates (no consumption). The final panel presents correlations from CEX model estimates (no wealth). Specific models (in header) are discussed in Section 4.

J What Drives the Distribution Dynamics?

Since the dynamic factor model we estimate is linear, it allows us to answer a natural question about the estimated system: how much of the movement in the distribution is spanned by aggregate dynamics, and how much is autonomous—variation in the cross-section that the aggregate history does not account for, such as changes in risk-sharing, in idiosyncratic risk, or in the incidence and timing of transfers?[51] Bayer, Born, and Luetticke (2024) analyze this question with a struc-

51. Formally, the autonomous component is the part of the one-step-ahead forecast error of the distributional factors that is orthogonal to the contemporaneous forecast errors of the entire aggregate block. It is defined relative to our aggregate factor set: forces imperfectly captured by that block, or reaching the cross-section before the measured aggregates, are attributed to the au-

Table 8: FEVD on the Distributional Factors, percent of forecast-error variance

Distributional Factor No.	1	2	3	4	5	6	7	8
Spanned by aggregates	15	23	31	15	20	21	25	69
Autonomous	85	77	69	85	80	79	75	31

Notes: Table reports the forecast error variance decomposition (FEVD) for the eight estimated distributional factors over a five-year horizon. The rows show the percentage of variation in each factor spanned by innovations to the aggregate block versus the autonomous remainder. Identification is via a blocked Cholesky with aggregates ordered first, which attributes all common variation to the aggregate block.

tural HANK model and selected distributional information, finding that aggregate shocks explain well the actual evolution of wealth inequality even at lower frequencies. In a more reduced-form manner, but with the same message, Kuhn, Schularick, and Steins (2020) argue that price changes are essential for understanding wealth dynamics.

We revisit this question within our much less structural DDFM, using two complementary decompositions: a forecast error variance decomposition (FEVD) of the distributional factors and, using the linear mapping from factors to observables, of observable distributional moments;[52] and a historical decomposition (HD) that attributes movements in the smoothed factors over time, with special attention to NBER recession windows.

Because the reduced-form innovations of the state equation are correlated (Ω is non-diagonal), both exercises require an orthogonalization, and we deliberately choose one that avoids structural labels: a *blocked* Cholesky factorization with the aggregate block ordered first, formalized below. The FEVD then asks how much of the forecast-error variance of each distributional factor is spanned by innovations to the aggregate block, and how much is orthogonal to all of them—autonomous in the sense above. No assumption is made about causal ordering within either block, and we do not interpret the orthogonalized innovations as structural shocks; where we write "shocks" below, orthogonalized innovations are meant.

Table 8 reports the results for the factors. Aggregate innovations span 15 to 31 percent of the variation of every factor, except the eighth factor which spans 69 percent. Thus, the autonomous component is not negligible—up to 85 percent

tonomous component—biasing its share upward—while ordering aggregates first in the orthogonalization attributes all common variation to aggregates—biasing it downward. Autonomous innovations may still affect aggregates at later horizons through the feedback matrix C.

52. We refer the reader to Appendix E for the transformations from factors to observables. In the appendix we use $\bar{g}_n(t)$, the average trend, rather than the non-parametric trend $g_n(t)$.

for factor 1 and 4.

Table 9: FEVD of group means and population shares. Percent of forecast-error variance

	Conditional means of ...					
	Consumption		*Income*		*Wealth*	
Group	Auton.	Agg.	Auton.	Agg.	Auton.	Agg.
Bottom 20%	82	18	75	25	70	30
Bottom 20 – 40%	83	17	71	29	75	25
Middle 40 – 80%	78	22	69	31	78	22
Top 20%	75	25	69	31	81	19
	Population shares with ...					
	Low Consumption		*High Consumption*			
Low Y, Low W	43	57	41	59		
Low Y, High W	42	58	40	60		
High Y, Low W	41	59	40	60		
High Y, High W	40	60	46	54		

Notes: Table reports the forecast error variance decomposition (FEVD, five-year horizon) of Table 8 mapped to observables consumption, income, and wealth. The top panel looks at consumption, income, and wealth group means across four household groups given by the bottom 20%, the next 20%, the next 40% and the top 20% of the distribution of the respective variable (consumption (C), income (Y), and wealth (W)). The bottom panel reports the corresponding FEVD for the population shares that have any combination of above/below median consumption, income, and wealth. Identification is via a blocked Cholesky with aggregates ordered first, which attributes all common variation to the aggregate block; autonomous shares (Auton.) are therefore lower bounds.

While these results for the factors are informative, it is not clear what this means for moments of the data that one would typically look at, like consumption of the poor or the rich. As these moments are linear in the factors, we can directly apply the FEVD. Table 9 reports the aggregate-spanned share for these better interpretable moments, decomposing the marginal distributions of consumption, income, and wealth for four household groups: the bottom 20%, the bottom 20 – 40%, the middle 40 – 80%, and the top 20% (top panel). We find that, typically, aggregate innovations span 17-31 percent of the variation in consumption, income and wealth. Looking at changes of the copula, here represented as the change in the share of four population groups within the joint distribution—for example, all observations with below median consumption, income, and wealth ("Low Consumption, Low Y, Low W")—the autonomous component is less important, accounting for around 40 percent of the fluctuations. At the same time, this means that even for the dependence structure, aggregate innovations span

the majority of fluctuations.

Two insights follow. First, similar to Angeletos, Collard, and Dellas (2020), we find evidence of a single main business-cycle factor: aggregate innovations account for 69 percent of the forecast-error variance of factor 8. For every other factor, they account for at most 31 percent, and on average around 25 percent. Second, the autonomous component is nevertheless smallest for the dependence structure—the copula—at around 40 percent, against around 70-80 percent for the marginals. Aggregate fluctuations thus matter little for the shape of the marginal distributions, but considerably for who sits where in the joint distribution.

The remainder of this appendix formalizes the identification and complements the FEVD with the historical decomposition of the smoothed factor paths.

On the Block Cholesky. Let $f_t \in \mathbb{R}^r$ denote the vector of estimated distributional factors and let $Y_t \in \mathbb{R}^q$ denote the aggregate observables. We work with a companion-state representation

$$x_t \;=\; \Phi x_{t-1} + \epsilon_t, \qquad \epsilon_t \sim (0, \Omega), \qquad x_t \in \mathbb{R}^n, \;\; n = rp + q, \tag{35}$$

where x_t stacks $(f_t, f_{t-1}, ..., f_{t-p+1}, Y_t)$ for the p lags and $\Phi \in \mathbb{R}^{n \times n}$ is the corresponding companion transition matrix.

Define $J \in \mathbb{R}^{r \times n}$ as the selector that extracts the contemporaneous factors from the state,

$$f_t \;=\; J x_t, \qquad J = \begin{bmatrix} I_r & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{36}$$

Let $S_f \in \mathbb{R}^{n \times r}$ and $S_y \in \mathbb{R}^{n \times q}$ embed factor and aggregate innovations, respectively, into the companion innovation vector:

$$S_f = \begin{bmatrix} I_r \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad S_y = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ I_q \end{bmatrix}. \tag{37}$$

We collect shocks using $S \in \mathbb{R}^{n \times (r+q)}$. Under the blocked ordering with aggregates first,

$$S \;=\; \begin{bmatrix} S_y & S_f \end{bmatrix}, \tag{38}$$

and define the selected innovation subvector $\epsilon_t^{(S)} \equiv S^\top \epsilon_t \in \mathbb{R}^{r+q}$ with covariance

$$\Omega_{\text{sub}} \;\equiv\; S^\top \Omega S \;\in\; \mathbb{R}^{(r+q)\times(r+q)}. \tag{39}$$

Identification. Partition Ω_{sub} conformably as

$$\Omega_{\text{sub}} = \begin{bmatrix} \Omega_{YY} & \Omega_{YF} \\ \Omega_{FY} & \Omega_{FF} \end{bmatrix}, \tag{40}$$

where the Y block corresponds to aggregate shocks and the F block to distributional shocks. The blocked Cholesky factorization constructs an impact matrix

$$B_{\text{sub}} = \begin{bmatrix} B_{YY} & 0 \\ B_{FY} & B_{FF} \end{bmatrix}, \qquad \Omega_{\text{sub}} = B_{\text{sub}} B_{\text{sub}}^\top, \tag{41}$$

with

$$B_{YY} = \text{chol}(\Omega_{YY}), \tag{42}$$

$$B_{FY} = \Omega_{FY} B_{YY}^{-\top}, \tag{43}$$

$$B_{FF} = \text{chol}\big(\Omega_{FF} - B_{FY} B_{FY}^\top\big). \tag{44}$$

This yields orthogonalized innovations $e_t \in \mathbb{R}^{r+q}$ satisfying

$$\epsilon_t^{(S)} = B_{\text{sub}} e_t, \qquad \mathbb{E}[e_t e_t^\top] = I_{r+q}, \qquad e_t = B_{\text{sub}}^{-1} \epsilon_t^{(S)}. \tag{45}$$

Embedding back into the full innovation space gives

$$B \;\equiv\; S B_{\text{sub}} \;\in\; \mathbb{R}^{n\times(r+q)}. \tag{46}$$

J.1 Historical Decomposition

To characterize the historical drivers of distributional dynamics, we implement a historical decomposition that attributes the time path of each estimated factor to aggregate versus distributional shocks. The decomposition is computed from the state transition equation in companion form and uses the same blocked-Cholesky identification as in the FEVD, ensuring that the two shock blocks are orthogonal and that the aggregate block is ordered first.

Let ϵ_t denote the smoothed reduced-form innovations from the Kalman smoother. Because $\hat{\Omega} = \mathbb{E}[\epsilon_t \epsilon_t^\top]$ is non-diagonal, we first map ϵ_t into orthonormal struc-

tural innovations e_t via the blocked-Cholesky impact matrix. We then propagate the contribution of each "shock" block through the transition equation to obtain time-t contributions to each factor. This yields an additive decomposition of the smoothed factor path into aggregate and distributional components (plus an initial zero condition term).

Formally, given smoothed reduced-form innovations $\{\epsilon_t\}_{t=1}^{T}$ (e.g., from the Kalman smoother), compute structural innovations

$$e_t \;=\; B_{\text{sub}}^{-1} S^{\top} \epsilon_t. \tag{47}$$

Let $e_{Y,t}$ and $e_{F,t}$ denote the aggregate and distributional blocks of e_t under the aggregate-first ordering. Define the block-attributed innovations

$$\epsilon_t^{\text{agg}} \;=\; B_{[:,Y]} e_{Y,t}, \qquad \epsilon_t^{\text{dist}} \;=\; B_{[:,F]} e_{F,t}, \tag{48}$$

and propagate contribution states

$$z_t^{\text{agg}} = \Phi z_{t-1}^{\text{agg}} + \epsilon_t^{\text{agg}}, \qquad z_t^{\text{dist}} = \Phi z_{t-1}^{\text{dist}} + \epsilon_t^{\text{dist}}. \tag{49}$$

The contributions to the factors are then

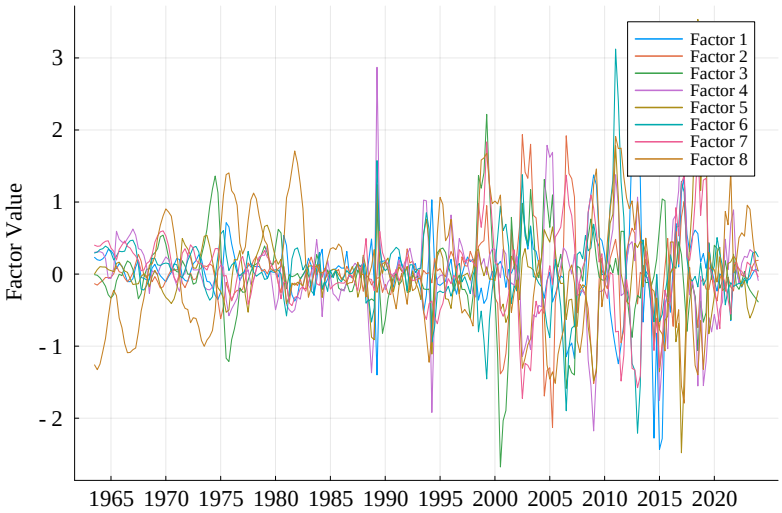
$$c_t^{\text{agg}} = J z_t^{\text{agg}}, \qquad c_t^{\text{dist}} = J z_t^{\text{dist}}, \qquad \hat{f}_t = J x_t^{\text{smoothed}}, \tag{50}$$

so that the smoothed factors admit the additive decomposition

$$\hat{f}_t = c_t^{\text{agg}} + c_t^{\text{dist}} \tag{51}$$

Figure 20 summarizes the historical decomposition by NBER recession window (peak to trough) by reporting *absolute contribution shares*—the share of the total absolute contribution accounted for by the aggregate block versus the distributional block. Level paths of the factors are shown in Figure 19. While recessions naturally vary in duration, magnitude, and context, a clear pattern emerges: in most cases, contributions from the distributional block exceeds that of the aggregate block. This means that during each recession window, the total magnitude of movements attributed to aggregate shocks is about less than the magnitude attributed to distributional shocks. So recessions are periods where both channels are comparably active, distributional slightly more, though we cannot comment (yet) on whether they are possibly offsetting one another. This is not at odds with the variance decomposition above: the FEVD measures unconditional forecast-

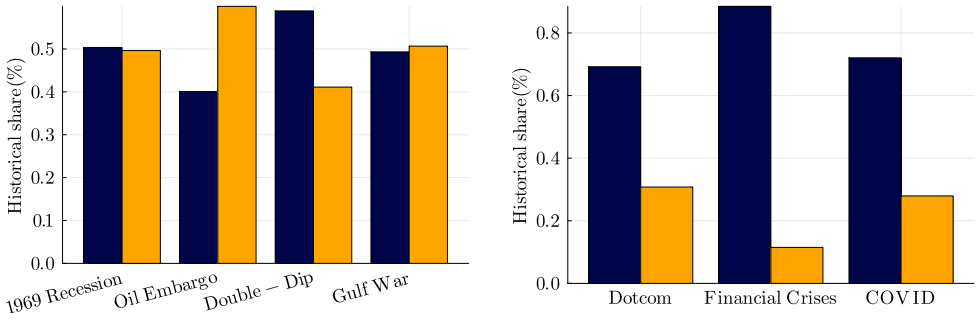
Figure 19: Distributional Factors



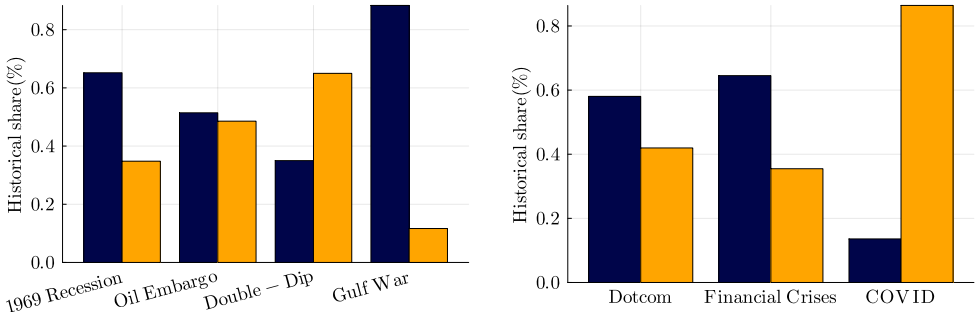
Notes: Raw factor estimates from the baseline estimation of the model. Estimates run from 1962Q3 until 2024Q1.

error variance at a five-year horizon, whereas the recession windows condition on particular episodes and compare absolute contributions to the realized paths. Taken together, the two results say that autonomous innovations contribute substantially to distributional fluctuations on average and during recessions.

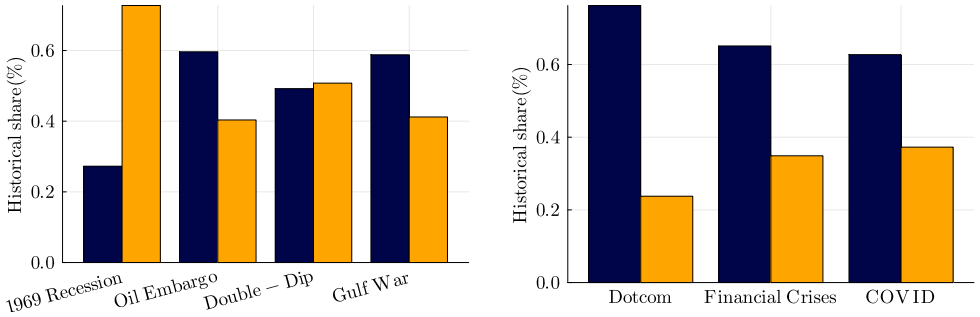
Figure 20: Historical Decomposition of Distributional Factors



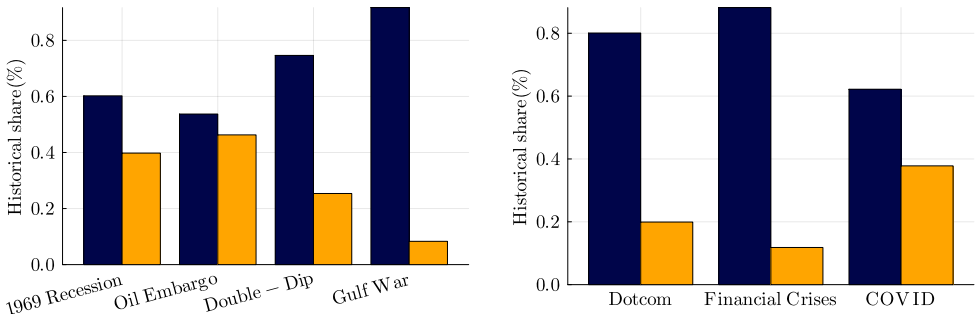
Factor #1



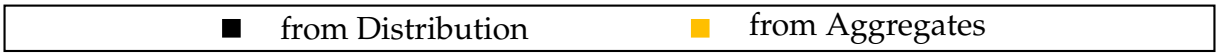
Factor #2

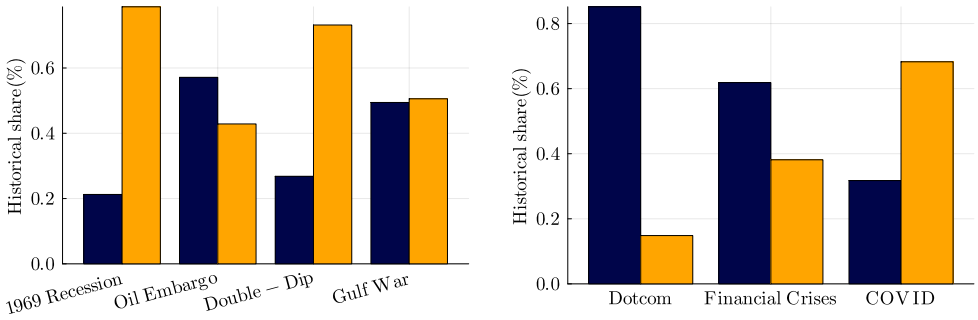


Factor #3

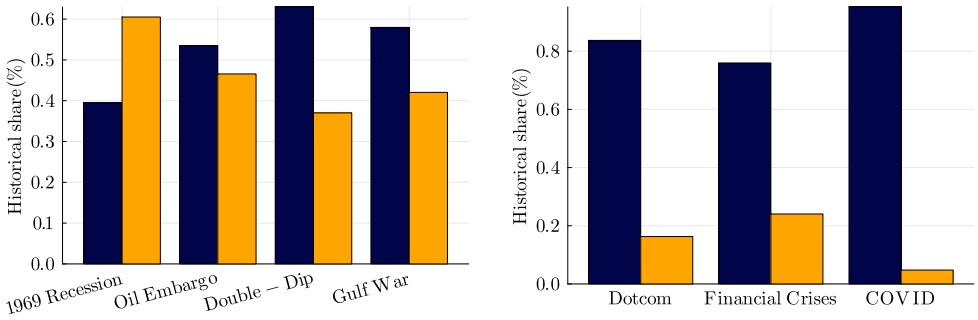


Factor #4

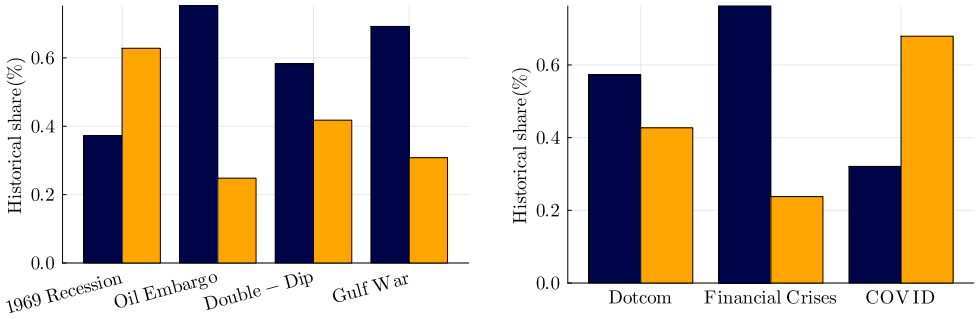




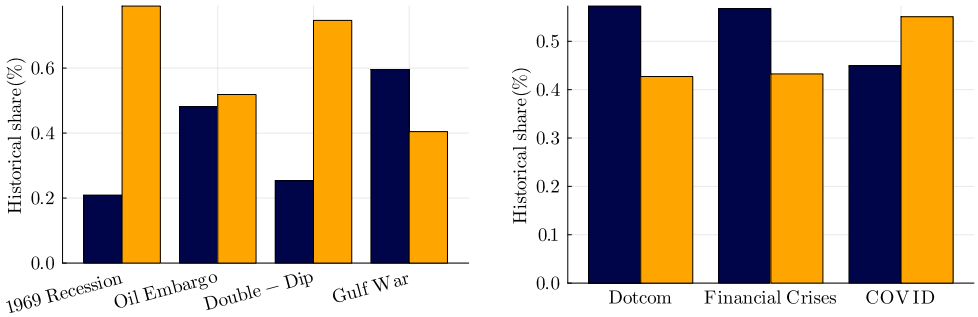
Factor #5



Factor #6



Factor #7



Factor #8

Notes: Event windows are from the official NBER recession dates, from peak to trough (1969 Unemployment Recession: 1969Q4–1970Q4, Oil Embargo: 1973Q4–1975Q1, Double-Dip Recession: 1981Q3–1982Q4, Gulf War: 1990Q3–1991Q1, Dot-com Recession: 2001Q1–2001Q4, Global Financial Crisis: 2007Q4–2009Q2, and COVID-19: 2019Q4-2020Q2).

J.2 Forecast Error Variance Decomposition

Table 8 reports the FEVD at the factor level, quantifying the share of each distributional factors forecast error variation explained by distributional shocks and aggregate shocks. Again, this is identified via a block cholesky with the aggregates ordered first. These contributions are then mapped to the level of observables, whose results are shown in Table 10, decomposing the marginal distributions of consumption, income, and wealth for three conventionally used household groups: the bottom 50%, the next 40%, and the top 10%.

Orthogonalized FEVD Define impulse responses of the factors to the orthogonal shocks:

$$\Psi_s \equiv J\Phi^s B \in \mathbb{R}^{r\times(r+q)}, \qquad s = 0, 1, \ldots, H-1. \tag{52}$$

For factor i and horizon H, the FEVD share attributable to distributional shocks (index set $\mathcal{D}$) and aggregate shocks (index set $\mathcal{A}$) is

$$\mathrm{FEVD}_i^{\mathrm{dist}}(H) = \frac{\sum_{s=0}^{H-1}\sum_{j\in\mathcal{D}}\Psi_s(i,j)^2}{\sum_{s=0}^{H-1}\sum_{j=1}^{r+q}\Psi_s(i,j)^2}, \qquad \mathrm{FEVD}_i^{\mathrm{agg}}(H) = \frac{\sum_{s=0}^{H-1}\sum_{j\in\mathcal{A}}\Psi_s(i,j)^2}{\sum_{s=0}^{H-1}\sum_{j=1}^{r+q}\Psi_s(i,j)^2}. \tag{53}$$

In summary, we believe two key insights can be inferred from the results presented here. From Table 10, we find substantial heterogeneity both across marginals and across household groups. Consumption exhibit the strongest exposure to distributional shocks. Income behaves similarly for most groups, while wealth scales with the conditional mean. This pattern suggests that aggregate shocks map more directly into income and wealth, whereas consumption reflects heterogeneous smoothing, insurance, and behavioral responses. The larger distributional component in consumption relative to wealth indicates that heterogeneity (constraints, insurance, preferences) may matter more for consumption dynamics than for wealth dynamics.

Lastly, looking across marginals, the distributional component is largest for the bottom 40 percent of consumption, the bottom 20 percent of income, and the top 20 percent of wealth, although the differences across groups are modest. This pattern is consistent with heterogeneity mattering most where (i) idiosyncratic labor-market risk and borrowing constraints dominate outcomes, at the bottom of the consumption and income distributions, and (ii) wealth is driven by idiosyncratic returns on concentrated holdings such as private businesses, at the top of the wealth distribution. In contrast, the top of the income distribution is comparatively more aggregate-driven, consistent with a larger role for common

shocks to capital and business income there.

Table 10: Decomposition of the Observables

Group	Distributional (%)	Aggregate (%)
	Consumption	
Bottom 50%	81.85	18.15
Next 40%	74.12	25.88
Top 10%	80.79	19.21
	Income	
Bottom 50%	72.15	27.85
Next 40%	70.79	29.21
Top 10%	64.21	35.79
	Wealth	
Bottom 50%	70.04	29.96
Next 40%	79.60	20.40
Top 10%	81.73	18.27

Notes: Table reports the forecast error variance decomposition (FEVD) of Table 8 mapped to observables consumption, income, and wealth, across three household groups. Identification is via a blocked cholesky with aggregates ordered first. Entries show the share of variation explained by distributional shocks versus aggregate shocks. Household groups are defined as the bottom 50%, bottom 50-90%, and top 10% of the respective marginal distribution, which is italicized. This is over a five-year horizon.

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